



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES
STA. MESA MANILA
COLLEGE OF ENGINEERING

Self-Study Report (SSR)

Bachelor of Science in Civil Engineering

Criterion 5 Curriculum

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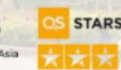
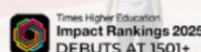
A LEADING COMPREHENSIVE
POLYTECHNIC UNIVERSITY IN ASIA



QUALITY MANAGEMENT
SYSTEM IS CERTIFIED
ACCORDING TO ISO 9001



EDUCATIONAL ORGANIZATION
MANAGEMENT SYSTEM IS CERTIFIED
ACCORDING TO ISO 21001



CRITERION 5. CURRICULUM

The curriculum of the **Bachelor of Science in Civil Engineering (BSCE)** program at the Polytechnic University of the Philippines is systematically designed and implemented in accordance with institutional policies, Commission on Higher Education (CHED) requirements, professional regulatory standards, and the curriculum criteria prescribed by the **Philippine Technological Council–Accreditation and Certification Board for Engineering and Technology (PTC-ACBET)**. The program has been granted **Certificate of Program Compliance (COPC) No. 108, Series of 2021** by CHED, affirming that the BSCE curriculum complies with the requirements prescribed under **CHED Memorandum Order No. 92, Series of 2017**, otherwise known as the Policies, Standards, and Guidelines for the Bachelor of Science in Civil Engineering Program. The issuance of the COPC provides institutional evidence that the program satisfies CHED standards on curriculum structure, faculty qualifications, learning resources, laboratory provisions, and overall program implementation.

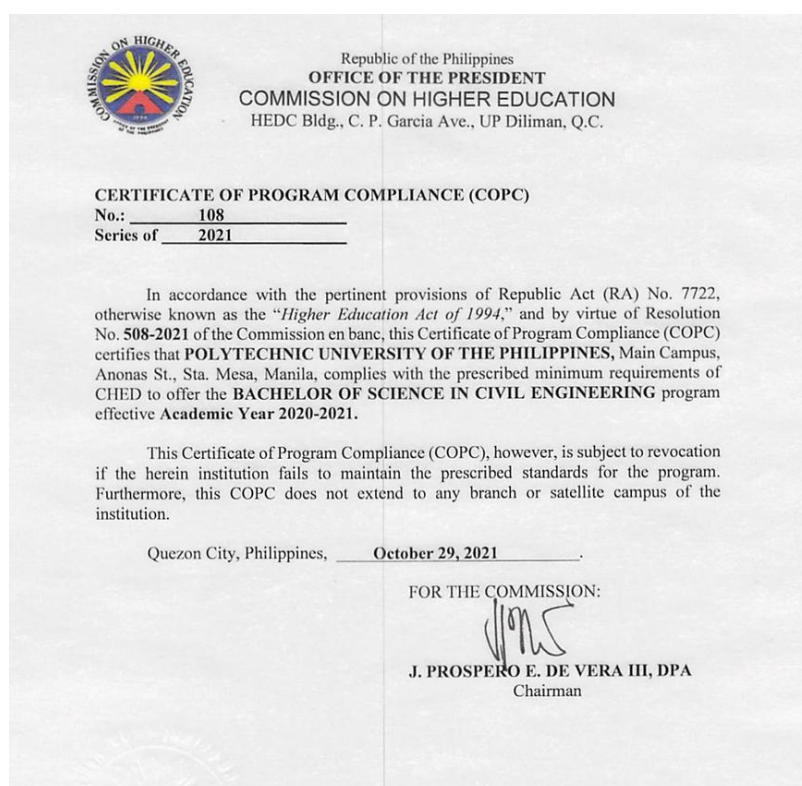


Figure 1. Certificate of Program Compliance (COPC) No. 108, Series of 2021

In addition to CHED recognition, the BSCE program has also received a **Certificate of Compliance from the Professional Regulation Commission (PRC)**. This certification signifies that the program is aligned with the professional requirements established under

Republic Act No. 544, otherwise known as the Civil Engineering Law. Together, these certifications demonstrate that the curriculum is implemented within a quality-assured framework that meets both academic and professional standards, thereby supporting the preparation of students for civil engineering licensure, professional practice, and ethical service to society.



Figure 2. Certificate of Compliance from the Professional Regulation Commission (PRC)

The BSCE program is also accredited by the **Accrediting Agency of Chartered Colleges and Universities in the Philippines (AACCUP)**. As shown in the accreditation certificate, the program has been granted **Level IV Re-accredited status** for the period **November 1, 2021 to October 31, 2026**, after having satisfactorily met the standards and requirements under the New Guidelines of 2020 for Levels III and IV Program Accreditation. This recognition provides additional evidence that the program has undergone external quality assurance evaluation covering areas such as curriculum and instruction, faculty, research, extension, student support, library and learning resources, laboratories, facilities, administration, and other institutional mechanisms that support program delivery.

The AACCUP accreditation complements the program's CHED COPC and PRC compliance by demonstrating that the BSCE program is not only compliant with minimum

regulatory requirements but is also subjected to continuous external review and quality assurance. The Level IV Re-accredited status reflects the maturity of the program's academic systems, the strength of its support mechanisms, and its commitment to sustained improvement in curriculum implementation, student development, research, extension, and professional preparation.



Figure 3. AACCUP Level IV Re-accreditation Certificate of the BS Civil Engineering Program

Together, the CHED COPC, PRC Certificate of Compliance, and AACCUP Level IV Re-accreditation demonstrate that the BS Civil Engineering curriculum is implemented within a quality-assured framework that meets academic, regulatory, professional, and accreditation standards. These recognitions provide evidence that the program is continuously reviewed, externally validated, and maintained to support the development of competent, ethical, and practice-ready civil engineering graduates.

Delivered under a semestral academic system, the curriculum provides an organized sequence of general education courses, mathematics and basic sciences, engineering sciences, professional civil engineering courses, laboratory courses, design experiences,

research, and industry-based learning activities. This sequencing ensures that students acquire foundational knowledge before progressing to advanced civil engineering analysis, design, laboratory, research, and professional practice courses. The recommended plan of study, course offerings, prerequisite structure, and average section enrollments are presented in **Table 5-1**.

The curriculum is aligned with the **Program Educational Objectives (PEOs)** and structured to support the progressive attainment of the **Student Outcomes (SOs)**. **Course Intended Learning Outcomes (CILOs)**, **Teaching-Learning Activities (TLAs)**, and **Assessment Tasks (ATs)** are mapped to the applicable SOs and performance indicators to ensure that students develop the required knowledge, technical competence, engineering judgment, professional responsibility, communication skills, teamwork, ethical awareness, and capacity for lifelong learning. The prerequisite structure further supports coherent knowledge progression by ensuring that students acquire the necessary competencies before undertaking higher-level and integrative courses.

Curriculum mapping, review, and revision are undertaken through a systematic and consultative process involving faculty members, students, alumni, industry and government representatives, and other relevant stakeholders. Assessment results, stakeholder feedback, developments in civil engineering practice, regulatory requirements, and accreditation standards are analyzed to identify necessary curricular and course-level improvements. These processes support **Continuous Quality Improvement (CQI)** and ensure that the program remains responsive to emerging technologies, industry expectations, national development priorities, and the evolving needs of the civil engineering profession.

This criterion also presents how the curriculum addresses the required **Knowledge Profile**, the **Range of Complex Problem Solving**, and **Complex Engineering Activities**. Particular attention is given to the major design experience, industry immersion and internship, laboratory instruction, application of engineering standards, and consideration of multiple technical and non-technical constraints. Supporting evidence—including curriculum maps, prerequisite flowcharts, course syllabi, textbooks and references, assessment instruments, laboratory activities, samples of student work, CHED COPC, and PRC Certificate of Compliance—is provided to demonstrate the adequacy, implementation, and effectiveness of the curriculum. The applicable course syllabi for mathematics, basic sciences, engineering sciences, and discipline-specific courses are included in the Appendix.

A. Program Curriculum

Complete Table 5-1 that describes the plan of study for students in this program including information on course offerings in the form of a recommended schedule by year and term along with average section enrollments for all courses in the program over the two years immediately preceding the visit. State whether you are on quarters or semesters and complete a separate table for each option (if any) in the program.

A.1 Plan of Study

Tables 5.1.1 and 5.1.2 depict the plan of study of the Polytechnic University of the Philippines (PUP-Sta Mesa) BSCE program for the 2018 and 2022 curricula, respectively. The plan includes the list of all course offerings, the number of credit hours in the subject area, year, and term the courses are offered, and the average section enrollment across all courses in the program over the last two years.

Both 2018 and 2022 curricula are 4-year programs with two semesters per year level and two mandatory summer/third semester during the second- and third-year levels. The 2018 curriculum was first implemented on the first-year level during A.Y. 2018 – 2019 and lastly implemented on the fourth-year level during A.Y. 2024 – 2025. On the other hand, the 2022 curriculum was first implemented on the first-year level during A.Y. 2022 – 2023 and is in full implementation across all four-year levels as of this current 2nd Semester of A.Y. 2025 – 2026.

The 2018 BSCE curriculum has a total of 211 credit units as shown in Table 5-1.1. The 152 technical course credit units are distributed as 15 units of Mathematics, 10 units of Natural and Physical Sciences, 28 units of Basic Engineering Sciences, 6 units of Allied Courses, 4 units of On-the-Job Training program, 74 units of Common CE Professional courses, 15 units of Specialized (Construction Engineering and Management, Geotechnical Engineering, Structural Engineering, Transportation Engineering and Water Resources Engineering) CE Professional courses. The 59 non-technical course credit units are distributed as 24 units of General Education courses, 12 units of GE Elective and Mandated courses, 9 units of PUP Institutional GE courses, 8 units of Physical Education and 6 units for the NSTP program.

The 2022 BSCE curriculum has a total of 195 credit units. The 136 technical course credit units are distributed as 15 units of Mathematics, 10 units of Natural and Physical Sciences, 25 units of Basic Engineering Sciences, 6 units of Allied Courses, 4 units of On-the-Job Training program, 61 units of Common CE Professional courses, 15 units of Specialized (Construction Engineering and Management, Geotechnical Engineering, Structural Engineering, Transportation Engineering and Water Resources Engineering) CE Professional

courses .The 59 non-technical course credit units are distributed as 24 units of General Education courses, 12 units of GE Elective and Mandated courses, 9 units of PUP Institutional GE courses, 8 units of Physical Education and 6 units for the NSTP program.

Table 5-1.1 Plan of Study
Bachelor of Science in Civil Engineering – 2018 Curriculum

Course Code	Course Description	Indicate Whether Course is Required, Elective or a Selected Elective by an R, an E or an SE.	Curricular Area (Credit Hours)				Last Two Terms the Course was Offered: Year and Semester	Average section enrolment for the last two terms the course was offered
			Math and Basic Sciences	Engineering Topics and Electives (Check if contains significant design)	General Education	Others		
FIRST YEAR - First Semester								
CHEM 20024	Chemistry for Engineers	R	4				A.Y. 2020-2021, 1st Sem	48.80
							A.Y. 2021-2022, 1st Sem	56.40
CIEN 20012	Civil Engineering Orientation 1	R		2			A.Y. 2020-2021, 1st Sem	52.60
							A.Y. 2021-2022, 1st Sem	52.60
GEED 10033	Readings in Philippine History	R			3		A.Y. 2020-2021, 1st Sem	50.60
							A.Y. 2021-2022, 1st Sem	55.60
GEED 10043	The Contemporary World	R			3		A.Y. 2020-2021, 1st Sem	51.00
							A.Y. 2021-2022, 1st Sem	52.00
GEED 10053	Mathematics in the Modern World	R			3		A.Y. 2020-2021, 1st Sem	50.40
							A.Y. 2021-2022, 1st Sem	50.00
GEED 10103	Filipinolohiya at Pambansang Kaunlaran	R			3		A.Y. 2020-2021, 1st Sem	50.80
							A.Y. 2021-2022, 1st Sem	56.60
MATH 20043	Calculus 1	R	3				A.Y. 2020-2021, 1st Sem	48.60
							A.Y. 2021-2022, 1st Sem	56.00
PHED 01	Physical Education 1	R				2	A.Y. 2020-2021, 1st Sem	50.40
							A.Y. 2021-2022, 1st Sem	50.00
NSTP 10013	CWTS/ROTC	R				3	A.Y. 2020-2021, 1st Sem	50.20
							A.Y. 2021-2022, 1st Sem	47.60
TOTAL UNITS			7	2	12	5		

Course Code	Course Description	Indicate Whether Course is Required, Elective or a Selected Elective by an R, an E or an SE.	Curricular Area (Credit Hours)				Last Two Terms the Course was Offered: Year and Semester	Average section enrolment for the last two terms the course was offered
			Math and Basic Sciences	Engineering Topics and Electives (Check if contains significant design)	General Education	Others		
FIRST YEAR - Second Semester								
CIEN 20022	Civil Engineering Orientation 2 (CE Tech 2)	R		2			A.Y. 2020-2021, 2nd Sem	46.80
							A.Y. 2021-2022, 2nd Sem	52.00
CMPE 20012	Computer Fundamentals and Programming	R		2			A.Y. 2020-2021, 2nd Sem	46.40
							A.Y. 2021-2022, 2nd Sem	45.00
ENSC 20112	Engineering Drawing and Plans	R		2			A.Y. 2020-2021, 2nd Sem	48.80
							A.Y. 2021-2022, 2nd Sem	48.20
GEED 10063	Purposive Communication	R			3		A.Y. 2020-2021, 2nd Sem	45.60
							A.Y. 2021-2022, 2nd Sem	44.60
GEED 10073	Art Appreciation	R			3		A.Y. 2020-2021, 2nd Sem	46.00
							A.Y. 2021-2022, 2nd Sem	45.20
GEED 10113	Pagsasalin sa Kontekstong Filipino	R			3		A.Y. 2020-2021, 2nd Sem	46.80
							A.Y. 2021-2022, 2nd Sem	49.20
MATH 20053	Calculus 2	R	3				A.Y. 2020-2021, 2nd Sem	43.60
							A.Y. 2021-2022, 2nd Sem	47.60
PHED 02	Physical Education 2	R				2	A.Y. 2020-2021, 2nd Sem	45.80
							A.Y. 2021-2022, 2nd Sem	42.60
PHYS 20034	Physics for Engineers	R	4				A.Y. 2020-2021, 2nd Sem	44.00
							A.Y. 2021-2022, 2nd Sem	47.60
NSTP 10023	CWTS/ROTC	R				3	A.Y. 2020-2021, 2nd Sem	45.40
							A.Y. 2021-2022, 2nd Sem	41.60
TOTAL UNITS			7	6	9	5		

Course Code	Course Description	Indicate Whether Course is Required, Elective or a Selected Elective by an R, an E or an SE.	Curricular Area (Credit Hours)				Last Two Terms the Course was Offered: Year and Semester	Average section enrolment for the last two terms the course was offered
			Math and Basic Sciences	Engineering Topics and Electives (Check if contains significant design)	General Education	Others		
SECOND YEAR - First Semester								
CIEN 20032	Geology for Engineers	R		2			A.Y. 2021-2022, 1st Sem	44.60
							A.Y. 2022-2023, 1st Sem	50.60
CIEN 30013	Fundamentals of Surveying 1	R		3			A.Y. 2021-2022, 1st Sem	44.40
							A.Y. 2022-2023, 1st Sem	47.80
ELEN 20023	Engineering Utilities 1 (Basic Electrical Engineering)	R		3			A.Y. 2021-2022, 1st Sem	44.20
							A.Y. 2022-2023, 1st Sem	47.40
ENSC 20013	Computer Aided Drafting	R		3			A.Y. 2021-2022, 1st Sem	45.20
							A.Y. 2022-2023, 1st Sem	47.40
ENSC 20043	Statics of Rigid Bodies	R		3			A.Y. 2021-2022, 1st Sem	44.00
							A.Y. 2022-2023, 1st Sem	48.20
ENSC 20093	Engineering Economics	R		3			A.Y. 2021-2022, 1st Sem	44.60
							A.Y. 2022-2023, 1st Sem	46.80
GEED 10013	Buhay at Mga Sinulat ni Rizal	R			3		A.Y. 2021-2022, 1st Sem	46.60
							A.Y. 2022-2023, 1st Sem	52.80
GEED 10133	Panitikang Filipino	R			3		A.Y. 2021-2022, 1st Sem	48.80
							A.Y. 2022-2023, 1st Sem	51.40
MATH 20063	Differential Equations	R	3				A.Y. 2021-2022, 1st Sem	45.40
							A.Y. 2022-2023, 1st Sem	52.40
PHED 03	Physical Education 3	R				2	A.Y. 2021-2022, 1st Sem	47.80
							A.Y. 2022-2023, 1st Sem	43.40
TOTAL UNITS			3	17	6	2		

Course Code	Course Description	Indicate Whether Course is Required, Elective or a Selected Elective by an R, an E or an SE.	Curricular Area (Credit Hours)				Last Two Terms the Course was Offered: Year and Semester	Average section enrolment for the last two terms the course was offered
			Math and Basic Sciences	Engineering Topics and Electives (Check if contains significant design)	General Education	Others		
SECOND YEAR - Second Semester								
CIEN 30023	Construction Materials and Testing	R		3			A.Y. 2021-2022, 2nd Sem	42.60
							A.Y. 2022-2023, 2nd Sem	46.40
CIEN 30033	Fundamentals of Surveying 2	R		3			A.Y. 2021-2022, 2nd Sem	42.80
							A.Y. 2022-2023, 2nd Sem	46.00
ENSC 20042	Engineering Management	R		2			A.Y. 2021-2022, 2nd Sem	43.60
							A.Y. 2022-2023, 2nd Sem	46.20
ENSC 20072	Dynamics of Rigid Bodies	R		2			A.Y. 2021-2022, 2nd Sem	42.60
							A.Y. 2022-2023, 2nd Sem	46.20
ENSC 20084	Mechanics of Deformable Bodies	R		4			A.Y. 2021-2022, 2nd Sem	45.60
							A.Y. 2022-2023, 2nd Sem	46.80
GEED 10023	Understanding the Self	R		3			A.Y. 2021-2022, 2nd Sem	43.00
							A.Y. 2022-2023, 2nd Sem	41.20
MATH 20093	Numerical Solutions to CE Problems	R	3				A.Y. 2021-2022, 2nd Sem	42.60
							A.Y. 2022-2023, 2nd Sem	49.60
MEEN 20013	Engineering Utilities 2 (Basic Mechanical Engineering)	R		3			A.Y. 2021-2022, 2nd Sem	44.20
							A.Y. 2022-2023, 2nd Sem	46.00
STAT 20023	Engineering Data Analysis	R	3				A.Y. 2021-2022, 2nd Sem	46.20
							A.Y. 2022-2023, 2nd Sem	46.20
PHED 04	Physical Education 4	R				2	A.Y. 2021-2022, 2nd Sem	43.00
							A.Y. 2022-2023, 2nd Sem	43.80
TOTAL UNITS			6	20	0	2		
SECOND YEAR - Third Semester								

Course Code	Course Description	Indicate Whether Course is Required, Elective or a Selected Elective by an R, an E or an SE.	Curricular Area (Credit Hours)				Last Two Terms the Course was Offered: Year and Semester	Average section enrolment for the last two terms the course was offered
			Math and Basic Sciences	Engineering Topics and Electives (Check if contains significant design)	General Education	Others		
CIEN 40012	On-The-Job Training 1 (300 hours)	R		2			A.Y. 2021-2022, 3rd Sem	43.00
							A.Y. 2022-2023, 3rd Sem	45.00
TOTAL UNITS			0	2	0	0		
THIRD YEAR - First Semester								
CIEN 30043	Structural Theory 1	R		3			A.Y. 2022-2023, 1st Sem	46.80
							A.Y. 2023-2024, 1st Sem	30.60
CIEN 30053	Route Surveying and Earthworks	R		3			A.Y. 2022-2023, 1st Sem	43.60
							A.Y. 2023-2024, 1st Sem	49.20
CIEN 30063	Hydraulics 1	R		3			A.Y. 2022-2023, 1st Sem	45.40
							A.Y. 2023-2024, 1st Sem	30.00
CIEN 30073	Building System Design	R		3			A.Y. 2022-2023, 1st Sem	45.20
							A.Y. 2023-2024, 1st Sem	45.20
CIEN 30083	Geotechnical Engineering 1	R		3			A.Y. 2022-2023, 1st Sem	45.20
							A.Y. 2023-2024, 1st Sem	32.40
CIEN 30092	Civil Engineering Research	R		2			A.Y. 2022-2023, 1st Sem	42.40
							A.Y. 2023-2024, 1st Sem	45.20
ENSC 20103	Technopreneurship 101	R		3			A.Y. 2022-2023, 1st Sem	42.40
							A.Y. 2023-2024, 1st Sem	44.00
GEED 10083	Science, Technology and Society	R			3		A.Y. 2022-2023, 1st Sem	45.80
							A.Y. 2023-2024, 1st Sem	39.00
GEED 10093	Ethics	R			3		A.Y. 2022-2023, 1st Sem	44.40
							A.Y. 2023-2024, 1st Sem	44.00
TOTAL UNITS			0	20	6	0		

Course Code	Course Description	Indicate Whether Course is Required, Elective or a Selected Elective by an R, an E or an SE.	Curricular Area (Credit Hours)				Last Two Terms the Course was Offered: Year and Semester	Average section enrolment for the last two terms the course was offered
			Math and Basic Sciences	Engineering Topics and Electives (Check if contains significant design)	General Education	Others		
THIRD YEAR - Second Semester								
CIEN 30103	Structural Theory 2	R		3			A.Y. 2022-2023, 2nd Sem	45.60
							A.Y. 2023-2024, 2nd Sem	26.40
CIEN 30113	Highway and Railroad Engineering	R		3			A.Y. 2022-2023, 2nd Sem	35.00
							A.Y. 2023-2024, 2nd Sem	41.00
CIEN 30122	Geotechnical Engineering 2	R		2			A.Y. 2022-2023, 2nd Sem	46.60
							A.Y. 2023-2024, 2nd Sem	30.20
CIEN 30132	Hydrology	R		2			A.Y. 2022-2023, 2nd Sem	36.00
							A.Y. 2023-2024, 2nd Sem	25.60
CIEN 30143	Hydraulics 2	R		3			A.Y. 2022-2023, 2nd Sem	36.20
							A.Y. 2023-2024, 2nd Sem	28.20
CIEN 30152	CE Law, Ethics and Contracts	R		2			A.Y. 2022-2023, 2nd Sem	41.40
							A.Y. 2023-2024, 2nd Sem	45.20
CIEN 30162	Matrix Analysis of Structures	R		2			A.Y. 2022-2023, 2nd Sem	44.00
							A.Y. 2023-2024, 2nd Sem	24.80
CIEN 30171	Integrative Course 1	R		1			A.Y. 2022-2023, 2nd Sem	40.20
							A.Y. 2023-2024, 2nd Sem	38.80
CIEN 30183	Principles of Reinforced Concrete	R		3			A.Y. 2022-2023, 2nd Sem	46.20
							A.Y. 2023-2024, 2nd Sem	25.60
GEED 20023	Politics, Governance and Citizenship	R			3		A.Y. 2022-2023, 2nd Sem	45.80
							A.Y. 2023-2024, 2nd Sem	41.40
TOTAL UNITS			0	21	3	0		
THIRD YEAR - Third Semester								
		R		2			A.Y. 2022-2023, 3rd Sem	40.00

Course Code	Course Description	Indicate Whether Course is Required, Elective or a Selected Elective by an R, an E or an SE.	Curricular Area (Credit Hours)				Last Two Terms the Course was Offered: Year and Semester	Average section enrolment for the last two terms the course was offered	
			Math and Basic Sciences	Engineering Topics and Electives (Check if contains significant design)	General Education	Others			
CIEN 40022	On The Job Training 2 (300 hours)						A.Y. 2023-2024, 3rd Sem	46.00	
TOTAL UNITS			0	2	0	0			
FOURTH YEAR - First Semester									
CIEN 30193	Reinforced Concrete Design	R		3			A.Y. 2023-2024, 1st Sem	26.20	
							A.Y. 2024-2025, 1st Sem	20.00	
CIEN 30203	Principles of Transportation Engineering	R		3			A.Y. 2023-2024, 1st Sem	42.00	
							A.Y. 2024-2025, 1st Sem	48.60	
CIEN 30212	CE Project 1	R		2			A.Y. 2023-2024, 1st Sem	39.40	
							A.Y. 2024-2025, 1st Sem	23.20	
CIEN 30222	Timber Design	R		2			A.Y. 2023-2024, 1st Sem	37.20	
							A.Y. 2024-2025, 1st Sem	37.60	
CIEN 30231	Integrative Course 2	R		1			A.Y. 2023-2024, 1st Sem	24.20	
							A.Y. 2024-2025, 1st Sem	30.20	
CIEN 30242	Foundation Engineering	R		2			A.Y. 2023-2024, 1st Sem	28.00	
							A.Y. 2024-2025, 1st Sem	20.20	
CIEN 30253	Principles of Steel Design	R		3			A.Y. 2023-2024, 1st Sem	29.80	
							A.Y. 2024-2025, 1st Sem	43.60	
CMEN 41013	Project Construction and Management	E		3			A.Y. 2023-2024, 1st Sem	54.00	
							A.Y. 2024-2025, 1st Sem	40.00	
GTEN 42013	Geosynthetics in Geotechnical Engineering	E						A.Y. 2023-2024, 1st Sem	46.00
								A.Y. 2024-2025, 1st Sem	40.00

Course Code	Course Description	Indicate Whether Course is Required, Elective or a Selected Elective by an R, an E or an SE.	Curricular Area (Credit Hours)				Last Two Terms the Course was Offered: Year and Semester	Average section enrolment for the last two terms the course was offered
			Math and Basic Sciences	Engineering Topics and Electives (Check if contains significant design)	General Education	Others		
STEN 43013	Computer Software in Structural Analysis	E		3			A.Y. 2023-2024, 1st Sem	45.00
							A.Y. 2024-2025, 1st Sem	43.00
TREN 44013	Transportation Systems Planning and Design	E					A.Y. 2023-2024, 1st Sem	56.00
							A.Y. 2024-2025, 1st Sem	42.00
WREN 45013	Water Resources Engineering	E					A.Y. 2023-2024, 1st Sem	46.00
							A.Y. 2024-2025, 1st Sem	42.00
CMEN 41023	Advanced Construction Methods and Equipment	E				A.Y. 2023-2024, 1st Sem	53.00	
						A.Y. 2024-2025, 1st Sem	40.00	
GTEN 42023	Geotechnical Earthquake Engineering	E				A.Y. 2023-2024, 1st Sem	45.00	
						A.Y. 2024-2025, 1st Sem	40.00	
STEN 43023	Prestressed Concrete Design	E				A.Y. 2023-2024, 1st Sem	43.00	
						A.Y. 2024-2025, 1st Sem	47.00	
TREN 44023	Highway Engineering	E				A.Y. 2023-2024, 1st Sem	54.00	
						A.Y. 2024-2025, 1st Sem	42.00	
WREN 45023	Flood Control and Drainage Design	E				A.Y. 2023-2024, 1st Sem	45.00	
						A.Y. 2024-2025, 1st Sem	38.00	
GEED 20093	Reading Visual Arts	R			3		A.Y. 2023-2024, 1st Sem	41.80
							A.Y. 2024-2025, 1st Sem	43.20
TOTAL UNITS			0	22	3	0		
FOURTH YEAR - Second Semester								
CIEN 30262	CE Project 2	R		2			A.Y. 2023-2024, 2nd Sem	38.60
							A.Y. 2024-2025, 2nd Sem	23.00
	Quantity Surveying	R		2			A.Y. 2023-2024, 2nd Sem	29.20

Course Code	Course Description	Indicate Whether Course is Required, Elective or a Selected Elective by an R, an E or an SE.	Curricular Area (Credit Hours)				Last Two Terms the Course was Offered: Year and Semester	Average section enrolment for the last two terms the course was offered
			Math and Basic Sciences	Engineering Topics and Electives (Check if contains significant design)	General Education	Others		
CIEN 30272							A.Y. 2024-2025, 2nd Sem	24.40
CIEN 30282	Earthquake Engineering	R		2			A.Y. 2023-2024, 2nd Sem	37.80
							A.Y. 2024-2025, 2nd Sem	36.60
CIEN 30293	Construction Methods and Project Management	R		3			A.Y. 2023-2024, 2nd Sem	32.00
							A.Y. 2024-2025, 2nd Sem	31.60
CIEN 30301	Integrative Course 3	R		1			A.Y. 2023-2024, 2nd Sem	23.40
							A.Y. 2024-2025, 2nd Sem	24.40
CIEN 30311	CE Field Trip	R		1			A.Y. 2023-2024, 2nd Sem	50.60
							A.Y. 2024-2025, 2nd Sem	53.40
CMEN 41033	Construction Cost Engineering	E		3			A.Y. 2023-2024, 2nd Sem	55.00
							A.Y. 2024-2025, 2nd Sem	34.00
GTEN 42033	Geotechnical Aspects of Landfill Design	E					A.Y. 2023-2024, 2nd Sem	51.00
							A.Y. 2024-2025, 2nd Sem	31.00
STEN 43033	Structural Design of Towers and Other Vertical Structures	E					A.Y. 2023-2024, 2nd Sem	50.00
							A.Y. 2024-2025, 2nd Sem	35.00
TREN 44033	Railroad Engineering	E					A.Y. 2023-2024, 2nd Sem	52.00
							A.Y. 2024-2025, 2nd Sem	35.00
WREN 45033	Irrigation Engineering	E					A.Y. 2023-2024, 2nd Sem	53.00
							A.Y. 2024-2025, 2nd Sem	27.00
CMEN 41043	Database Management in Construction	E		3			A.Y. 2023-2024, 2nd Sem	55.00
							A.Y. 2024-2025, 2nd Sem	30.00
		E					A.Y. 2023-2024, 2nd Sem	53.00

Course Code	Course Description	Indicate Whether Course is Required, Elective or a Selected Elective by an R, an E or an SE.	Curricular Area (Credit Hours)				Last Two Terms the Course was Offered: Year and Semester	Average section enrolment for the last two terms the course was offered
			Math and Basic Sciences	Engineering Topics and Electives (Check if contains significant design)	General Education	Others		
GTEN 42043	Geotechnical Engineering (Rock Mechanics)						A.Y. 2024-2025, 2nd Sem	32.00
STEN 43043	Bridge Engineering	E					A.Y. 2023-2024, 2nd Sem	50.00
							A.Y. 2024-2025, 2nd Sem	34.00
TREN 44043	Airport Design	E					A.Y. 2023-2024, 2nd Sem	52.00
							A.Y. 2024-2025, 2nd Sem	35.00
WREN 45043	Water Supply Planning and Development	E		3			A.Y. 2023-2024, 2nd Sem	52.00
							A.Y. 2024-2025, 2nd Sem	28.00
CMEN 41053	Construction Occupational Safety and Health (COSH)	E					A.Y. 2023-2024, 2nd Sem	55.00
							A.Y. 2024-2025, 2nd Sem	32.00
GTEN 42053	Ground Improvement	E					A.Y. 2023-2024, 2nd Sem	52.00
							A.Y. 2024-2025, 2nd Sem	31.00
STEN 43053	Dam Engineering	E					A.Y. 2023-2024, 2nd Sem	50.00
							A.Y. 2024-2025, 2nd Sem	34.00
TREN 44053	Ports and Harbors	E					A.Y. 2023-2024, 2nd Sem	52.00
							A.Y. 2024-2025, 2nd Sem	35.00
WREN 45053	River Engineering	E					A.Y. 2023-2024, 2nd Sem	53.00
							A.Y. 2024-2025, 2nd Sem	27.00
GEED 20033	Gender and Society	R			3		A.Y. 2023-2024, 2nd Sem	35.00
							A.Y. 2024-2025, 2nd Sem	16.80
TOTAL UNITS			0	20	3	0		

Course Code	Course Description	Indicate Whether Course is Required, Elective or a Selected Elective by an R, an E or an SE.	Curricular Area (Credit Hours)				Last Two Terms the Course was Offered: Year and Semester	Average section enrolment for the last two terms the course was offered
			Math and Basic Sciences	Engineering Topics and Electives (Check if contains significant design)	General Education	Others		
OVERALL TOTAL CREDIT HOURS FOR THE DEGREE		211	23	132	42	14		
Semester Credit Hours or percentage of total			10.90%	62.56%	19.91%	6.64%		
Suggested PTC-CACBET-EAC credit hours or percentage		Semester Units	25	96	36	14		
		Percentage	14.62%	56.14%	21.05%	8.19%		

Note:

1. For courses that include multiple elements (lecture, laboratory, recitation, etc), indicate the average enrollment in each element.
2. Required courses are required of all students in the program, elective course are optional for students, and selected electives are courses where students must take one or more courses from a specific group. Instructional materials and student work verifying compliance with PTC-ACBET-EAC criteria for the categories indicated above will be required during the campus visit.

Table 5-1.2 Plan of Study
Bachelor of Science in Civil Engineering – 2022 Curriculum

Course Code	Course Description	Indicate Whether Course is Required, Elective or a Selected Elective by an R, an E or an SE.	Curricular Area (Credit Hours)				Last Two Terms the Course was Offered: Year and Semester	Average section enrolment for the last two terms the course was offered
			Math and Basic Sciences	Engineering Topics and Electives (Check if contains significant design)	General Education	Others		
FIRST YEAR - First Semester								
CHEM 015	Chemistry for Engineers	R	4				A.Y. 2024-2025, 1st Sem	56.57
							A.Y. 2025-2026, 1st Sem	51.00
CIEN 101	Civil Engineering Orientation	R		2			A.Y. 2024-2025, 1st Sem	57.43
							A.Y. 2025-2026, 1st Sem	55.43
GEED 002	Readings in Philippine History	R			3		A.Y. 2024-2025, 1st Sem	56.00
							A.Y. 2025-2026, 1st Sem	53.00
GEED 004	Mathematics in the Modern World	R			3		A.Y. 2024-2025, 1st Sem	55.86
							A.Y. 2025-2026, 1st Sem	52.71
GEED 005	Purposive Communication	R			3		A.Y. 2024-2025, 1st Sem	56.57
							A.Y. 2025-2026, 1st Sem	54.00
GEED 007	Science, Technology and Society	R			3		A.Y. 2024-2025, 1st Sem	56.43
							A.Y. 2025-2026, 1st Sem	54.00
MATH 101	Calculus 1	R	3				A.Y. 2024-2025, 1st Sem	55.29
							A.Y. 2025-2026, 1st Sem	51.29
PATHFit 1	Physical Activity Towards Health and Fitness (PATHFit1): Movement Competency Training	R				2	A.Y. 2024-2025, 1st Sem	55.57
							A.Y. 2025-2026, 1st Sem	52.00
	CWTS 1/ ROTC 1	R				3	A.Y. 2024-2025, 1st Sem	55.86

Course Code	Course Description	Indicate Whether Course is Required, Elective or a Selected Elective by an R, an E or an SE.	Curricular Area (Credit Hours)				Last Two Terms the Course was Offered: Year and Semester	Average section enrolment for the last two terms the course was offered
			Math and Basic Sciences	Engineering Topics and Electives (Check if contains significant design)	General Education	Others		
CWTS 001/ ROTC 001							A.Y. 2025-2026, 1st Sem	52.14
TOTAL UNITS			7	2	12	5		
FIRST YEAR - Second Semester								
CETE 011	Civil Engineering Technology	R		2			A.Y. 2024-2025, 2nd Sem	44.56
							A.Y. 2025-2026, 2nd Sem	42.67
CMPE 011	Computer Fundamentals and Programming	R		2			A.Y. 2024-2025, 2nd Sem	43.56
							A.Y. 2025-2026, 2nd Sem	41.78
ENSC 011	Engineering Drawing and Plans	R		1			A.Y. 2024-2025, 2nd Sem	43.22
							A.Y. 2025-2026, 2nd Sem	41.56
GEED 003	The Contemporary World/Ang Kasalukuyang Daigdig	R			3		A.Y. 2024-2025, 2nd Sem	44.11
							A.Y. 2025-2026, 2nd Sem	41.78
GEED 006	Art Appreciation/Pagpapahalaga sa Sining	R			3		A.Y. 2024-2025, 2nd Sem	43.78
							A.Y. 2025-2026, 2nd Sem	41.78
MATH 103	Calculus 2	R	3				A.Y. 2024-2025, 2nd Sem	43.33
							A.Y. 2025-2026, 2nd Sem	32.67
PATHFit 2	Physical Activity Towards Health and Fitness (PATHFit 2)	R				2	A.Y. 2024-2025, 2nd Sem	43.22
							A.Y. 2025-2026, 2nd Sem	40.22
PHYS 013	Physics for Engineers (Calculus-based)	R	4				A.Y. 2024-2025, 2nd Sem	42.89
							A.Y. 2025-2026, 2nd Sem	32.67

Course Code	Course Description	Indicate Whether Course is Required, Elective or a Selected Elective by an R, an E or an SE.	Curricular Area (Credit Hours)				Last Two Terms the Course was Offered: Year and Semester	Average section enrolment for the last two terms the course was offered
			Math and Basic Sciences	Engineering Topics and Electives (Check if contains significant design)	General Education	Others		
CWTS 002/ ROTC 002	CWTS 2/ ROTC 2	R				3	A.Y. 2024-2025, 2nd Sem	43.44
							A.Y. 2025-2026, 2nd Sem	40.00
TOTAL UNITS			7	5	6	5		
SECOND YEAR - First Semester								
CIEN 201	Fundamentals of Surveying 1	R		2			A.Y. 2024-2025, 1st Sem	44.89
							A.Y. 2025-2026, 1st Sem	45.78
CIEN 203	Geology for Civil Engineers	R		2			A.Y. 2024-2025, 1st Sem	42.44
							A.Y. 2025-2026, 1st Sem	43.00
ELEN 011	Engineering Utilities 1	R		3			A.Y. 2024-2025, 1st Sem	43.22
							A.Y. 2025-2026, 1st Sem	42.78
ENSC 014	Computer-Aided Drafting	R		1			A.Y. 2024-2025, 1st Sem	39.44
							A.Y. 2025-2026, 1st Sem	44.00
ENSC 016	Statics of Rigid Bodies	R		3			A.Y. 2024-2025, 1st Sem	44.67
							A.Y. 2025-2026, 1st Sem	45.00
ENSC 023	Engineering Economics	R		3			A.Y. 2024-2025, 1st Sem	40.67
							A.Y. 2025-2026, 1st Sem	47.78
ENSC 024	Engineering Management	R		2			A.Y. 2024-2025, 1st Sem	42.67
							A.Y. 2025-2026, 1st Sem	46.33
GEED 037	Life and Works of Rizal	R			3		A.Y. 2024-2025, 1st Sem	40.33
							A.Y. 2025-2026, 1st Sem	42.78
MATH 209	Differential Equations	R	3				A.Y. 2024-2025, 1st Sem	43.67
							A.Y. 2025-2026, 1st Sem	42.67
PATHFit 3		R				2	A.Y. 2024-2025, 1st Sem	40.56

Course Code	Course Description	Indicate Whether Course is Required, Elective or a Selected Elective by an R, an E or an SE.	Curricular Area (Credit Hours)				Last Two Terms the Course was Offered: Year and Semester	Average section enrolment for the last two terms the course was offered
			Math and Basic Sciences	Engineering Topics and Electives (Check if contains significant design)	General Education	Others		
	Physical Activity Towards Health and Fitness (PATHFit						A.Y. 2025-2026, 1st Sem	42.56
TOTAL UNITS			3	16	3	2		
SECOND YEAR - Second Semester								
CIEN 201	Fundamentals of Surveying 2 (Applied Surveying)	R		2			A.Y. 2024-2025, 2nd Sem	44.56
							A.Y. 2025-2026, 2nd Sem	47.44
CIEN 204	Construction Materials and Testing	R		3			A.Y. 2024-2025, 2nd Sem	31.89
							A.Y. 2025-2026, 2nd Sem	24.44
CIEN 205	Numerical Solutions to CE Problems	R		3			A.Y. 2024-2025, 2nd Sem	46.89
							A.Y. 2025-2026, 2nd Sem	45.56
STAT 012	Engineering Data Analysis	R		3			A.Y. 2024-2025, 2nd Sem	42.33
							A.Y. 2025-2026, 2nd Sem	42.89
ENSC 017	Dynamics of Rigid Bodies	R		2			A.Y. 2024-2025, 2nd Sem	48.00
							A.Y. 2025-2026, 2nd Sem	41.22
ENSC 019	Mechanics of Deformable Bodies	R		4			A.Y. 2024-2025, 2nd Sem	24.56
							A.Y. 2025-2026, 2nd Sem	27.78
MEEN 011	Engineering Utilities 2	R		3			A.Y. 2024-2025, 2nd Sem	42.89
							A.Y. 2025-2026, 2nd Sem	41.89
GEED 008	Ethics/Etika	R			3		A.Y. 2024-2025, 2nd Sem	43.88
							A.Y. 2025-2026, 2nd Sem	41.11
PATHFIT 4		R				2	A.Y. 2024-2025, 2nd Sem	43.25

Course Code	Course Description	Indicate Whether Course is Required, Elective or a Selected Elective by an R, an E or an SE.	Curricular Area (Credit Hours)				Last Two Terms the Course was Offered: Year and Semester	Average section enrolment for the last two terms the course was offered
			Math and Basic Sciences	Engineering Topics and Electives (Check if contains significant design)	General Education	Others		
	Physical Activity Towards Health and Fitness 4						A.Y. 2025-2026, 2nd Sem	41.33
TOTAL UNITS			0	20	3	2		
SECOND YEAR - Third Semester								
GEED 001	Understanding the Self/Pag-unawa sa Sarili	R			3		A.Y. 2024-2025, 3rd Sem	42.63
							A.Y. 2025-2026, 3rd Sem	43.26
GEED 032	Filipinolohiya at Pambansang Kaunlaran	R			3		A.Y. 2024-2025, 3rd Sem	42.75
							A.Y. 2025-2026, 3rd Sem	43.55
TOTAL UNITS			0	0	6	0		
THIRD YEAR - First Semester								
CIEN 301	Geotechnical Engineering 1 (Soil Mechanics)	R		2			A.Y. 2024-2025, 1st Sem	35.40
							A.Y. 2025-2026, 1st Sem	27.50
CIEN 303	Highway and Railroad Engineering	R		3			A.Y. 2024-2025, 1st Sem	37.60
							A.Y. 2025-2026, 1st Sem	45.71
CIEN 305	Building Systems Design	R		3			A.Y. 2024-2025, 1st Sem	48.00
							A.Y. 2025-2026, 1st Sem	48.00
CIEN 306	Structural Theory	R		4			A.Y. 2024-2025, 1st Sem	35.20
							A.Y. 2025-2026, 1st Sem	29.88
CIEN 310	Hydraulics 1	R		2			A.Y. 2024-2025, 1st Sem	27.40
							A.Y. 2025-2026, 1st Sem	24.63
ENSC 029	Technopreneurship 101	R		3			A.Y. 2024-2025, 1st Sem	50.20

Course Code	Course Description	Indicate Whether Course is Required, Elective or a Selected Elective by an R, an E or an SE.	Curricular Area (Credit Hours)				Last Two Terms the Course was Offered: Year and Semester	Average section enrolment for the last two terms the course was offered
			Math and Basic Sciences	Engineering Topics and Electives (Check if contains significant design)	General Education	Others		
							A.Y. 2025-2026, 1st Sem	41.75
GEED 018	Gender and Society	R			3		A.Y. 2024-2025, 1st Sem	49.60
							A.Y. 2025-2026, 1st Sem	46.75
GEED 028	Reading Visual Arts	R			3		A.Y. 2024-2025, 1st Sem	46.60
							A.Y. 2025-2026, 1st Sem	40.63
GEED 035	Panitikang Filipino	R			3		A.Y. 2024-2025, 1st Sem	44.60
							A.Y. 2025-2026, 1st Sem	41.00
TOTAL UNITS			0	17	9	0		
THIRD YEAR - Second Semester								
CIEN 302	Geotechnical Engineering 2	R		2			A.Y. 2024-2025, 2nd Sem	35.29
							A.Y. 2025-2026, 2nd Sem	24.13
CIEN 304	Principles of Transportation Engineering	R		3			A.Y. 2024-2025, 2nd Sem	39.57
							A.Y. 2025-2026, 2nd Sem	43.63
CIEN 307	Matrix Analysis of Structures	R		2			A.Y. 2024-2025, 2nd Sem	29.86
							A.Y. 2025-2026, 2nd Sem	25.50
CIEN 308	Principles of Reinforced Concrete	R		4			A.Y. 2024-2025, 2nd Sem	37.00
							A.Y. 2025-2026, 2nd Sem	31.38
CIEN 309	CE Integration 1	R		1			A.Y. 2024-2025, 2nd Sem	40.43
							A.Y. 2025-2026, 2nd Sem	43.38
CIEN 311	Hydraulics 2	R		3			A.Y. 2024-2025, 2nd Sem	27.43
							A.Y. 2025-2026, 2nd Sem	24.38
CIEN 312	Hydrology	R		2			A.Y. 2024-2025, 2nd Sem	27.14

Course Code	Course Description	Indicate Whether Course is Required, Elective or a Selected Elective by an R, an E or an SE.	Curricular Area (Credit Hours)				Last Two Terms the Course was Offered: Year and Semester	Average section enrolment for the last two terms the course was offered
			Math and Basic Sciences	Engineering Topics and Electives (Check if contains significant design)	General Education	Others		
							A.Y. 2025-2026, 2nd Sem	23.88
CIEN 313	Civil Engineering Research	R		2			A.Y. 2024-2025, 2nd Sem	50.00
							A.Y. 2025-2026, 2nd Sem	39.25
ENGL 012	Technical Communication	R			3		A.Y. 2024-2025, 2nd Sem	43.43
							A.Y. 2025-2026, 2nd Sem	43.50
GEED 020	Politics, Governance and Citizenship	R			3		A.Y. 2024-2025, 2nd Sem	47.00
							A.Y. 2025-2026, 2nd Sem	41.25
TOTAL UNITS			0	19	6	0		
THIRD YEAR - Third Semester								
CIEN 314	On-The-Job Training 1 (300 Hours)	R		3			A.Y. 2024-2025, 3rd Sem	48.50
							N/A	N/A
TOTAL UNITS			0	3	0	0		
FOURTH YEAR - First Semester								
CIEN 401	CE Project 1	R		2			N/A	N/A
							A.Y. 2025-2026, 1st Sem	51.67
CIEN 403	Foundation Engineering	R		2			N/A	N/A
							A.Y. 2025-2026, 1st Sem	21.83
CIEN 404	CE Law, Ethics and Contracts	R		2			N/A	N/A
							A.Y. 2025-2026, 1st Sem	41.17
CIEN 407	Principles of Steel Design	R		3			N/A	N/A
							A.Y. 2025-2026, 1st Sem	31.67
CIEN 408	Earthquake Engineering	R		2			N/A	N/A

Course Code	Course Description	Indicate Whether Course is Required, Elective or a Selected Elective by an R, an E or an SE.	Curricular Area (Credit Hours)				Last Two Terms the Course was Offered: Year and Semester	Average section enrolment for the last two terms the course was offered
			Math and Basic Sciences	Engineering Topics and Electives (Check if contains significant design)	General Education	Others		
							A.Y. 2025-2026, 1st Sem	49.00
CIEN 410	CE Integration 2	R		1			N/A	N/A
							A.Y. 2025-2026, 1st Sem	29.83
CMEN 451	Project Construction and Management	E		3			N/A	N/A
							A.Y. 2025-2026, 1st Sem	24.00
GTEN 451	Geosynthetics in Geotechnical Engineering	E					N/A	N/A
							A.Y. 2025-2026, 1st Sem	22.00
STEN 451	Computer Software in Structural Analysis	E					N/A	N/A
							A.Y. 2025-2026, 1st Sem	33.00
TREN 451	Transportation Systems Planning and Design	E					N/A	N/A
							A.Y. 2025-2026, 1st Sem	25.00
WREN 451	Water Resources Engineering	E					N/A	N/A
							A.Y. 2025-2026, 1st Sem	15.00
CMEN 461	Advanced Construction Methods and Equipment	E		3			N/A	N/A
							A.Y. 2025-2026, 1st Sem	23.50
GTEN 461	Geotechnical Earthquake Engineering	E					N/A	N/A
							A.Y. 2025-2026, 1st Sem	21.00
STEN 461	Prestressed Concrete Design	E					N/A	N/A
							A.Y. 2025-2026, 1st Sem	36.00
TREN 461	Highway Engineering	E					N/A	N/A
							A.Y. 2025-2026, 1st Sem	29.00
WREN 461	Flood Control and Drainage Design	E					N/A	N/A
							A.Y. 2025-2026, 1st Sem	13.00

Course Code	Course Description	Indicate Whether Course is Required, Elective or a Selected Elective by an R, an E or an SE.	Curricular Area (Credit Hours)				Last Two Terms the Course was Offered: Year and Semester	Average section enrolment for the last two terms the course was offered
			Math and Basic Sciences	Engineering Topics and Electives (Check if contains significant design)	General Education	Others		
CMEN 472	Construction Procurement	E		3			N/A	N/A
							A.Y. 2025-2026, 1st Sem	35.00
GTEN 472	Slope Stability and Earthworks	E					N/A	N/A
							A.Y. 2025-2026, 1st Sem	23.00
STEN 472	Design of Steel Structures	E					N/A	N/A
							A.Y. 2025-2026, 1st Sem	31.00
TREN 471	Railroad Engineering	E					N/A	N/A
							A.Y. 2025-2026, 1st Sem	27.00
WREN 471	Irrigation Engineering	E					N/A	N/A
							A.Y. 2025-2026, 1st Sem	46.00
TOTAL UNITS			0	21	0	0		
FOURTH YEAR - Second Semester								
CIEN 402	CE Project 2	R		2			N/A	N/A
							A.Y. 2025-2026, 2nd Sem	50.50
CIEN 405	Construction Methods and Project Management	R		3			N/A	N/A
							A.Y. 2025-2026, 2nd Sem	28.67
CIEN 406	Quantity Surveying	R		2			N/A	N/A
							A.Y. 2025-2026, 2nd Sem	26.67
CIEN 409	CE Site Inspection and Seminars	R		1			N/A	N/A
							A.Y. 2025-2026, 2nd Sem	45.17
CIEN 411	CE Integration 3	R		1			N/A	N/A
							A.Y. 2025-2026, 2nd Sem	27.00
		R		1			N/A	N/A

Course Code	Course Description	Indicate Whether Course is Required, Elective or a Selected Elective by an R, an E or an SE.	Curricular Area (Credit Hours)				Last Two Terms the Course was Offered: Year and Semester	Average section enrolment for the last two terms the course was offered
			Math and Basic Sciences	Engineering Topics and Electives (Check if contains significant design)	General Education	Others		
CIEN 412	On-The-Job Training 2 (200 Hours)						A.Y. 2025-2026, 2nd Sem	50.67
CMEN 481	Database Management in Construction	E		3			N/A	N/A
							A.Y. 2025-2026, 2nd Sem	44.00
GTEN 481	Geotechnical Engineering (Rock Mechanics)	E					N/A	N/A
							A.Y. 2025-2026, 2nd Sem	48.00
STEN 481	Bridge Engineering	E					N/A	N/A
							A.Y. 2025-2026, 2nd Sem	45.00
TREN 481	Airport Design	E					N/A	N/A
							A.Y. 2025-2026, 2nd Sem	48.00
WREN 481	Water Supply Planning and Development	E		3			N/A	N/A
							A.Y. 2025-2026, 2nd Sem	43.00
CMEN 491	Construction Occupational Safety and Health (COSH)	E					N/A	N/A
							A.Y. 2025-2026, 2nd Sem	51.50
GTEN 491	Ground Improvement	E					N/A	N/A
							A.Y. 2025-2026, 2nd Sem	43.00
STEN 491	Structural Design of Towers and Other Vertical Structures	E					N/A	N/A
							A.Y. 2025-2026, 2nd Sem	46.00
TREN 491	Ports and Harbors	E		3			N/A	N/A
							A.Y. 2025-2026, 2nd Sem	45.00
WREN 491	River Engineering	E					N/A	N/A

Course Code	Course Description	Indicate Whether Course is Required, Elective or a Selected Elective by an R, an E or an SE.	Curricular Area (Credit Hours)				Last Two Terms the Course was Offered: Year and Semester	Average section enrolment for the last two terms the course was offered
			Math and Basic Sciences	Engineering Topics and Electives (Check if contains significant design)	General Education	Others		
							A.Y. 2025-2026, 2nd Sem	42.00
TOTAL UNITS			0	16	0	0		
OVERALL TOTAL CREDIT HOURS FOR THE DEGREE		195	17	119	45	14		
Semester Credit Hours or percentage of total			8.72%	61.03%	23.08%	7.18%		
Suggested PTC-CACBET-EAC credit hours or percentage		Semester Units	25	96	36	14		
		Percentage	14.62%	56.14%	21.05%	8.19%		

Describe how the curriculum aligns with the program educational objectives.

A.2 Alignment of the Curriculum with the Program Educational Objectives

The **Program Educational Objectives (PEOs)** describe the expected professional accomplishments of graduates of the Bachelor of Science in Civil Engineering (BSCE) program three to five years after graduation. These objectives reflect the program's expectations regarding graduates' engagement in civil engineering practice, demonstration of social and ethical responsibility, and participation in lifelong learning and continuing professional development. To support their attainment, the BSCE curriculum is systematically designed to provide students with progressively developed technical knowledge, engineering competencies, professional values, and transferable skills required for responsible civil engineering practice.

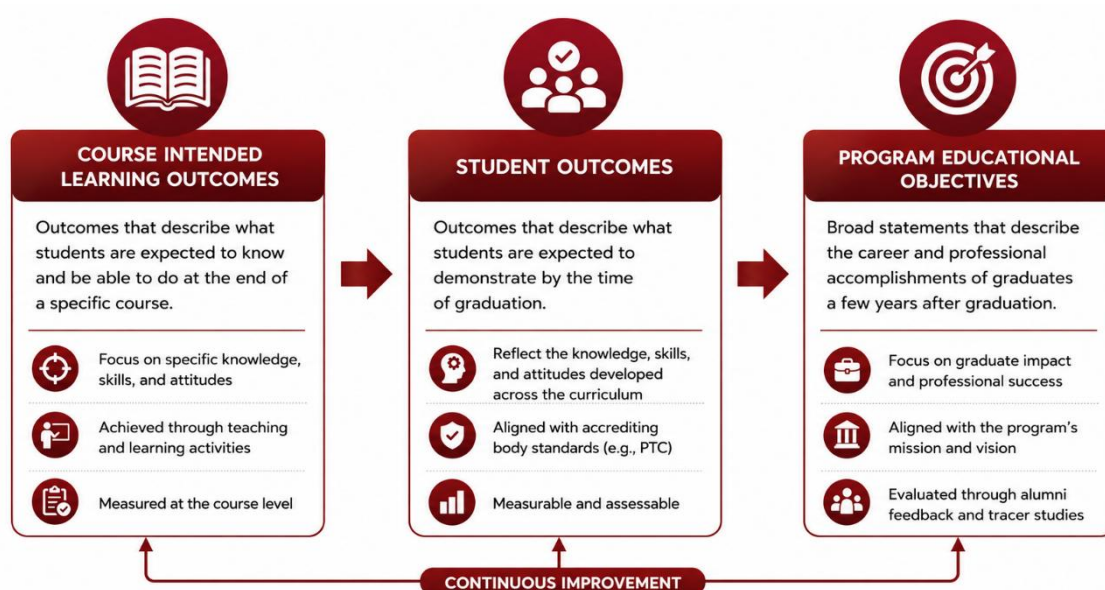


Figure 5.2.1. CILO–SO–PEO Alignment Framework of the BS Civil Engineering Program. The framework illustrates how course-level learning outcomes contribute to the attainment of Student Outcomes and the longer-term professional accomplishments defined in the Program Educational Objectives.

The alignment of the curriculum with the PEOs follows an outcomes-based framework in which learning is developed and assessed at different but interconnected levels. Each course contains clearly defined **Course Intended Learning Outcomes (CILOs)** that specify the knowledge, skills, and professional attributes students are expected to demonstrate upon successful completion of the course. The CILOs are mapped to the applicable **Student Outcomes (SOs)**, which represent the competencies students are expected to attain by the

time of graduation. The SOs, in turn, provide the educational foundation for the longer-term professional accomplishments expressed in the PEOs.

Through this hierarchical alignment, student achievement at the course level contributes to the attainment of program-level competencies and, subsequently, to the professional development of graduates. The relationship among the CILOs, SOs, and PEOs is documented through curriculum mapping matrices that identify how individual courses contribute to the intended graduate attributes. These mappings enable the Department to verify that the curriculum provides sufficient and coherent learning opportunities for the attainment of each PEO and that no essential professional competency is inadequately addressed.

The curriculum supports the attainment of the PEOs through the balanced integration of **mathematics and basic sciences, engineering sciences, professional civil engineering courses, general education courses, institutional courses, laboratory instruction, design experiences, research, internship, and field-based learning activities**. Foundational courses in mathematics, physics, chemistry, and engineering sciences develop the analytical and scientific competencies necessary for advanced civil engineering applications. These courses provide students with the ability to apply mathematical models, scientific principles, and engineering fundamentals in analyzing and solving civil engineering problems.

Professional courses progressively develop competencies in the principal areas of civil engineering practice, namely **Construction Engineering and Management, Geotechnical Engineering, Structural Engineering, Transportation Engineering, and Water Resources Engineering**. Through specialization and professional courses, students are exposed to engineering analysis, planning, design, construction, operation, maintenance, and management of civil infrastructure systems. These curricular components support **PEO 1** by preparing graduates for employment and professional practice in civil engineering and allied fields.

Laboratory courses reinforce the theoretical concepts introduced in lecture courses by requiring students to conduct experiments, operate engineering instruments and equipment, analyze measured data, interpret results, and prepare technical reports ([Attachment C5-01 – Technical Laboratory Manual and Reports](#)). Laboratory activities in materials testing, soil mechanics, hydraulics, surveying, structural engineering, and related areas develop students'

ability to investigate engineering problems and apply appropriate standards and procedures ([Attachment C5-02 – Laboratory and Field-Based Learning Activities](#)).

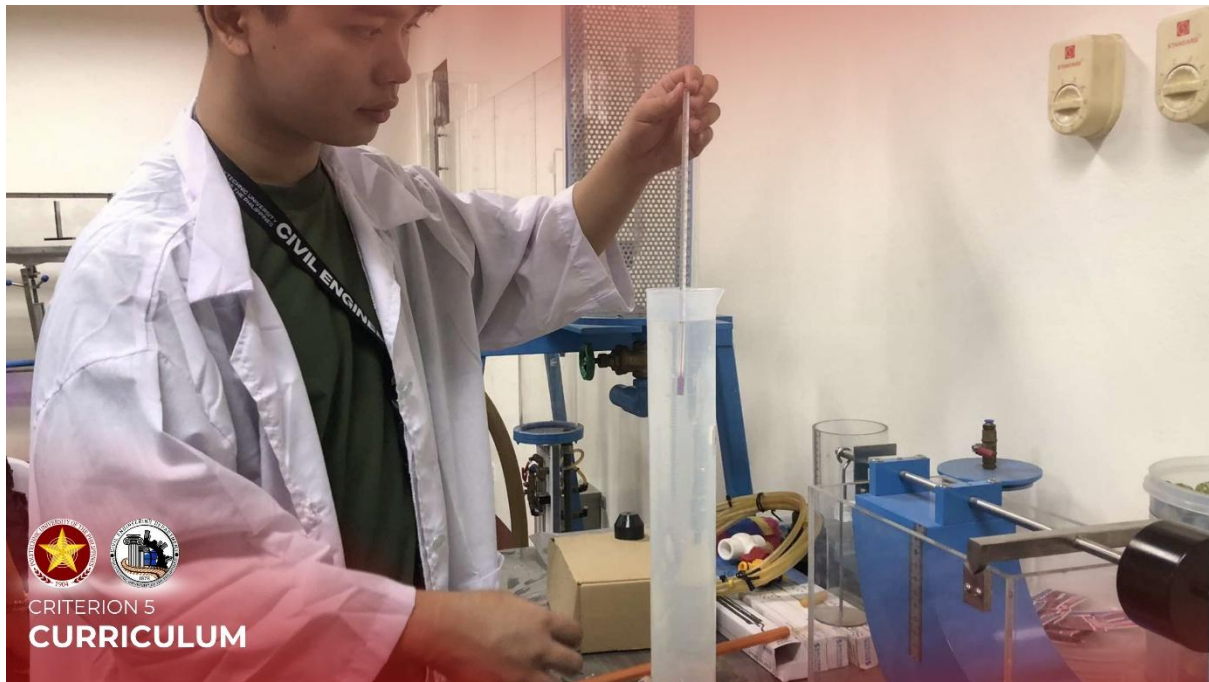


Figure 5.2.2 Third-year BS Civil Engineering students conduct hydraulics laboratory experiments at the PUP Engineering and Science Research Center. The activity enables students to apply fluid mechanics principles, operate specialized laboratory equipment, and analyze experimental data in support of the applicable Course Intended Learning Outcomes and Student Outcomes.

Laboratory instruction is structured to provide students with progressive experience in experimental planning, equipment preparation, data acquisition, quality control, engineering computation, and interpretation of results. Students are required to relate observed laboratory behavior to the governing theories discussed in lecture courses and to evaluate possible sources of error, limitations of testing procedures, and the reliability of measured data. Through these activities, students develop technical judgment and the ability to formulate evidence-based conclusions relevant to civil engineering analysis and design.

Building on the competencies developed through mathematics, basic sciences, engineering sciences, laboratory instruction, and specialization courses, the curriculum provides a progressive sequence of design-oriented learning experiences that culminates in the major design and capstone requirements. This sequence enables students to move from the application of established principles in well-defined problems toward the analysis and resolution of complex engineering problems requiring the integration of technical knowledge, engineering judgment, professional standards, and multidisciplinary considerations.

Major design and capstone experiences provide students with opportunities to integrate knowledge and skills acquired from foundational, engineering science, and professional courses. Students are required to formulate and evaluate engineering solutions while considering applicable codes and standards, technical feasibility, public health and safety, environmental sustainability, economic limitations, constructability, and societal requirements. These activities provide evidence that students can synthesize multidisciplinary knowledge and exercise engineering judgment in addressing complex civil engineering problems ([Attachment C5-03 – Major Design or Capstone Project Activities](#)).



Figure 5.2.3 Fourth-year BS Civil Engineering students defend their capstone projects before a panel of faculty members and evaluators. The activity enables students to integrate knowledge from foundational, engineering science, and professional courses, formulate engineering solutions, and justify their design decisions based on applicable codes and standards, technical feasibility, safety, sustainability, economic considerations, constructability, and societal requirements.

The curriculum also includes integrative and review courses that consolidate competencies acquired across the major civil engineering fields ([Attachment C5-04 – Conduct of Integrative and In-House Review Classes](#)). These courses support students in identifying relationships among structural, geotechnical, transportation, water resources, and construction engineering concepts. They likewise strengthen preparation for professional practice, technical competency assessments, the Civil Engineering Licensure Examination, and other professional qualification requirements. Through these courses, students are provided with opportunities to identify competency gaps and reinforce areas requiring further development.

Internship and industry immersion constitute significant components of the curriculum. Through these activities, students apply engineering principles in actual workplace settings, observe professional practices, participate in project-related assignments, and develop familiarity with organizational procedures, safety requirements, technical documentation, and professional conduct. Internship assignments may be aligned with the students' preferred or assigned areas of civil engineering specialization, enabling them to connect academic preparation with actual industry practice ([Attachment C5-05 – Internship and Industry Immersion Activities](#)).



Figure 5.2.4 Third-year BS Civil Engineering students undergo internship and industry immersion at the Department of Transportation (DOTr). The activity enables students to apply engineering principles in actual workplace settings, observe professional practices, participate in project-related assignments, and develop familiarity with organizational procedures, safety requirements, technical documentation, and professional conduct relevant to civil engineering practice.

Research courses further contribute to the attainment of the PEOs by developing students' ability to identify relevant civil engineering problems, review technical literature, formulate appropriate methodologies, collect and analyze data, and communicate findings. Research and thesis activities may address concerns related to infrastructure development, disaster resilience, transportation systems, construction materials, environmental sustainability, water resources, and community needs. These experiences strengthen analytical thinking, innovation, technical communication, and evidence-based decision-making, which are essential for professional practice and advanced studies.

The curriculum incorporates learning experiences that develop communication, leadership, teamwork, project management, and organizational competencies. Students prepare technical reports, design documents, research manuscripts, laboratory reports, project proposals, and oral presentations. Group-based activities require them to define responsibilities, coordinate tasks, manage schedules, resolve differences, and contribute effectively to multidisciplinary teams. These competencies support graduates' transition into professional practice and their capacity to assume increasing levels of responsibility within engineering organizations.

Courses addressing **engineering ethics, professional responsibility, occupational health and safety, environmental sustainability, legal requirements, and the societal implications of engineering solutions** contribute to the attainment of **PEO 2**. These courses emphasize that civil engineering decisions must consider not only technical requirements but also their effects on individuals, communities, the environment, and the broader public interest. Students are required to evaluate engineering problems using ethical, legal, safety, environmental, and societal considerations, thereby developing accountability and integrity in professional decision-making.



Figure 5.2.5 Fourth-year BS Civil Engineering students conduct a community outreach program in compliance with the requirements of their Earthquake Engineering course. The activity enables students to apply earthquake engineering concepts in a community-based setting, promote awareness on seismic safety and disaster preparedness, and recognize the role of civil engineers in addressing infrastructure concerns, community needs, and nation-building.

Community-based activities, extension-related learning experiences, field studies, and stakeholder-responsive projects further strengthen students' awareness of the role of civil

engineers in community development and nation-building. These activities expose students to actual infrastructure concerns and allow them to recognize the needs of intended beneficiaries, government agencies, industry partners, and local communities ([C5-06 – Community-Based and Extension-Related Learning Activities](#)).

The curriculum likewise supports **PEO 3** by promoting lifelong learning and continuing professional development. Students are exposed to research, emerging technologies, contemporary engineering issues, technical seminars, professional organizations, certification programs, and opportunities for advanced studies. They are encouraged to independently acquire new knowledge, evaluate technological developments, use modern engineering and digital tools responsibly, and recognize the need for continuous updating of professional competencies.

Participation in seminars, conferences, technical workshops, review programs, professional organization activities, and certification-related training provides students with additional opportunities to broaden their knowledge beyond formal classroom instruction. These activities support adaptability to changing professional standards, emerging construction technologies, digital engineering applications, sustainability requirements, and evolving industry practices ([Attachment C5-07 – Professional Development, Certification, and Lifelong Learning Activities](#)).



Figure 5.2.6 Fourth-year BS Civil Engineering students participate in and organize the Pylon Convention as part of the requirements of CE Site Inspection and Seminars. The activity provides students with professional development exposure through seminars, technical discussions, event coordination, and industry-oriented learning sessions that broaden their

understanding of emerging construction technologies, digital engineering applications, sustainability requirements, professional standards, and evolving civil engineering practices.

The alignment of the curriculum with the PEOs is periodically reviewed through curriculum mapping, outcomes assessment, faculty evaluation, and consultation with students, alumni, employers, government representatives, and industry stakeholders. Feedback from these groups is used to determine whether the curriculum continues to provide relevant preparation for professional practice, social responsibility, and lifelong learning. Findings from curriculum review and stakeholder consultation are incorporated into course enhancement and curriculum revision as part of the program's **Continuous Quality Improvement (CQI)** process.

Student performance in the identified SO-contributing courses is evaluated using standardized assessment rubrics aligned with the corresponding **Course Intended Learning Outcomes (CILOs), Student Outcomes (SOs), and SO performance indicators**. The rubrics establish explicit criteria and measurable performance levels for assessing outputs such as laboratory reports, engineering designs, case studies, research papers, oral presentations, internship reports, and major design or capstone projects. This alignment ensures that assessment results provide valid, consistent, and traceable evidence of students' achievement of course-level and program-level outcomes.

The BS Civil Engineering curriculum integrates research as a structured academic component designed to develop students' ability to identify civil engineering problems, formulate research questions, review relevant literature, apply appropriate methodologies, analyze data, and communicate findings in a technical and professional manner. Research courses provide students with opportunities to connect theoretical and applied learning with contemporary issues in infrastructure development, transportation systems, railway engineering, sustainable construction, environmental resilience, and other areas relevant to civil engineering practice.

As part of this curricular component, the Department conducts **TransConVerge**, a culminating research pitching activity for third-year Civil Engineering students enrolled in the CE Research course. The activity serves as an academic platform where students present and defend research proposals before faculty members and invited professionals. Through poster exhibits, oral presentations, and research pitching, students demonstrate their ability to define engineering problems, establish research objectives, justify the relevance of their

studies, and communicate proposed technical solutions in a clear and evidence-based manner.

TransConVerge strengthens the research sequence of the curriculum by allowing students to receive formative feedback before proceeding to more advanced research and project courses. The activity enables faculty members to evaluate the quality, feasibility, relevance, and technical soundness of proposed research topics. It also provides evidence of students' development in critical thinking, technical writing, oral communication, innovation, and problem formulation, which are essential competencies for succeeding research, thesis, capstone, and professional design activities.

The activity is aligned with the applicable **Course Intended Learning Outcomes (CILOs)** and **Student Outcomes (SOs)**, particularly those related to problem analysis, investigation, design thinking, communication, teamwork, ethics, contemporary issues, and lifelong learning. Student outputs are assessed through presentation rubrics, poster evaluation tools, and faculty or panel feedback, which provide evidence of students' readiness to undertake research-based civil engineering projects. The results and documentation of the activity are used to support curriculum evaluation and Continuous Quality Improvement (CQI) initiatives related to research instruction and student preparedness.



Figure 5.2.7. TransConVerge Research Pitching Activity. TransConVerge serves as a culminating venue for third-year Civil Engineering students to present and communicate their proposed research studies through poster displays, oral presentations, and structured research pitching. The activity strengthens students' competencies in research conceptualization, critical analysis, technical communication, innovation, and professional presentation.

The curriculum also provides opportunities for student design and research outputs to be externally presented and evaluated through University-level research competitions. One relevant mechanism is the **PUP Undergraduate Research Competition in Science and Technology**, organized by the **Research Institute for Science and Technology (RIST)**. This annual competition showcases undergraduate research outputs from different PUP colleges, branches, and campuses, thereby providing a venue for students to present innovative, research-based, and design-oriented solutions before a broader academic and technical audience.

Participation in the RIST undergraduate research competition supports the curriculum's emphasis on inquiry-based learning, engineering design, research dissemination, and evidence-based problem solving. For BS Civil Engineering students, outputs from research, capstone, and design-related courses may be developed into competition entries that address civil engineering concerns such as infrastructure systems, construction materials, transportation, water resources, geotechnical applications, environmental sustainability, disaster resilience, and community-responsive engineering solutions. These activities demonstrate how course-based outputs are extended beyond classroom assessment and subjected to external review and comparative evaluation.



Figure 5.2.8. BS Civil Engineering Students at the 2026 RIST Undergraduate Research Competition in Science and Technology. BS Civil Engineering students participated in the PUP Undergraduate Research Competition in Science and Technology organized by the Research Institute for Science and Technology and placed 3rd Overall in the 2026 competition. The recognition demonstrates the curriculum's support for research-based design, innovation, technical communication, and external evaluation of student outputs.

Through this mechanism, the Department is able to further evaluate the quality and relevance of student outputs generated from the curriculum. Feedback, results, and recognitions from research competitions serve as supplementary evidence for assessing student readiness in research, design, communication, and professional presentation. These outcomes also inform the program's Continuous Quality Improvement (CQI) process by identifying strengths and areas for enhancement in research mentoring, project development, technical documentation, and presentation preparation.

Student research outputs developed through the curriculum may also lead to participation in national and international conferences, research dissemination activities, and publication opportunities. An example is the participation of BS Civil Engineering students in the **9th International Conference on Green Urbanism** organized by IEREK in collaboration with Sapienza University of Rome. The conference provided students with an international platform to present research aligned with sustainable and resilient urban development, including themes on green urban systems, smart and green mobility, climate adaptation, nature-based solutions, urban regeneration, and social sustainability.



Figure 5.2.9. Student Research Presentation at the 9th International Conference on Green Urbanism. PUP BS Civil Engineering students disseminated their research work in an international academic conference centered on sustainable, resilient, and future-oriented urban development. This engagement highlights the students' developing competencies in research presentation, technical communication, global academic participation, and the application of civil engineering concepts to current urban and environmental challenges.

Participation in international research conferences demonstrates how the curriculum extends research learning beyond classroom instruction and exposes students to global scholarly standards. Through these experiences, students strengthen their technical communication, research presentation skills, professional confidence, and awareness of emerging civil engineering and urban development issues. The opportunity for publication under an international academic publisher further supports the development of research competence, lifelong learning, and professional engagement.

The results generated through these rubrics are analyzed to determine the extent to which students attain the identified CILOs and SO performance indicators. Although the **Program Educational Objectives (PEOs)** represent graduate accomplishments expected three to five years after graduation and are assessed primarily through alumni, employer, and stakeholder feedback, the rubric-based assessment of CILOs and SOs provides essential curricular evidence that students are developing the competencies, values, and professional attributes necessary to support the eventual attainment of the PEOs. Identified performance gaps are used to improve Teaching-Learning Activities, Assessment Tasks, course content, and other curricular components through the program's Continuous Quality Improvement process ([Attachment C5-08 – Assessment Rubrics](#)).

Portions of **Table 5-2.1** and **Table 5-2.2** presented in the succeeding pages illustrate the alignment of the professional courses in the 2018 and 2022 BSCE curricula respectively with the PEOs. The complete mappings of the technical and non-technical courses in the 2018 and 2022 BSCE curricula are provided in the corresponding attachments. These mappings, together with the documented laboratory, design, internship, research, community engagement, and professional development activities, demonstrate how the curriculum systematically contributes to the development of graduates who are prepared for professional civil engineering practice, social and ethical responsibility, and continuous professional advancement.

[\(Attachment C5-09 – Table 5-2.1. Relationship of Civil Engineering Courses to the Program Educational Objectives - 2018 BSCE Curriculum\)](#)

[\(Attachment C5-10 – Table 5-2.2. Relationship of Civil Engineering Courses to the Program Educational Objectives - 2022 BSCE Curriculum\)](#)

Portion of Table 5-2.1. Relationship of Civil Engineering Courses to the Program Educational Objectives - 2018 BSCE Curriculum

Course	Course Intended Learning Outcome (CILO)		<i>Three to five years after graduation, the Bachelor of Science in Civil Engineering alumni shall:</i>		
(Course Code and Title)	#	CILO	PEO 1. Demonstrating Engineering Profession Employed and/or practiced in the field of Civil Engineering, such as but not limited to Construction Engineering and Management, Geotechnical Engineering, Structural Engineering, Transportation Engineering, and Water Resources Engineering, or other related fields.	PEO 2. Social and Ethical Responsibility Committed members of professional and other allied organizations engaging in community development or nation-building.	PEO 3. Lifelong Learning Involved in continuous training and development in current trends and advancements in their fields of specialization.
E. Professional Courses - Common					
(CIEN 30013) Fundamentals of Surveying 1 LEC	1	Comprehend and compute the length and direction of a line, the elevation and/or difference in elevation between two points, and an area of a lot.	✓	✓	
	2	Prepare and explain how each method is applied in surveying practice.	✓	✓	
	3	Perform the computation and analyze the significance of each.	✓	✓	
(CIEN 30013) Fundamentals of Surveying 1 - Field	1	Demonstrate knowledge on the technical know-how of the basics of surveying like taping, determining elevation and/or difference in elevation between two points, setting up and manipulation of the Engineer's Transit, angle by repetition and traversing.	✓	✓	
	2	Begin to show willingness to perform the assigned task/s.	✓	✓	
	3	Develop confidence in performing each field.	✓	✓	
(CIEN 30033) Fundamentals of Surveying 2 LEC	1	Comprehend and compute for the omitted information of a closed traverse, partitioning of a lot, principle of the stadia, contouring, triangulation, sextant, hydrographic surveying and astronomical observation.	✓	✓	

Course	Course Intended Learning Outcome (CILO)		Three to five years after graduation, the Bachelor of Science in Civil Engineering alumni shall:		
(Course Code and Title)	#	CILO	PEO 1. Demonstrating Engineering Profession Employed and/or practiced in the field of Civil Engineering, such as but not limited to Construction Engineering and Management, Geotechnical Engineering, Structural Engineering, Transportation Engineering, and Water Resources Engineering, or other related fields.	PEO 2. Social and Ethical Responsibility Committed members of professional and other allied organizations engaging in community development or nation-building.	PEO 3. Lifelong Learning Involved in continuous training and development in current trends and advancements in their fields of specialization.
	2	Prepare and explain how each method is applied in surveying practice.	✓	✓	
	3	Perform the computation and discuss the information of a closed traverse, partitioning of a lot, principle of the stadia, contouring, triangulation, sextant, hydrographic surveying and astronomical observation.	✓	✓	
(CIEN 30033) Fundamentals of Surveying 2 - Field	1	Comprehend and compute for the omitted information of a closed traverse, partitioning of a lot, principle of the stadia, contouring, triangulation, sextant, hydrographic surveying and astronomical observation.	✓	✓	
	2	Prepare and explain how each method is applied in surveying practice.	✓	✓	
	3	Perform the computation and discuss the significance of each.	✓	✓	
(CIEN 30053) Route Surveying and Earthworks LEC	1	Comprehend and compute problems on topics like simple curve, compound curves, reversed curves, symmetrical and unsymmetrical parabolic curve, passing sight distance, spiral curve, volumes of earthworks, and haul and mass diagram	✓	✓	
	2	Prepare and explain how each method is applied in surveying practice	✓	✓	

Course	Course Intended Learning Outcome (CILO)		Three to five years after graduation, the Bachelor of Science in Civil Engineering alumni shall:		
(Course Code and Title)	#	CILO	PEO 1. Demonstrating Engineering Profession Employed and/or practiced in the field of Civil Engineering, such as but not limited to Construction Engineering and Management, Geotechnical Engineering, Structural Engineering, Transportation Engineering, and Water Resources Engineering, or other related fields.	PEO 2. Social and Ethical Responsibility Committed members of professional and other allied organizations engaging in community development or nation-building.	PEO 3. Lifelong Learning Involved in continuous training and development in current trends and advancements in their fields of specialization.
	3	Perform the necessary calculations and discuss their significance in surveying practice	✓	✓	
(CIEN 30053) Route Surveying and Earthworks LAB	1	Demonstrate knowledge on the technical know-how of engineering surveys like computations and/or laying out of simple curves, compound curves, reversed curves, symmetrical and unsymmetrical parabolic curve, passing sight distance, spiral curve and volumes of earth.	✓	✓	
	2	Exemplify initiative and leadership in performing surveying works	✓	✓	
	3	Exemplify sound knowledge in performing each fieldwork	✓	✓	
(CIEN 30023) Construction Materials and Testing LEC	1	Explain relevant properties of common construction materials.	✓	✓	
	2	Assess the qualities / properties of different construction materials	✓	✓	
	3	Apply appropriate procedures on testing based on local or international standards	✓	✓	
(CIEN 30023) Construction Materials and Testing LAB	1	List the commonly used construction materials and their uses	✓	✓	
	2	Compute the properties of materials based on the data taken from laboratory experiment	✓	✓	
	3	Assess the qualities / properties of different construction materials	✓	✓	

Course	Course Intended Learning Outcome (CILO)		Three to five years after graduation, the Bachelor of Science in Civil Engineering alumni shall:		
(Course Code and Title)	#	CILO	PEO 1. Demonstrating Engineering Profession Employed and/or practiced in the field of Civil Engineering, such as but not limited to Construction Engineering and Management, Geotechnical Engineering, Structural Engineering, Transportation Engineering, and Water Resources Engineering, or other related fields.	PEO 2. Social and Ethical Responsibility Committed members of professional and other allied organizations engaging in community development or nation-building.	PEO 3. Lifelong Learning Involved in continuous training and development in current trends and advancements in their fields of specialization.
	4	Demonstrate appropriate procedures on testing based on local or international standards	✓	✓	
	5	Apply the safety procedures in the laboratory	✓	✓	
	6	Formulate concrete and asphalt mixtures based on actual condition of materials	✓	✓	
(CIEN 30043) Structural Theory 1	1	To determine the fundamental principles in the analysis of structures.	✓		
	2	Analyze stresses in statically determinate structures.	✓		
	3	Analyze stresses in statically indeterminate structures using approximate methods.	✓		
(CIEN 30043) Structural Theory 1 - Computational	1	To determine the fundamental principles in the analysis of structures.	✓		
	2	Analyze stresses in statically determinate structures.	✓		
	3	Analyze stresses in statically indeterminate structures using approximate methods.	✓		
(CIEN 30103) Structural Theory 2 Lec	1	Solve deflections of statically determinate and indeterminate structural members.	✓	✓	
	2	Analyze forces stresses in statically indeterminate structures using precise methods.	✓	✓	
	3	Calculate structural responses using the basic concepts of matrix structural analysis.	✓	✓	

Course	Course Intended Learning Outcome (CILO)		Three to five years after graduation, the Bachelor of Science in Civil Engineering alumni shall:		
(Course Code and Title)	#	CILO	PEO 1. Demonstrating Engineering Profession Employed and/or practiced in the field of Civil Engineering, such as but not limited to Construction Engineering and Management, Geotechnical Engineering, Structural Engineering, Transportation Engineering, and Water Resources Engineering, or other related fields.	PEO 2. Social and Ethical Responsibility Committed members of professional and other allied organizations engaging in community development or nation-building.	PEO 3. Lifelong Learning Involved in continuous training and development in current trends and advancements in their fields of specialization.
(CIEN 30103) Structural Theory 2 Computational	1	Solve deflections of statically determinate and indeterminate structural members.	✓	✓	
	2	Analyze forces stresses in statically indeterminate structures using precise methods.	✓	✓	
	3	Calculate structural responses using the basic concepts of matrix structural analysis.	✓	✓	
(CIEN 30162) Matrix Analysis of Structures	1	Calculate deflections, reactions and internal forces for trusses, beams and frames using analytical and computer-based methods	✓	✓	
	2	Extend the study of linear elastic analysis to include nonlinear aspects of structure behavior	✓	✓	
	3	Interpret computer output and validate results using simplified models and hand calculations	✓	✓	
(CIEN 30113) Highway and Railroad Engineering	1	Explain the principles underlying highway design	✓	✓	
	2	Assess alternative pavement designs and understand their deterioration and maintenance schemes for highways.	✓	✓	
	3	Design a highway allowing for different terrains, horizontal and vertical curves.	✓	✓	
	4	Develop basic traffic stream parameters and models, traffic flow models, and apply the queuing theory	✓	✓	

Course	Course Intended Learning Outcome (CILO)		Three to five years after graduation, the Bachelor of Science in Civil Engineering alumni shall:		
(Course Code and Title)	#	CILO	PEO 1. Demonstrating Engineering Profession Employed and/or practiced in the field of Civil Engineering, such as but not limited to Construction Engineering and Management, Geotechnical Engineering, Structural Engineering, Transportation Engineering, and Water Resources Engineering, or other related fields.	PEO 2. Social and Ethical Responsibility Committed members of professional and other allied organizations engaging in community development or nation-building.	PEO 3. Lifelong Learning Involved in continuous training and development in current trends and advancements in their fields of specialization.
(CIEN 30073) Building Systems Design LEC	1	Identify appropriate provisions of the National Building Code and its referral codes as these apply to buildings.	✓	✓	
	2	Ability to design sites, facilities, and system principles that are responsive to relevant codes and regulations, and include the principle of life safety and accessibility standards.	✓	✓	
	3	Ability to make technically clear drawings, prepare outline specifications, and construct models illustrating and identifying the assembly of materials, systems, and components appropriate for a building design.	✓	✓	
	4	Understanding of the basic principles involved in the appropriate selection and application of building envelope systems relative to fundamental performance, aesthetics, moisture transfer, durability, and energy and material resources.	✓	✓	
	5	Understanding of the basic principles used in the appropriate selection of interior and exterior construction, finishes, products, components, and assemblies based on their inherent performance, including environmental impact and reuse.	✓	✓	
	6	Understanding of the basic principles and appropriate application and performance of	✓	✓	

Course	Course Intended Learning Outcome (CILO)		Three to five years after graduation, the Bachelor of Science in Civil Engineering alumni shall:		
(Course Code and Title)	#	CILO	PEO 1. Demonstrating Engineering Profession Employed and/or practiced in the field of Civil Engineering, such as but not limited to Construction Engineering and Management, Geotechnical Engineering, Structural Engineering, Transportation Engineering, and Water Resources Engineering, or other related fields.	PEO 2. Social and Ethical Responsibility Committed members of professional and other allied organizations engaging in community development or nation-building.	PEO 3. Lifelong Learning Involved in continuous training and development in current trends and advancements in their fields of specialization.
(CIEN 30073) Building Systems Design LAB		building service systems, including lighting, mechanical, plumbing, electrical, communication, vertical transportation, security, and fire protection systems			
	7	Understanding of the fundamentals of building costs, which must include life cycle costs.	✓	✓	
	1	Identify appropriate provisions of the National Building Code and its referral codes as these apply to buildings.	✓	✓	
	2	Ability to design sites, facilities, and system principles that are responsive to relevant codes and regulations, and include the principle of life safety and accessibility standards.	✓	✓	
	3	Ability to make technically clear drawings, prepare outline specifications, and construct models illustrating and identifying the assembly of materials, systems, and components appropriate for a building design.	✓	✓	
	4	Understanding of the basic principles involved in the appropriate selection and application of building envelope systems relative to fundamental performance, aesthetics, moisture transfer, durability, and energy and material resources.	✓	✓	

Course	Course Intended Learning Outcome (CILO)		Three to five years after graduation, the Bachelor of Science in Civil Engineering alumni shall:		
(Course Code and Title)	#	CILO	PEO 1. Demonstrating Engineering Profession Employed and/or practiced in the field of Civil Engineering, such as but not limited to Construction Engineering and Management, Geotechnical Engineering, Structural Engineering, Transportation Engineering, and Water Resources Engineering, or other related fields.	PEO 2. Social and Ethical Responsibility Committed members of professional and other allied organizations engaging in community development or nation-building.	PEO 3. Lifelong Learning Involved in continuous training and development in current trends and advancements in their fields of specialization.
	5	Understanding of the basic principles used in the appropriate selection of interior and exterior construction, finishes, products, components, and assemblies based on their inherent performance, including environmental impact and reuse.	✓	✓	
	6	Understanding of the basic principles and appropriate application and performance of building service systems, including lighting, mechanical, plumbing, electrical, communication, vertical transportation, security, and fire protection systems	✓	✓	
	7	Understanding of the fundamentals of building costs, which must include life cycle costs.	✓	✓	
(CIEN 30253) Principles of Steel Design LEC	1	Understand the stress-strain characteristics of steel in both elastic and plastic stages of deformation.	✓		
	2	Be able to determine the strength of structural steel members in axial loading, bending, shear, and torsion for various cross-sectional shapes in both elastic and plastic conditions	✓		
	3	Be able to apply allowable strength and load-and-resistance factor design methods of steel design.	✓		
	4	Be able to determine the strength of bolted/riveted and welded connections.	✓		

Course	Course Intended Learning Outcome (CILO)		Three to five years after graduation, the Bachelor of Science in Civil Engineering alumni shall:		
(Course Code and Title)	#	CILO	PEO 1. Demonstrating Engineering Profession Employed and/or practiced in the field of Civil Engineering, such as but not limited to Construction Engineering and Management, Geotechnical Engineering, Structural Engineering, Transportation Engineering, and Water Resources Engineering, or other related fields.	PEO 2. Social and Ethical Responsibility Committed members of professional and other allied organizations engaging in community development or nation-building.	PEO 3. Lifelong Learning Involved in continuous training and development in current trends and advancements in their fields of specialization.
(CIEN 30253) Principles of Steel Design Computational	1	At the end of the course, the students must be able to Demonstrate by numerical calculations and other methods the concepts, principles and theories obtained from the lecture portion of the course	✓		
(CIEN 30183) Principles of Reinforced Concrete LEC	1	Understand the in-depth stress-strain characteristics of concrete for all stages of loading.	✓	✓	
	2	Evaluate the strength of structural plain, reinforced, or prestressed concrete elements in axial loading, bending, shear, and torsion under pseudo-elastic and inelastic stages of loading.	✓	✓	
	3	Apply allowable strength and load-and-resistance factor design methods for reinforced and prestressed concrete.	✓	✓	
(CIEN 30183) Principles of Reinforced Concrete Computational	1	Demonstrate by numerical calculations and other methods the concepts, principles, and design of plain theories obtained from the lecture portion of the course.	✓	✓	
(CIEN 30193) Reinforced Concrete Design LEC	1	Understand the process involved in the design of reinforced concrete frame structures	✓	✓	
	2	Demonstrate how to design a reinforced concrete structure such as building	✓	✓	

Course	Course Intended Learning Outcome (CILO)		Three to five years after graduation, the Bachelor of Science in Civil Engineering alumni shall:		
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(CIEN 30193) Reinforced Concrete Design Computational	1	Solve theoretical and situational problems in the analysis and design of reinforced concrete beams.	✓	✓	
	2	Solve theoretical and situational problems in the analysis and design of reinforced concrete columns.	✓	✓	
(CIEN 30222) Timber Design LEC	1	Know the basic concepts and design techniques of structural timber design	✓		
	2	Solve theoretical and situational problems in the analysis and design of timber beams, columns, and other structures	✓	✓	
	3	Create structural plans, detailed drawings, and calculations for at least two storey timber structure	✓		
(CIEN 30222) Timber Design Computational	1	Solve theoretical and situational problems in the analysis and design of timber beams, columns, and other structures.	✓		
	2	Create structural plans, detailed drawings, and calculations for at least two-storey timber structure.	✓	✓	
	3	Evaluate laboratory and field data to select appropriate parameters to use in timber designs.	✓		
(CIEN 30063) Hydraulics 1 (LEC)	1	Understand basic properties of fluids and its behavior.	✓	✓	
	2	Compute hydrostatic pressure on different type of surface.	✓	✓	

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	3	Understand relative equilibrium of liquids.	✓	✓	
(CIEN 30063) Hydraulics 1 (LAB)	1	Understand basic properties of fluids and its behavior including the measurement of Pressure	✓	✓	
	2	Compute hydrostatic pressure on different type of surface	✓	✓	
	3	Understand buoyancy and stability of floating bodies	✓	✓	
(CIEN 30143) Hydraulics 2 LEC	1	To apply the principle of continuity in closed conduits in determining the discharge	✓		
	2	To apply the Bernoulli's principle in calculating the flow energy in closed conduits and open channels	✓		
	3	To apply momentum equations in solving for forces on fixed and moving objects	✓		
(CIEN 30143) Hydraulics 2 LAB	1	To apply the principle of continuity in closed conduits in determining the discharge	✓	✓	
	2	To apply the Bernoulli's principle in calculating the flow energy in closed conduits and open channels	✓	✓	
	3	To apply momentum equations in solving for forces on fixed and moving objects	✓	✓	
(CIEN 30132) Hydrology	1	Discuss with appropriate diagrams the hydrologic cycle and the different processes and storages within the cycle	✓	✓	
	2	Perform calculation related to measurements, movement, and storages in	✓	✓	

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		the different processes of the hydrologic cycle.			
(CIEN 30152) CE Law, Ethics and Contracts LEC	1	Explain the Civil Engineering Code of Ethics which includes the fundamental principles and canon.		✓	
	2	Explain the laws governing the practice of civil engineering profession in the Philippines.		✓	
	3	Evaluate the results of the test of common construction materials		✓	
	4	Appraise the civil engineering fundamental principles and canons in oral presentations		✓	
(CIEN 30083) Geotechnical Engineering 1 LEC	1	To explain the concept of soil as an engineering material and the properties and methods used to characterize soil for geotechnical analysis and design.	✓	✓	
	2	To interpret terminology and parameters used to characterize and classify soil; stresses and stress conditions in soils; factors affecting soil strength and stress-strain behavior.	✓	✓	
	3	To classify the seepage and water flow through soils and their effects on soil stresses and strength; deformation and settlement characteristics of soils; lateral earth pressure, bearing capacity and slope stability concepts.	✓	✓	

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(CIEN 30083) Geotechnical Engineering 1 FIELD	1	To utilize the concept of soil as an engineering material and the properties and methods used to characterize soil for geotechnical analysis and design.	✓	✓	
	2	To model the parameters used to characterize and classify soils; stresses and stress conditions in soils; factors affecting soil strength and stress-strain behavior.	✓	✓	
	3	To experiment with the seepage and water flow through soils and their effects on soil stresses and strength; deformation and settlement characteristics of soils.	✓	✓	
(CIEN 30122) Geotechnical Engineering 2 LEC	1	Identify the objectives of geotechnical data collection and rock mass classification methods, and successfully collect and analyze a range of geotechnical datasets for design purposes	✓	✓	
	2	describe the theory and analysis of in situ and induced stresses in a rock mass and structurally controlled failure	✓	✓	
	3	apply the principles of rock mechanics and excavation design to develop excavation proposals for given geologic environments (e.g stratified, massive, blocky, or faulted lithologies)	✓	✓	
	4	explain the principles and techniques of reinforcement design for the primary failure modes in underground rock excavations,	✓	✓	

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		and apply these principles and techniques to inform design for tunnels and large excavations			
	5	describe the theory, analysis, and control of rock and soil slope stability and rockfall hazards	✓	✓	
	6	calculate the Factor of Safety of rock slopes and underground excavations	✓	✓	
	7	describe the mechanical implications of various blast types and patterns for a rock mass, and characterize blasting control systems	✓	✓	
(CIEN 30242) Foundation Engineering LEC	1	Plan a subsurface exploration, and select appropriate drilling, sampling, and field property measurement tools for different soil profiles.	✓	✓	
	2	Create an idealized soil profile for analysis and design from collected soil properties.	✓	✓	
	3	Evaluate laboratory and field data to select appropriate parameters to use in foundation analysis.	✓	✓	
	4	Determine the lateral loads from soils.	✓	✓	
	5	Design earth retaining structures.			
	6	Apply shallow and deep bearing capacity and settlement equation for design with applicable correction factors with a	✓	✓	

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		consideration of water effects and layering.			
(CIEN 30242) Foundation Engineering LAB	1	Plan a subsurface exploration, and select appropriate drilling, sampling, and field property measurement tools for different soil profiles.	✓	✓	
	2	Create an idealized soil profile for analysis and design from collected soil properties.	✓	✓	
	3	Evaluate laboratory and field data to select appropriate parameters to use in foundation analysis.	✓	✓	
	4	Determine the lateral loads from soils.	✓	✓	
	5	Design earth retaining structures.			
	6	Apply shallow and deep bearing capacity and settlement equation for design with applicable correction factors with a consideration of water effects and layering.	✓	✓	
(CIEN 30203) Principles of Transportation Engineering	1	Define concepts of transportation systems analysis and planning	✓	✓	
	2	Apply travel demand forecasting computation and analysis	✓	✓	
	3	Prepare traffic impact assessment reports	✓	✓	
	4	Recognize current transportation issues and policies, economic, safety and environmental concerns	✓	✓	
	1	Determine quantity of various building materials	✓	✓	✓

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(CIEN 30272) Quantity Surveying LEC	2	Develop bid costs for materials, labor and equipment for several construction project elements	✓	✓	✓
	3	Evaluate total cost of engineering projects, including labor fringes, taxes, bonding fees, insurance, overhead and profit.	✓	✓	✓
(CIEN 30272) Quantity Surveying LAB	1	Define quantity of various building materials	✓	✓	✓
	2	Develop bid costs for materials, labor and equipment for several construction project elements.	✓	✓	✓
	3	Evaluate total cost of engineering projects, including labor fringes, taxes, bonding fees, insurance, overhead and profit.	✓	✓	✓
(CIEN 30293) Construction Methods and Project Management	1	Understand the construction project cycle from inception, planning, execution, testing and certification.	✓		
	2	Understand construction project requirements such as permits and licenses, clearance, and compliance to all applicable laws and ordinances, etc	✓		
	3	Read and interpret construction plans, working drawings, and revise contract documents, estimates and technical specifications	✓		
	4	Understand the components of the Terms of Reference (TOR) for professional services	✓		

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		and bidding/tendering processes, award and acceptance			
	5	Prepare and update construction schedule and work programs,	✓		
	6	Understand construction methods, equipment, materials and manpower control:	✓		
	7	Understand temporary facilities requirements of a construction project,	✓		
	8	Understand the existing local labor laws and regulations	✓		
(CIEN 30212) CE Project 1 LEC	1	Critically evaluate a range of research methods and techniques and be able to create a convincing written rationale for selecting them.	✓	✓	
	2	Craft research questions pertinent to the fields of engineering and related disciplines, based in an extensive knowledge of the relevant literature.	✓	✓	
	3	Design a research project of appropriate scope for a higher degree by research candidature.	✓	✓	
	4	Write a literature review which supports a research project and states clearly the project's contribution to knowledge	✓	✓	
(CIEN 30212) CE Project 1 LAB	1	Critically evaluate a range of research methods and techniques and be able to	✓	✓	

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		create a convincing written rationale for selecting them.			
	2	Craft research questions pertinent to the fields of engineering and related disciplines, based in an extensive knowledge of the relevant literature.	✓	✓	
	3	Design a research project of appropriate scope for a higher degree by research candidature.	✓	✓	
	4	Write a literature review which supports a research project and states clearly the project's contribution to knowledge	✓	✓	
(CIEN 30262) CE Project 2 LEC	1	Critically evaluate a range of research methods and techniques and be able to create a convincing written rationale for selecting them.	✓	✓	
	2	Craft research questions pertinent to the fields of engineering and related disciplines, based in an extensive knowledge of the relevant literature.	✓	✓	
	3	Design a research project of appropriate scope for a higher degree by research candidature.	✓	✓	
	4	Write a literature review which supports a research project and states clearly the project's contribution to knowledge	✓	✓	

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(CIEN 30262) CE Project 2 LAB	1	Critically evaluate a range of research methods and techniques and be able to create a convincing written rationale for selecting them.	✓	✓	
	2	Craft research questions pertinent to the fields of engineering and related disciplines, based in an extensive knowledge of the relevant literature.	✓	✓	
	3	Design a research project of appropriate scope for a higher degree by research candidature.	✓	✓	
	4	Write a literature review which supports a research project and states clearly the project's contribution to knowledge	✓	✓	
(CIEN 30171) Integrative Course 1	1	Gain thorough understanding of all early math courses (Algebra, Plane and Spherical Trigonometry, Solid Mensuration, and Analytic Geometry) completed.	✓		
	2	Gain thorough understanding of all late math courses (Calculus 1 and 2, Differential Equations, Engineering Data Analysis, Numerical SOLUTIONS to CE Problems, Engineering Economics) and physics course (Physics for Engineers) completed.	✓		
	3	Gain thorough understanding of all surveying courses (Fundamentals of Surveying, Route Surveying, Highway	✓		

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		Engineering and Principles of Transportation Engineering) completed.			
(CIEN 30231) Integrative Course 2	1	Gain thorough understanding of all Hydraulics and Water Resources Engineering courses completed.	✓		
	2	Gain thorough understanding of all Geotechnical Engineering courses completed.	✓		
(CIEN 30301) Integrative Course 3	1	Gain a thorough understanding of all analysis of systems and structures taken as rigid and motion bodies considering the effects of forces.	✓		
	2	Gain a thorough understanding of all different types of structural systems, loads and its combinations and placement, applicable codes and specifications, and methods of analysis for statically determinate structures..	✓		
	3	Gain a thorough understanding of all fundamental concepts, principles, and theories in the structural strength analysis and design of plain, reinforced, prestressed concrete, and elements in a structure.	✓		
	4	Gain a thorough understanding of the methods used to prepare a building and/or bridge construction cost estimate, earthwork costs, and to prepare quantity takeoff utilizing plans and specifications	✓		

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(CIEN 30311) CE Field Trip	1	To assess and evaluate the current situation of the construction in the country.	✓	✓	✓
	2	To demonstrate the actual works in different civil works.	✓	✓	✓
	3	To have a knowledge on various discipline in construction.	✓	✓	✓
	4	To have an idea on the current problems and issues facing the existing civil engineering in the country.	✓	✓	✓
(CIEN 30092) Civil Engineering Research LEC	1	Critically evaluate a range of research methods and techniques and be able to create a convincing written rationale for selecting them.	✓	✓	
	2	Craft research questions pertinent to the fields of engineering and related disciplines, based in an extensive knowledge of the relevant literature.	✓	✓	
	3	Design a research project of appropriate scope for a higher degree by candidature. research	✓	✓	
	4	Write a literature review which supports a research project and states clearly the project's contribution to knowledge	✓	✓	
(CIEN 30092) Civil Engineering Research LAB	1	Critically evaluate a range of research methods and techniques and be able to create a convincing written rationale for selecting them.	✓	✓	

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	2	Craft research questions pertinent to the fields of engineering and related disciplines, based in an extensive knowledge of the relevant literature.	✓	✓	
	3	Design a research project of appropriate scope for a higher degree by candidature. research	✓	✓	
	4	Write a literature review which supports a research project and states clearly the project's contribution to knowledge	✓	✓	
(CIEN 30282) Earthquake Engineering LEC	1	Understand the occurrence and origin of earthquakes.	✓	✓	
	2	Understand structural dynamics and appreciate the response of structures to seismic excitation.	✓	✓	
	3	Apply the latest developments in earthquake and seismic design relevance to the National Structural Code of the Philippines.	✓	✓	
F. Professional Courses - Specialized					
(CMEN 41013) Professional Course – Specialized 1 (Project Construction Management)	1	Describe the management process and explain the different functions of management.	✓	✓	✓
	2	Craft a personal mission and vision statement.	✓	✓	✓
	3	Perform a SWOT analysis.	✓	✓	✓
	4	Identify the types of organizational structures.	✓	✓	✓

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	5	Identify management practices applied in civil engineering.	✓	✓	✓
(GTEN 42013) Professional Course – Specialized 1 (Geosynthetics in Geotechnical Engineering)	1	Evaluate the fundamental concepts of geosynthetics.	✓	✓	
	2	Evaluate the geotechnical applications of geosynthetics.	✓	✓	
	3	Evaluate the hydraulic and geoenvironmental applications of geosynthetics.	✓	✓	
	4	Evaluate the transportation applications of geosynthetics.	✓	✓	
	5	Evaluate the mining, agricultural and aquacultural applications of geosynthetics.	✓	✓	
	6	Create geosynthetic application designs considering the general guidelines and installation survivability requirements.	✓	✓	
	7	Create geosynthetic application designs using fiber-reinforced soils.	✓	✓	
(STEN 43013) Professional Course – Specialized 1 (Computer Software in Structural Analysis)	1	Familiar with computer software commercially available for structural analysis and knowledgeable in their use and application.	✓		
(TREN 44103) Professional Course – Specialized 1	1	Describe the elements of the Transportation, evolution of urban transportation and its policy.	✓	✓	

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(Transportation Systems Planning and Design)	2	Explain the Modelling Transportation Systems, Urban Travel & Land Transportation system characteristics, travel demand with Land use.	✓	✓	
	3	Make a modeling mode of mode choice and traffic Assignment.	✓	✓	
	4	Investigate and analyze value pricing, ITS and its policy.	✓	✓	
(WREN 45301) Professional Course – Specialized 1 (Water Resources Engineering)	1	Estimate peak runoff, determine the appropriate pump rating to deliver water from identified sources to a receiving tank/reservoir, determine the sizes of pipes given flow rate, and determine the appropriate size of open channel given discharge.	✓		✓
	2	Determine the parameters of different probability distributions commonly used in flood frequency analysis.	✓		✓
	3	Design a basic drainage system and/or simple water supply system.	✓		✓
(CMEN 41023) Professional Course – Specialized 2 (Advanced Construction Methods and Equipment)	1	Draw a network diagram using Activity on Arrow (AOA) and Activity on Node (AON) methods.	✓	✓	✓
	2	Prepare and update construction estimates, schedule, and material and manpower report.	✓	✓	✓
	3	Calculate labor and equipment productivity.	✓	✓	✓

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	4	Read and interpret construction plans, working drawings, contract documents and technical specifications in the preparation of a complete bid for a construction project.	✓	✓	✓
	5	Identify and present a traditional and new method/theory/practice applied in construction projects.	✓	✓	✓
	6	Investigate, report and propose a solution to a safety or quality issue in a construction project.	✓	✓	✓
(GTEN 42023) Professional Course – Specialized 2 (Geotechnical Earthquake Engineering)	1	Classify and evaluate earthquake sources/loads and associated geotechnical earthquake hazards.	✓	✓	
	2	Explain key concepts in the seismic assessment of engineering structures.	✓	✓	
	3	Interpret the behavior of soils and geotechnical structures during earthquakes.	✓	✓	
	4	Apply simplified design-oriented methods for liquefaction analysis, and seismic analysis and design of retaining walls, embankments, bridges and building foundations.	✓	✓	
	5	Analyze the response of sites, earth structures and soil-structure systems under earthquake loads.	✓	✓	
(STEN 43023) Professional Course – Specialized 2	1	Explain the principle of prestressing.	✓	✓	
	2	Analyze prestressed concrete beams, columns, piles, slabs and other structures.	✓	✓	

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(Prestressed Concrete Design)	3	Design Structural plans, shop drawings, and computation of at least two – storey prestressed concrete structure.	✓	✓	
(TREN 44023) Professional Course – Specialized 2 (Highway Engineering)	1	Discuss the development of early roads and highway in the Philippines from the early part of 1900 to the present era.	✓	✓	
	2	Recognize the importance of planning stage of a proposed road construction project and be familiar with the items confronting to highway design standard.	✓	✓	
	3	Apply the methods of analysis and design relevant to highway engineering.	✓	✓	
	4	Define the factors of affect the performance of highways and related structures.	✓	✓	
(WREN 45023) Professional Course – Specialized 2 (Flood Control and Drainage Design)	1	Identify and discuss the different components of rainwater harvesting system.	✓	✓	
	2	Identify and discuss the different components of storm water drainage system.	✓	✓	
	3	Identify and discuss the different components of flood gates and pumping systems.	✓	✓	
	4	Perform calculation related to sizing of gutters, inlets, culverts and channels needed to convey storm water as well as sizing of retention and detention ponds.	✓	✓	

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(CMEN 41033) Professional Course – Specialized 3 (Construction Cost Engineering)	1	Apply the project management information system, cost engineering and accounting concepts in construction projects.	✓	✓	✓
	2	To analyze and design an improved cost engineering and accounting system for a construction project.	✓	✓	✓
(GTEN 42033) Professional Course – Specialized 3 (Slope Stability and Earth Dams)	1	Evaluate the geotechnical aspects of the causes of slope stability failure.	✓	✓	
	2	Evaluate the structural stability of reinforced slopes and embankments in Earth dams.	✓	✓	
	3	Evaluate the seismic safety and reliability of slope stabilization and repair.	✓	✓	
	4	Evaluate the geotechnical considerations on slope stability in designing Earth dams.	✓	✓	
(STEN 43033) Professional Course – Specialized 3 (Structural Design of Towers and other Vertical Structures)	1	Understand the process involved about the design of towers and other vertical structures.	✓	✓	
	2	Demonstrate how to design towers structure such as transmission towers.	✓	✓	
(TREN 44033) Professional Course – Specialized 3 (Railroad Engineering)	1	Discuss the development of early railroads and railways in the Philippines from the early part of 1900 to the present era.	✓	✓	
	2	Recognize the importance of planning stage of a proposed railroad construction project and be familiar with the items confronting to railway design standard.	✓	✓	

Course	Course Intended Learning Outcome (CILO)		Three to five years after graduation, the Bachelor of Science in Civil Engineering alumni shall:		
(Course Code and Title)	#	CILO	PEO 1. Demonstrating Engineering Profession Employed and/or practiced in the field of Civil Engineering, such as but not limited to Construction Engineering and Management, Geotechnical Engineering, Structural Engineering, Transportation Engineering, and Water Resources Engineering, or other related fields.	PEO 2. Social and Ethical Responsibility Committed members of professional and other allied organizations engaging in community development or nation-building.	PEO 3. Lifelong Learning Involved in continuous training and development in current trends and advancements in their fields of specialization.
	3	Apply the methods of analysis and design relevant to railroad engineering.	✓	✓	
	4	Define the factors of affect the performance of railroads and related structures.	✓	✓	
(WREN 45033) Professional Course – Specialized 3 (Irrigation Engineering)	1	Discuss soil-water elation, enumerate and discuss methods of measuring soil moisture.	✓	✓	
	2	Enumerate and discuss the different types of irrigation systems.	✓	✓	
	3	Perform calculation related to movement of water into the soil.	✓	✓	
	4	Perform calculation to determine the size of irrigation canals/pipes.	✓	✓	
	5	Estimate consumptive use of water using evaporation and evapotranspiration formulas.	✓	✓	
(CMEN 41043) Professional Course – Specialized 4 (Database Management and Construction)	1	Identify project information systems.	✓	✓	✓
	2	Recognize structures and functions of project components.	✓	✓	✓
	3	Write and present reports in the proper format and in correct English using word processor, spreadsheets.	✓	✓	✓
	4	Develop teamwork and cooperative learning through group exercises.	✓	✓	✓
	5	Design and propose a new database management system in an existing information system in the construction industry.	✓	✓	✓

Course	Course Intended Learning Outcome (CILO)		Three to five years after graduation, the Bachelor of Science in Civil Engineering alumni shall:		
(Course Code and Title)	#	CILO	PEO 1. Demonstrating Engineering Profession Employed and/or practiced in the field of Civil Engineering, such as but not limited to Construction Engineering and Management, Geotechnical Engineering, Structural Engineering, Transportation Engineering, and Water Resources Engineering, or other related fields.	PEO 2. Social and Ethical Responsibility Committed members of professional and other allied organizations engaging in community development or nation-building.	PEO 3. Lifelong Learning Involved in continuous training and development in current trends and advancements in their fields of specialization.
(GTEN 42043) Professional Course – Specialized 4 (Rock Mechanics)	1	Identify the objectives of geotechnical data collection and rock mass classification methods, and successfully collect and analyze a range of geotechnical datasets for design purposes.	✓	✓	
	2	Describe the theory and analysis of in situ and induced stresses in a rock mass and structurally controlled failure.	✓	✓	
	3	Apply the principles of rock mechanics and excavation design to develop excavation proposals for a given geologic environment (e.g. stratified massive, blocky, or faulted lithology).	✓	✓	
	4	Explain the principles and techniques of reinforcement design for the primary failure modes in underground rock excavations, and apply these principles and techniques to inform design for tunnels and large excavations.	✓	✓	
	5	Describe the theory, analysis, and control of rock and soil slope stability and rock fall hazards.	✓	✓	
	6	Calculate the Factor of Safety of rock slopes and underground excavations.	✓	✓	
	7	Describe the mechanical implications of various blast types and patterns for a rock mass, and characterize blasting control	✓	✓	

Course	Course Intended Learning Outcome (CILO)		Three to five years after graduation, the Bachelor of Science in Civil Engineering alumni shall:		
(Course Code and Title)	#	CILO	PEO 1. Demonstrating Engineering Profession Employed and/or practiced in the field of Civil Engineering, such as but not limited to Construction Engineering and Management, Geotechnical Engineering, Structural Engineering, Transportation Engineering, and Water Resources Engineering, or other related fields.	PEO 2. Social and Ethical Responsibility Committed members of professional and other allied organizations engaging in community development or nation-building.	PEO 3. Lifelong Learning Involved in continuous training and development in current trends and advancements in their fields of specialization.
		systems for surface and underground excavations.			
(STEN 43043) Professional Course – Specialized 4 (Bridge Engineering)	1	To understand different types of bridges and its importance to the transportation sector.	✓	✓	
	2	To select appropriate type of bridge considering different factors such as span length, volume of traffic and design requirements that affecting the design	✓	✓	
	3	To understand the principles in designing concrete and steel bridges.	✓	✓	
	4	To design bridge pier and its foundation .	✓	✓	
(TREN 44043) Professional Course – Specialized 4 (Airport Design)	1	State the history of aviation and requirements of airport design.	✓	✓	
	2	Explain the runway length, airspace requirements, navigation and air traffic control.	✓	✓	
	3	Illustrate airfield capacity and delay runway and taxiway Geometrics, Passenger Terminal Capacity.	✓	✓	
	4	Explain the Air Traffic Flow Management, Airport Management and economics and airport noise.	✓	✓	
(WREN 45043) Professional Course – Specialized 4 (Water Supply	1	Identify different water sources and discuss the different treatments.	✓	✓	
	2	Discuss the environmental impacts of developing the different water sources.	✓	✓	
	3	Estimate water demands for different users.	✓	✓	

Course	Course Intended Learning Outcome (CILO)		Three to five years after graduation, the Bachelor of Science in Civil Engineering alumni shall:		
(Course Code and Title)	#	CILO	PEO 1. Demonstrating Engineering Profession Employed and/or practiced in the field of Civil Engineering, such as but not limited to Construction Engineering and Management, Geotechnical Engineering, Structural Engineering, Transportation Engineering, and Water Resources Engineering, or other related fields.	PEO 2. Social and Ethical Responsibility Committed members of professional and other allied organizations engaging in community development or nation-building.	PEO 3. Lifelong Learning Involved in continuous training and development in current trends and advancements in their fields of specialization.
Planning and Development)	4	Provide a typical layout of the water supply system.	✓	✓	
(CMEN 41053) Professional Course – Specialized 5 (Construction Occupational Safety and Health)	1	Discuss and explain the importance and value of safety.	✓	✓	✓
	2	Describe the different health and hazards and their prevention.	✓	✓	✓
	3	Apply the concepts and principles of safety in engineering practice	✓	✓	✓
	4	Explain “Safety as the engineer’s professional and moral responsibility.”	✓	✓	✓
(GTEN 42053) Professional Course – Specialized 5 (Geotechnical and Geoenvironmental Engineering)	1	Evaluate the fundamental geotechnical concepts and implications of environmental engineering.	✓	✓	
	2	Evaluate the geotechnical and geoenvironmental aspects of ground investigation and compaction/earthwork design.	✓	✓	
	3	Evaluate the geotechnical and geoenvironmental aspects of groundwater, soil permeability, consolidation and settlement.	✓	✓	
	4	Evaluate the geotechnical and geoenvironmental aspects of slope stability and retaining structures.	✓	✓	
	5	Evaluate the geotechnical and geoenvironmental aspects of landfills and waste disposal.	✓	✓	

Course	Course Intended Learning Outcome (CILO)		Three to five years after graduation, the Bachelor of Science in Civil Engineering alumni shall:		
(Course Code and Title)	#	CILO	PEO 1. Demonstrating Engineering Profession Employed and/or practiced in the field of Civil Engineering, such as but not limited to Construction Engineering and Management, Geotechnical Engineering, Structural Engineering, Transportation Engineering, and Water Resources Engineering, or other related fields.	PEO 2. Social and Ethical Responsibility Committed members of professional and other allied organizations engaging in community development or nation-building.	PEO 3. Lifelong Learning Involved in continuous training and development in current trends and advancements in their fields of specialization.
	6	Create a noise and ground vibration proposal plan considering the geotechnical and geoenvironmental conditions of a site.	✓	✓	
(STEN 43053) Professional Course – Specialized 5 (Dam Engineering)	1	Create a gravity dam structural design using the appropriate calculations.	✓	✓	
	2	Evaluate the structural integrity of a gravity dam.	✓	✓	
	3	Create an arch dam structural design using the appropriate calculations	✓	✓	
	4	Evaluate the structural integrity of an arch dam.	✓	✓	
	5	Evaluate the structural integrity of a dam using structural dynamics.	✓	✓	
(TREN 44053) Professional Course – Specialized 5 (Ports and Harbors)	1	Show familiarity in the of concept and importance port operation in the economic development of a nation.	✓	✓	
	2	Knowledgeably plan ports and harbors.	✓	✓	
	3	Knowledgeably discuss the developments and operation of ports and harbors.	✓	✓	
	4	Investigate and analyze areas of improvement in ports and harbors.	✓	✓	
(WREN 45053) Professional Course – Specialized 5 (River Engineering)	1	Discuss the contribution of rivers in flood control, water supply, environmental conservation and economic activities of the area.	✓	✓	
	2	Perform calculation related to measurements, movement, and storages in	✓	✓	

Course	Course Intended Learning Outcome (CILO)		Three to five years after graduation, the Bachelor of Science in Civil Engineering alumni shall:		
(Course Code and Title)	#	CILO	PEO 1. Demonstrating Engineering Profession Employed and/or practiced in the field of Civil Engineering, such as but not limited to Construction Engineering and Management, Geotechnical Engineering, Structural Engineering, Transportation Engineering, and Water Resources Engineering, or other related fields.	PEO 2. Social and Ethical Responsibility Committed members of professional and other allied organizations engaging in community development or nation-building.	PEO 3. Lifelong Learning Involved in continuous training and development in current trends and advancements in their fields of specialization.
		the different processes of the hydrologic cycle.			
G. On-The-Job Training					
(CIEN 40012) On-the-Job Training 1	1	To bridge the gap between school and the world of work.	✓	✓	✓
	2	To incorporate the engineering principles to applied civil engineering	✓	✓	✓
	3	To address the issues and problems confronting the issues and problems confronting the industry manpower particularly the construction-related professionals	✓	✓	✓
(CIEN 40022) On-the-Job Training 2	1	To bridge the gap between school and the world of work.	✓	✓	✓
	2	To incorporate the engineering principles to applied civil engineering	✓	✓	✓
	3	To address the issues and problems confronting the issues and problems confronting the industry manpower particularly the construction-related professionals	✓	✓	✓

A.3 Alignment of the Curriculum and Its Associated Prerequisite Structure Support the Attainment of the Student Outcomes

A.3.1 Mapping of the Curriculum to the Student Outcomes

The **Student Outcomes (SOs)** represent the knowledge, skills, abilities, and professional attributes that BS Civil Engineering students are expected to demonstrate by the time of graduation. These outcomes serve as program-level targets that guide curriculum implementation, instructional planning, assessment, and Continuous Quality Improvement (CQI). As shown in the curriculum mapping tables, each course in the BSCE curriculum is identified together with its corresponding **Course Intended Learning Outcomes (CILOs)**, which describe the specific competencies students are expected to attain upon successful completion of the course. Each CILO is then systematically mapped to the applicable SOs to ensure that course-level learning contributes directly to the broader program outcomes expected of graduating students.

The CILO–SO mapping demonstrates the logical relationship between individual course outcomes and the overall competencies required of BS Civil Engineering graduates. Through this alignment, the Department ensures that the learning objectives, teaching-learning activities, and assessment tasks in each course are not implemented in isolation but are intentionally connected to the development of student capabilities across the curriculum. This mechanism also allows the Department to verify whether the required SOs are adequately introduced, reinforced, and demonstrated through the prescribed sequence of courses.

In the mapping tables, each CILO and SO relationship is classified according to the level of attainment: **Introductory (I)**, **Enabling (E)**, and **Demonstrative (D)**. The **Introductory** level indicates that students are first exposed to the concepts, principles, and foundational competencies related to a particular SO. The **Enabling** level indicates that students are provided with opportunities to apply, reinforce, and develop these competencies through more advanced learning activities, problem-solving tasks, laboratory work, design exercises, and course outputs. The **Demonstrative** level indicates that students are expected to integrate and independently demonstrate the required competencies through higher-level professional courses, research, internship, major design, and capstone experiences.

This classification supports the formulation of appropriate **Teaching-Learning Activities (TLAs)** and **Assessment Tasks (ATs)** for each course. It also provides a clearer basis for determining which courses serve as primary sources of evidence for SO attainment,

particularly those identified at the Demonstrative level. Through this process, faculty members are guided in designing instructional strategies and assessment tools that are appropriate to the expected level of student performance. The mapping therefore strengthens the consistency of outcomes-based education implementation across the BSCE curriculum.

The curriculum mapping also provides evidence that the curriculum content is distributed across the curricular areas required by the Philippine Technological Council, including mathematics and basic sciences, engineering sciences, engineering design and synthesis, complementary studies, laboratory and field work, and practical training. The overall structure of the curriculum ensures that students progress from foundational knowledge to advanced application and integration, enabling them to demonstrate the SOs by graduation through appropriately sequenced and outcomes-aligned learning experiences.

The following pages present portions of **Table 5-3.1**, showing the SO alignment for selected professional courses in the 2018 BSCE curriculum. The complete alignment mapping for all technical and non-technical courses of the 2018 and 2022 BSCE curricula is provided in the attachments for reference and verification.

[\(Attachment C5-11 – Table 5-3.1 Mapping of Course Intended Learning Outcomes \(CILO\) and Student Outcomes \(SO\) - 2018 BSCE Curriculum\)](#)

[\(Attachment C5-12 – Table 5-3.2 Mapping of Course Intended Learning Outcomes \(CILO\) and Student Outcomes \(SO\) - 2022 BSCE Curriculum\)](#)

Portion of Table 5-3.1 Mapping of Course Intended Learning Outcomes (CILO) and Student Outcomes (SO) - 2018 BSCE Curriculum

Course	Pre-requisite	Units		Course Intended Learning Outcome (CILO)		Student Outcomes											
<i>(Course Code and Title)</i>	<i>(Course Code and Title)</i>	<i>Lec</i>	<i>Lab</i>	#	CILO	SO1	SO2	SO3	SO4	SO5	SO6	SO7	SO8	SO9	SO10	SO11	SO12
NON-TECHNICAL COURSES																	
(GEED 10023) Understanding the Self	None	3	0	1	Discuss the different representations and conceptualizations of the self from various disciplinary perspectives.												
				2	Demonstrate critical, reflective thought in integrating the various aspects of self												
				3	Apply these new skills to oneself and functioning for a better quality of life												
(GEED 10033) Readings in Philippine History	None	3	0	1	Distinguish different interpretations of and approaches to globalization												
				2	Describe the emergence of global economic, political, social, and cultural systems												
				3	Analyze the various contemporary drivers of globalization												
(GEED 10043) The Contemporary World	None	3	0	1	Assess the effects of globalization on different social units and their responses.												
				2	Distinguish different interpretations of and approaches to globalization.												
				3	Analyze the various contemporary drivers of globalization.												
(GEED 10053) Mathematics in the Modern World	None	3	0	1	Develop innovative methods and remedies that are guided by ethical standards.												
				2	Utilize computational tools to efficiently process information												
				3	Create solutions to problems in various fields												
(GEED 10063)	None	3	0	1	Describe the models, nature, elements, and functions of verbal and non-verbal												
				2	Determine culturally appropriate terms, expressions, and images												

Course	Pre-requisite	Units		Course Intended Learning Outcome (CILO)		Student Outcomes											
(Course Code and Title)	(Course Code and Title)	Lec	Lab	#	CILO	SO1	SO2	SO3	SO4	SO5	SO6	SO7	SO8	SO9	SO10	SO11	SO12
Purposive Communication				3	Create clear, coherent, and effective business correspondence and letters						I		I				
(GEED 10073) Art Appreciation	None	3	0	1	Demonstrate an understanding and appreciation of arts in general, including their function, value, and historical significance.				I				I				
				2	Discover their identity through art with respect to their nationality, culture, and religion				I				I				
				3	Create their own works of art and curate their own production or exhibit.				I				I				
(GEED 10083) Science, Technology & Society	None	3	0	1	Explain how science and technology affect society and the environment and their role in nation-building.									I	I		
				2	Analyze the human condition in order to deeply reflect and express philosophical ramifications that are meaningful to the students as part of society.									I	I		
				3	Critique human flourishing vis-à-vis the progress of science and technology such that the student may be able to define for himself/herself the meaning of good life.									I	I		
(GEED 10093) Ethics	None	3	0	1	Differentiate between moral and non-moral problems								I		I		
				2	Explain the influence of Filipino culture on the way students look at moral experiences and solve moral dilemmas								I				
				3	Internalize the ethical norms of contemporary society at the individual, societal, and in interactions with the environment and other shared resources levels.								I		I		
(GEED 20023) Politics, Governance and Citizenship	None	3	0	1	Discuss the overview of the nature of Philippine institutions and Philippine politics.				I				I		I		

Course	Pre-requisite	Units		Course Intended Learning Outcome (CILO)		Student Outcomes											
(Course Code and Title)	(Course Code and Title)	Lec	Lab	#	CILO	SO1	SO2	SO3	SO4	SO5	SO6	SO7	SO8	SO9	SO10	SO11	SO12
				2	Demonstrate the various political actors in the Philippine society and how they shape key political institutions.				I				I		I		
				3	Understanding of citizenship and territoriality beyond legal definitions				I				I		I		
(GEED 20093) Reading Visual Arts	None	3	0	1	Determine the different ways of reading visual arts according to experts and from the perspective of an Audience.				I				I		I		
				2	Interpret artworks based on the elements and principles of art organization.				I				I		I		
				3	Appreciate varied cultures, history, people's lives, and the experiences of others through arts				I				I		I		
(GEED 10103) Filipinolohiya at Pambansang Kaunlaran	None	3	0	1	Maipaliwanag nang may husay ang kahulugan at kalikasan ng Filipinolohiya na nakaugnay sa Wikang Filipino at nasyunalismo, siyensya at mamamayan							I					
				2	Makapaglahad ng kalagayan at halaga ng Filipinolohiya sa pamamagitan ng paglalatag ng sitwasyong pangkultura, pangwika at panlipunan sa bansa at ibang bansa.							I					
				3	Makapag-analisa ng kalagayan ng Pambansang Industriya sa Pilipinas at sa ibang Bansa.							I					
(GEED 10113) Pagsasalin sa Konseptong Filipino	None	3	0	1	Mailahad ang kronolohikal na kasaysayan ng pagsasalin sa bansa at daigdig							I					
				2	Maipaliwanag ang uri, kahulugan at katangian ng pagsasaling teknikal at pampanitikan.							I					
				3	Makapag-analisa ng kalagayan at halaga ng pagsasalin sa industriya ng bansa na nakasalig sa wikang katutubo							I					

Course	Pre-requisite	Units		Course Intended Learning Outcome (CILO)		Student Outcomes												
(Course Code and Title)	(Course Code and Title)	Lec	Lab	#	CILO	SO1	SO2	SO3	SO4	SO5	SO6	SO7	SO8	SO9	SO10	SO11	SO12	
(GEED 10133) Panitikang Pilipino	None	3	0	1	Makapagtamo ng kaalaman sa mga umiiral na panitikan sa Pilipinas ayon sa historikal na pag-unlad nito;							I						
				2	Nakapagtatalakay ng mga akdang Pampanitikan na sumasalamain sa iba't ibang aspekto ng pamumuhay, kultura, lipunan, pamahalaan at kasaysayan ng sambayanang Pilipino.							I						
				3	Nakapagpapahayag ng may kabisaan ukol sa pasalita at pasulat na tradisyon ng panitikang Pilipino.								I					
(GEED 10013) Life and Works of Rizal	None	3	0	1	Learn and understand the life of Dr. Jose P. Rizal from his birth to his martyrdom.								I		I			
				2	Critically understand Rizal's heroism and realize the significance of his ideas in the present time									I		I		
				3	Interpret the selected poem, essays and novels and its impact to socio-political situation during the past and present time.									I		I		
E. Professional Courses - Common																		
(CIEN 30013) Fundamentals of Surveying 1 LEC	(ENSC 20112) Engineering Drawing and Plans	2	0	1	Comprehend and compute the length and direction of a line, the elevation and/or difference in elevation between two points, and an area of a lot.				I	I		I				I		
				2	Prepare and explain how each method is applied in surveying practice.				I	I		I				I		
				3	Perform the computation and analyze the significance of each.				I	I		I				I		
(CIEN 30013) Fundamentals of Surveying 1 - Field	(ENSC 20112) Engineering Drawing and Plans	0	1	1	Demonstrate knowledge on the technical know-how of the basics of surveying like taping, determining elevation and/or difference in elevation between two points, setting up and manipulation of the Engineer's Transit, angle by repetition and traversing.				I	I		I				I		

Course	Pre-requisite	Units		Course Intended Learning Outcome (CILO)		Student Outcomes											
(Course Code and Title)	(Course Code and Title)	Lec	Lab	#	CILO	SO1	SO2	SO3	SO4	SO5	SO6	SO7	SO8	SO9	SO10	SO11	SO12
				2	Begin to show willingness to perform the assigned task/s.				I	I		I				I	
				3	Develop confidence in performing each field.				I	I		I				I	
(CIEN 30033) Fundamentals of Surveying 2 LEC	(CIEN 30013) Fundamentals of Surveying 1	2	0	1	Comprehend and compute for the omitted information of a closed traverse, partitioning of a lot, principle of the stadia, contouring, triangulation, sextant, hydrographic surveying and astronomical observation.				I	I		I				I	
				2	Prepare and explain how each method is applied in surveying practice.				I	I		I				I	
				3	Perform the computation and discuss the information of a closed traverse, partitioning of a lot, principle of the stadia, contouring, triangulation, sextant, hydrographic surveying and astronomical observation.				I	I		I				I	
(CIEN 30033) Fundamentals of Surveying 2 - Field	(CIEN 30013) Fundamentals of Surveying 1	0	1	1	Comprehend and compute for the omitted information of a closed traverse, partitioning of a lot, principle of the stadia, contouring, triangulation, sextant, hydrographic surveying and astronomical observation.				I	I		I				I	
				2	Prepare and explain how each method is applied in surveying practice.				I	I		I				I	
				3	Perform the computation and discuss the significance of each.				I	I		I				I	
(CIEN 30053) Route Surveying and Earthworks LEC	(CIEN 30033) Fundamentals of Surveying 2	2	0	1	Comprehend and compute problems on topics like simple curve, compound curves, reversed curves, symmetrical and unsymmetrical parabolic curve, passing sight distance, spiral curve, volumes of earthworks, and haul and mass diagram				E	I		E				E	
				2	Prepare and explain how each method is applied in surveying practice				E	I		E				E	

Course	Pre-requisite	Units		Course Intended Learning Outcome (CILO)		Student Outcomes											
(Course Code and Title)	(Course Code and Title)	Lec	Lab	#	CILO	SO1	SO2	SO3	SO4	SO5	SO6	SO7	SO8	SO9	SO10	SO11	SO12
				3	Perform the necessary calculations and discuss their significance in surveying practice				E	I		E				E	
(CIEN 30053) Route Surveying and Earthworks LAB	(CIEN 30033) Fundamentals of Surveying 2	0	1	1	Demonstrate knowledge on the technical know-how of engineering surveys like computations and/or laying out of simple curves, compound curves, reversed curves, symmetrical and unsymmetrical parabolic curve, passing sight distance, spiral curve and volumes of earth.				E	I		E				E	
				2	Exemplify initiative and leadership in performing surveying works				E	I		E				E	
				3	Exemplify sound knowledge in performing each fieldwork				E	I		E				E	
(CIEN 30023) Construction Materials and Testing LEC	(ENSC 016) Statics of Rigid Bodies	2	0	1	Explain relevant properties of common construction materials.	I			I			E					
				2	Assess the qualities / properties of different construction materials	I			I			E					
				3	Apply appropriate procedures on testing based on local or international standards	I			I			E					
(CIEN 30023) Construction Materials and Testing LAB	(ENSC 016) Statics of Rigid Bodies	0	1	1	List the commonly used construction materials and their uses	I			I			E			I	I	
				2	Compute the properties of materials based on the data taken from laboratory experiment	I			I			E			I	I	
				3	Assess the qualities / properties of different construction materials	I			I			E			I	I	
				4	Demonstrate appropriate procedures on testing based on local or international standards	I			I			E			I	I	
				5	Apply the safety procedures in the laboratory	I			I			E			I	I	
				6	Formulate concrete and asphalt mixtures based on actual condition of materials	I			I			E			I	I	

[illegible]

Course	Pre-requisite	Units		Course Intended Learning Outcome (CILO)		Student Outcomes											
(Course Code and Title)	(Course Code and Title)	Lec	Lab	#	CILO	SO1	SO2	SO3	SO4	SO5	SO6	SO7	SO8	SO9	SO10	SO11	SO12
				2	Extend the study of linear elastic analysis to include nonlinear aspects of structure behavior	E	E	E								E	E
				3	Interpret computer output and validate results using simplified models and hand calculations	E	E	E								E	E
(CIEN 30113) Highway and Railroad Engineering	(CIEN 30053) Route Surveying and Earthworks	3	0	1	Explain the principles underlying highway design	E		E			E		E				
				2	Assess alternative pavement designs and understand their deterioration and maintenance schemes for highways.	E		E			E		E				
				3	Design a highway allowing for different terrains, horizontal and vertical curves.	E		E			E		E				
				4	Develop basic traffic stream parameters and models, traffic flow models, and apply the queuing theory	E		E			E		E				
(CIEN 30073) Building Systems Design LEC	(ENSC 20112) Engineering Drawing and Plans	2	0	1	Identify appropriate provisions of the National Building Code and its referral codes as these apply to buildings.			E	E	E	I	I	I		I	E	
				2	Ability to design sites, facilities, and system principles that are responsive to relevant codes and regulations, and include the principle of life safety and accessibility standards.			E	E	E	I	I	I		I	E	
				3	Ability to make technically clear drawings, prepare outline specifications, and construct models illustrating and identifying the assembly of materials, systems, and components appropriate for a building design.			E	E	E	I	I	I		I	E	
				4	Understanding of the basic principles involved in the appropriate selection and application of building envelope systems relative to fundamental performance, aesthetics, moisture transfer, durability, and energy and material resources.			E	E	E	I	I	I		I	E	

Course	Pre-requisite	Units		Course Intended Learning Outcome (CILO)		Student Outcomes											
(Course Code and Title)	(Course Code and Title)	Lec	Lab	#	CILO	SO1	SO2	SO3	SO4	SO5	SO6	SO7	SO8	SO9	SO10	SO11	SO12
				5	Understanding of the basic principles used in the appropriate selection of interior and exterior construction, finishes, products, components, and assemblies based on their inherent performance, including environmental impact and reuse.			E	E	E	I	I	I		I	E	
				6	Understanding of the basic principles and appropriate application and performance of building service systems, including lighting, mechanical, plumbing, electrical, communication, vertical transportation, security, and fire protection systems			E	E	E	I	I	I		I	E	
				7	Understanding of the fundamentals of building costs, which must include life cycle costs.			E	E	E	I	I	I		I	E	
(CIEN 30073) Building Systems Design LAB	(ENSC 20112) Engineering Drawing and Plans	0	1	1	Identify appropriate provisions of the National Building Code and its referral codes as these apply to buildings.			E	E	E	I	I	I		I	E	
				2	Ability to design sites, facilities, and system principles that are responsive to relevant codes and regulations and include the principle of life safety and accessibility standards.			E	E	E	I	I	I		I	E	
				3	Ability to make technically clear drawings, prepare outline specifications, and construct models illustrating and identifying the assembly of materials, systems, and components appropriate for a building design.			E	E	E	I	I	I		I	E	
				4	Understanding of the basic principles involved in the appropriate selection and application of building envelope systems relative to fundamental performance, aesthetics, moisture transfer, durability, and energy and material resources.			E	E	E	I	I	I		I	E	

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Course	Pre-requisite	Units		Course Intended Learning Outcome (CILO)		Student Outcomes											
(Course Code and Title)	(Course Code and Title)	Lec	Lab	#	CILO	SO1	SO2	SO3	SO4	SO5	SO6	SO7	SO8	SO9	SO10	SO11	SO12
(CIEN 30183) Principles of Reinforced Concrete LEC	(CIEN 30043) Structural Theory 1	2	0	1	Understand the in-depth stress-strain characteristics of concrete for all stages of loading.	D		D		D		E				E	
				2	Evaluate the strength of structural plain, reinforced, or prestressed concrete elements in axial loading, bending, shear, and torsion under pseudo-elastic and inelastic stages of loading.	D		D		D		E				E	
				3	Apply allowable strength and load-and-resistance factor design methods for reinforced and prestressed concrete.	D		D		D		E				E	
(CIEN 30183) Principles of Reinforced Concrete Computational	(CIEN 30043) Structural Theory 1	0	1	1	Demonstrate by numerical calculations and other methods the concepts, principles, and design of plain theories obtained from the lecture portion of the course.	D		D		D		E				E	
(CIEN 30193) Reinforced Concrete Design LEC	(CIEN 30183) Principles of Reinforced Concrete	2	0	1	Understand the process involved in the design of reinforced concrete frame structures	D		D		D		D				D	
	(CIEN 30103) Structural Theory 2			2	Demonstrate how to design a reinforced concrete structure such as building	D		D		D		D				D	
(CIEN 30193) Reinforced Concrete Design Computational	(CIEN 30183) Principles of Reinforced Concrete	0	1	1	Solve theoretical and situational problems in the analysis and design of reinforced concrete beams.	D		D	D	D							
	(CIEN 30103) Structural Theory 2			2	Solve theoretical and situational problems in the analysis and design of reinforced concrete columns.	D		D	D	D							
(CIEN 30222) Timber Design LEC		1	0	1	Know the basic concepts and design techniques of structural timber design	E				E							

Course	Pre-requisite	Units		Course Intended Learning Outcome (CILO)		Student Outcomes											
(Course Code and Title)	(Course Code and Title)	Lec	Lab	#	CILO	SO1	SO2	SO3	SO4	SO5	SO6	SO7	SO8	SO9	SO10	SO11	SO12
	(CIEN 30103) Structural Theory 2			2	Solve theoretical and situational problems in the analysis and design of timber beams, columns, and other structures	D		D		D			D				
				3	Create structural plans, detailed drawings, and calculations for at least two storey timber structure	D		D		D						D	
(CIEN 30222) Timber Design Computational	(CIEN 30103) Structural Theory 2	0	1	1	Solve theoretical and situational problems in the analysis and design of timber beams, columns, and other structures.	E				E							
				2	Create structural plans, detailed drawings, and calculations for at least two-storey timber structure.	D		D		D			D				
				3	Evaluate laboratory and field data to select appropriate parameters to use in timber designs.	D		D		D						D	
(CIEN 30063) Hydraulics 1 (LEC)	(ENSC 20084) Mechanics of Deformable Bodies (ENSC 20072) Dynamics of Rigid Bodies	2	0	1	Understand basic properties of fluids and its behavior.	E	E		E	E		E					
				2	Compute hydrostatic pressure on different type of surface.	E	E		E	E		E					
				3	Understand relative equilibrium of liquids.	E	E		E	E		E					
(CIEN 30063) Hydraulics 1 (LAB)	(ENSC 20084) Mechanics of Deformable Bodies (ENSC 20072) Dynamics of Rigid Bodies	0	1	1	Understand basic properties of fluids and its behavior including the measurement of Pressure	E	E		E	E		E					
				2	Compute hydrostatic pressure on different type of surface	E	E		E	E		E					
				3	Understand buoyancy and stability of floating bodies	E	E		E	E		E					

Course	Pre-requisite	Units		Course Intended Learning Outcome (CILO)		Student Outcomes											
(Course Code and Title)	(Course Code and Title)	Lec	Lab	#	CILO	SO1	SO2	SO3	SO4	SO5	SO6	SO7	SO8	SO9	SO10	SO11	SO12
(CIEN 30143) Hydraulics 2 LEC	(CIEN 30063) Hydraulics 1	2	0	1	To apply the principle of continuity in closed conduits in determining the discharge	D		E		D						D	
				2	To apply the Bernoulli's principle in calculating the flow energy in closed conduits and open channels	D		E		D						D	
				3	To apply momentum equations in solving for forces on fixed and moving objects	D		E		D						D	
(CIEN 30143) Hydraulics 2 LAB	(CIEN 30063) Hydraulics 1	0	1	1	To apply the principle of continuity in closed conduits in determining the discharge	E	D		E	E							
				2	To apply the Bernoulli's principle in calculating the flow energy in closed conduits and open channels	E	D		E	E							
				3	To apply momentum equations in solving for forces on fixed and moving objects	E	D		E	E							
(CIEN 30132) Hydrology	(CIEN 30063) Hydraulics 1	2	0	1	Discuss with appropriate diagrams the hydrologic cycle and the different processes and storages within the cycle	E				E		E				E	
				2	Perform calculation related to measurements, movement, and storages in the different processes of the hydrologic cycle.	E				E		E				E	
(CIEN 30152) CE Law, Ethics and Contracts LEC		2	0	1	Explain the Civil Engineering Code of Ethics which includes the fundamental principles and canon.						D	E			D		
				2	Explain the laws governing the practice of civil engineering profession in the Philippines.						D	E			D		
				3	Evaluate the results of the test of common construction materials						D	E			D		
				4	Appraise the civil engineering fundamental principles and canons in oral presentations						D	E			D		

Course	Pre-requisite	Units		Course Intended Learning Outcome (CILO)		Student Outcomes											
(Course Code and Title)	(Course Code and Title)	Lec	Lab	#	CILO	SO1	SO2	SO3	SO4	SO5	SO6	SO7	SO8	SO9	SO10	SO11	SO12
(CIEN 30083) Geotechnical Engineering 1 LEC	(ENSC 20084) Mechanics of Deformable Bodies	2	0	1	To explain the concept of soil as an engineering material and the properties and methods used to characterize soil for geotechnical analysis and design.	E	E		E	E		E			E	E	
				2	To interpret terminology and parameters used to characterize and classify soil; stresses and stress conditions in soils; factors affecting soil strength and stress-strain behavior.	E	E		E	E		E			E	E	
				3	To classify the seepage and water flow through soils and their effects on soil stresses and strength; deformation and settlement characteristics of soils; lateral earth pressure, bearing capacity and slope stability concepts.	E	E		E	E		E			E	E	
(CIEN 30083) Geotechnical Engineering 1 FIELD	(ENSC 20084) Mechanics of Deformable Bodies	0	1	1	To utilize the concept of soil as an engineering material and the properties and methods used to characterize soil for geotechnical analysis and design.	E	E		E	E		E			E	E	
				2	To model the parameters used to characterize and classify soils; stresses and stress conditions in soils; factors affecting soil strength and stress-strain behavior.	E	E		E	E		E			E	E	
				3	To experiment with the seepage and water flow through soils and their effects on soil stresses and strength; deformation and settlement characteristics of soils.	E	E		E	E		E			E	E	
(CIEN 30122) Geotechnical Engineering 2 LEC	(CIEN 30083) Geotechnical Engineering 1	2	0	1	Identify the objectives of geotechnical data collection and rock mass classification methods, and successfully collect and analyze a range of geotechnical datasets for design purposes	E		E		E		E	E			E	
				2	describe the theory and analysis of in situ and induced stresses in a rock mass and structurally controlled failure	E		E		E		E	E			E	

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(Course Code and Title)	(Course Code and Title)	Lec	Lab	#	CILO	SO1	SO2	SO3	SO4	SO5	SO6	SO7	SO8	SO9	SO10	SO11	SO12
				3	apply the principles of rock mechanics and excavation design to develop excavation proposals for given geologic environments (e.g stratified, massive, blocky, or faulted lithologies)	E		E		E		E	E			E	
				4	explain the principles and techniques of reinforcement design for the primary failure modes in underground rock excavations, and apply these principles and techniques to inform design for tunnels and large excavations	E		E		E		E	E			E	
				5	describe the theory, analysis, and control of rock and soil slope stability and rockfall hazards	E		E		E		E	E			E	
				6	calculate the Factor of Safety of rock slopes and underground excavations	E		E		E		E	E			E	
				7	describe the mechanical implications of various blast types and patterns for a rock mass, and characterize blasting control systems	E		E		E		E	E			E	
(CIEN 30242) Foundation Engineering LEC	(CIEN 30122) Geotechnical Engineering 2 (CIEN 30183) Principles of Reinforced Concrete	1	0	1	Plan a subsurface exploration, and select appropriate drilling, sampling, and field property measurement tools for different soil profiles.	D		D		D		E	E			D	
				2	Create an idealized soil profile for analysis and design from collected soil properties.	D		D		D		E	E			D	
				3	Evaluate laboratory and field data to select appropriate parameters to use in foundation analysis.	D		D		D		E	E			D	
				4	Determine the lateral loads from soils.	D		D		D		E	E			D	
				5	Design earth retaining structures.												
				6	Apply shallow and deep bearing capacity and settlement equation for design with applicable correction factors with a consideration of water effects and layering.	D		D		D		E	E			D	

Course	Pre-requisite	Units		Course Intended Learning Outcome (CILO)		Student Outcomes											
(Course Code and Title)	(Course Code and Title)	Lec	Lab	#	CILO	SO1	SO2	SO3	SO4	SO5	SO6	SO7	SO8	SO9	SO10	SO11	SO12
(CIEN 30242) Foundation Engineering LAB	(CIEN 30122) Geotechnical Engineering 2 (CIEN 30183) Principles of Reinforced Concrete	0	1	1	Plan a subsurface exploration, and select appropriate drilling, sampling, and field property measurement tools for different soil profiles.	D		D		D		E	E			D	
				2	Create an idealized soil profile for analysis and design from collected soil properties.	D		D		D		E	E			D	
				3	Evaluate laboratory and field data to select appropriate parameters to use in foundation analysis.	D		D		D		E	E			D	
				4	Determine the lateral loads from soils.	D		D		D		E	E			D	
				5	Design earth retaining structures.	D		D		D		E	E			D	
				6	Apply shallow and deep bearing capacity and settlement equation for design with applicable correction factors with a consideration of water effects and layering.	D		D		D		E	E			D	
(CIEN 30203) Principles of Transportation Engineering	(CIEN 30113) Highway and Railroad Engineering	3	0	1	Define concepts of transportation systems analysis and planning	E		D			D						
				2	Apply travel demand forecasting computation and analysis	E		D			D						
				3	Prepare traffic impact assessment reports	E		D			D						
				4	Recognize current transportation issues and policies, economic, safety and environmental concerns	E		D			D						
(CIEN 30272) Quantity Surveying LEC	(CIEN 30073) Building Systems Design	1	0	1	Determine quantity of various building materials	E			E		D	E		D		D	E
				2	Develop bid costs for materials, labor and equipment for several construction project elements	E			E		D	E		D		D	E
				3	Evaluate total cost of engineering projects, including labor fringes, taxes, bonding fees, insurance, overhead and profit.	E			E		D	E		D		D	E
	(CIEN 30073) Building	0	1	1	Define quantity of various building materials	E			E		D	E		D		D	E

Course	Pre-requisite	Units		Course Intended Learning Outcome (CILO)		Student Outcomes											
(Course Code and Title)	(Course Code and Title)	Lec	Lab	#	CILO	SO1	SO2	SO3	SO4	SO5	SO6	SO7	SO8	SO9	SO10	SO11	SO12
(CIEN 30272) Quantity Surveying LAB	Systems Design			2	Develop bid costs for materials, labor and equipment for several construction project elements.	E			E		D	E		D		D	E
				3	Evaluate total cost of engineering projects, including labor fringes, taxes, bonding fees, insurance, overhead and profit.	E			E		D	E		D		D	E
(CIEN 30293) Construction Methods and Project Management		3	0	1	Understand the construction project cycle from inception, planning, execution, testing and certification.	E		E		E						D	
				2	Understand construction project requirements such as permits and licenses, clearance, and compliance to all applicable laws and ordinances, etc	E		E		E						D	
				3	Read and interpret construction plans, working drawings, and revise contract documents, estimates and technical specifications	E		E		E						D	
				4	Understand the components of the Terms of Reference (TOR) for professional services and bidding/tendering processes, award and acceptance	E		E		E						D	
				5	Prepare and update construction schedule and work programs,	E		E		E						D	
				6	Understand construction methods, equipment, materials and manpower control:	E		E		E						D	
				7	Understand temporary facilities requirements of a construction project,	E		E		E						D	
				8	Understand the existing local labor laws and regulations	E		E		E						D	
(CIEN 30212) CE Project 1 LEC	(CIEN 30092) Civil Engineering Research	1	0	1	Critically evaluate a range of research methods and techniques and be able to create a convincing written rationale for selecting them.	D	D	D	D	D	D	D	D			D	

Course	Pre-requisite	Units		Course Intended Learning Outcome (CILO)		Student Outcomes											
(Course Code and Title)	(Course Code and Title)	Lec	Lab	#	CILO	SO1	SO2	SO3	SO4	SO5	SO6	SO7	SO8	SO9	SO10	SO11	SO12
				2	Craft research questions pertinent to the fields of engineering and related disciplines, based in an extensive knowledge of the relevant literature.	D	D	D	D	D	D	D	D			D	
				3	Design a research project of appropriate scope for a higher degree by research candidature.	D	D	D	D	D	D	D	D			D	
				4	Write a literature review which supports a research project and states clearly the project's contribution to knowledge	D	D	D	D	D	D	D	D			D	
(CIEN 30212) CE Project 1 LAB	(CIEN 30092) Civil Engineering Research	0	1	1	Critically evaluate a range of research methods and techniques and be able to create a convincing written rationale for selecting them.	D	D	D	D	D	D	D	D			D	D
				2	Craft research questions pertinent to the fields of engineering and related disciplines, based in an extensive knowledge of the relevant literature.	D	D	D	D	D	D	D	D			D	D
				3	Design a research project of appropriate scope for a higher degree by research candidature.	D	D	D	D	D	D	D	D			D	
				4	Write a literature review which supports a research project and states clearly the project's contribution to knowledge	D	D	D	D	D	D	D	D			D	
(CIEN 30262) CE Project 2 LEC	(CIEN 30212) CE Project 1	1	0	1	Critically evaluate a range of research methods and techniques and be able to create a convincing written rationale for selecting them.	D	D	D	D	D	D	D	D			D	
				2	Craft research questions pertinent to the fields of engineering and related disciplines, based in an extensive knowledge of the relevant literature.	D	D	D	D	D	D	D	D			D	
				3	Design a research project of appropriate scope for a higher degree by research candidature.	D	D	D	D	D	D	D	D			D	

Course	Pre-requisite	Units		Course Intended Learning Outcome (CILO)		Student Outcomes											
(Course Code and Title)	(Course Code and Title)	Lec	Lab	#	CILO	SO1	SO2	SO3	SO4	SO5	SO6	SO7	SO8	SO9	SO10	SO11	SO12
				4	Write a literature review which supports a research project and states clearly the project's contribution to knowledge	D	D	D	D	D	D	D	D			D	
(CIEN 30262) CE Project 2 LAB	(CIEN 30212) CE Project 1	0	1	1	Critically evaluate a range of research methods and techniques and be able to create a convincing written rationale for selecting them.	D	D	D	D	D	D	D	D			D	
				2	Craft research questions pertinent to the fields of engineering and related disciplines, based in an extensive knowledge of the relevant literature.	D	D	D	D	D	D	D	D			D	
				3	Design a research project of appropriate scope for a higher degree by research candidature.	D	D	D	D	D	D	D	D			D	
				4	Write a literature review which supports a research project and states clearly the project's contribution to knowledge	D	D	D	D	D	D	D	D			D	
(CIEN 30171) Integrative Course 1	(ENSC 20093) Engineering Economics (MATH 20093) Numerical Solutions to CE Problems	0	1	1	Gain thorough understanding of all early math courses (Algebra, Plane and Spherical Trigonometry, Solid Mensuration, and Analytic Geometry) completed.	D				D							
				2	Gain thorough understanding of all late math courses (Calculus 1 and 2, Differential Equations, Engineering Data Analysis, Numerical Solutions to CE Problems, Engineering Economics) and physics course (Physics for Engineers) completed.	D				D							
				3	Gain thorough understanding of all surveying courses (Fundamentals of Surveying, Route Surveying, Highway Engineering and Principles of Transportation Engineering) completed.	D				D						I	

Course	Pre-requisite	Units		Course Intended Learning Outcome (CILO)		Student Outcomes											
(Course Code and Title)	(Course Code and Title)	Lec	Lab	#	CILO	SO1	SO2	SO3	SO4	SO5	SO6	SO7	SO8	SO9	SO10	SO11	SO12
	(CIEN 30231) CE Integration 2																
(CIEN 30311) CE Field Trip		0	1	1	To assess and evaluate the current situation of the construction in the country.				D		D		D				
				2	To demonstrate the actual works in different civil works.				D		D		D				
				3	To have a knowledge on various discipline in construction.				D		D		D				
				4	To have an idea on the current problems and issues facing the existing civil engineering in the country.				D		D		D				
(CIEN 30092) Civil Engineering Research LEC		1	0	1	Critically evaluate a range of research methods and techniques and be able to create a convincing written rationale for selecting them.	E	E	E	E	E	E	E	E			E	
				2	Craft research questions pertinent to the fields of engineering and related disciplines, based in an extensive knowledge of the relevant literature.	E	E	E	E	E	E	E	E			E	
				3	Design a research project of appropriate scope for a higher degree by candidature. research	E	E	E	E	E	E	E	E			E	
				4	Write a literature review which supports a research project and states clearly the project's contribution to knowledge	E	E	E	E	E	E	E	E			E	
(CIEN 30092) Civil Engineering Research LAB		0	1	1	Critically evaluate a range of research methods and techniques and be able to create a convincing written rationale for selecting them.	E	E	E	E	E	E	E	E			E	
				2	Craft research questions pertinent to the fields of engineering and related disciplines, based in an extensive knowledge of the relevant literature.	E	E	E	E	E	E	E	E			E	

Course	Pre-requisite	Units		Course Intended Learning Outcome (CILO)		Student Outcomes											
(Course Code and Title)	(Course Code and Title)	Lec	Lab	#	CILO	SO1	SO2	SO3	SO4	SO5	SO6	SO7	SO8	SO9	SO10	SO11	SO12
				3	Design a research project of appropriate scope for a higher degree by candidature. research	E	E	E	E	E	E	E	E			E	
				4	Write a literature review which supports a research project and states clearly the project's contribution to knowledge	E	E	E	E	E	E	E	E			E	
(CIEN 30282) Earthquake Engineering LEC	(CIEN 30103) Structural Theory 2	2	0	1	Understand the occurrence and origin of earthquakes.	E		E				E				E	
				2	Understand structural dynamics and appreciate the response of structures to seismic excitation.	E		E				E				E	
				3	Apply the latest developments in earthquake and seismic design relevance to the National Structural Code of the Philippines.	E		E				E				E	
F. Professional Courses - Specialized																	
(CMEN 41013) Professional Course – Specialized 1 (Project Construction Management)		3	0	1	Describe the management process and explain the different functions of management.							E		E	E	D	E
				2	Craft a personal mission and vision statement.							E		E	E	D	E
				3	Perform a SWOT analysis.							E		E	E	D	E
				4	Identify the types of organizational structures.							E		E	E	D	E
				5	Identify management practices applied in civil engineering.							E		E	E	D	E
(GTEN 42013) Professional Course – Specialized 1 (Geosynthetics in Geotechnical Engineering)				1	Evaluate the fundamental concepts of geosynthetics.	E		D		E		E	E			D	
				2	Evaluate the geotechnical applications of geosynthetics.	E		D		E		E	E			D	
				3	Evaluate the hydraulic and geoenvironmental applications of geosynthetics.	E		D		E		E	E			D	
				4	Evaluate the transportation applications of geosynthetics.	E		D		E		E	E			D	

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(Course Code and Title)	(Course Code and Title)	Lec	Lab	#	CILO	SO1	SO2	SO3	SO4	SO5	SO6	SO7	SO8	SO9	SO10	SO11	SO12	
				5	Evaluate the mining, agricultural and aquacultural applications of geosynthetics.	E		D		E		E	E			D		
				6	Create geosynthetic application designs considering the general guidelines and installation survivability requirements.	E		D		E		E	E			D		
				7	Create geosynthetic application designs using fiber-reinforced soils.	E		D		E		E	E			D		
(STEN 43013) Professional Course – Specialized 1 (Computer Software in Structural Analysis)				1	Familiar with computer software commercially available for structural analysis and knowledgeable in their use and application.											D		
(TREN 44103) Professional Course – Specialized 1 (Transportation Systems Planning and Design)				1	Describe the elements of the Transportation, evolution of urban transportation and its policy.	E		E		E		D					E	
				2	Explain the Modelling Transportation Systems, Urban Travel & Land Transportation system characteristics, travel demand with Land use.	E		E		E		D				E		
				3	Make a modeling mode of mode choice and traffic Assignment.	E		E		E		D				E		
				4	Investigate and analyze value pricing, ITS and its policy.	E		E		E		D				E		
(WREN 45301) Professional Course – Specialized 1 (Water Resources Engineering)				1	Estimate peak runoff, determine the appropriate pump rating to deliver water from identified sources to a receiving tank/reservoir, determine the sizes of pipes given flow rate, and determine the appropriate size of open channel given discharge.	D		D							E		E	
				2	Determine the parameters of different probability distributions commonly used in flood frequency analysis.	D		D						E		E		

Course	Pre-requisite	Units		Course Intended Learning Outcome (CILO)		Student Outcomes											
(Course Code and Title)	(Course Code and Title)	Lec	Lab	#	CILO	SO1	SO2	SO3	SO4	SO5	SO6	SO7	SO8	SO9	SO10	SO11	SO12
				3	Design a basic drainage system and/or simple water supply system.	D		D						E		E	
(CMEN 41023) Professional Course – Specialized 2 (Advanced Construction Methods and Equipment)		3	0	1	Draw a network diagram using Activity on Arrow (AOA) and Activity on Node (AON) methods.							E		E	E	D	E
				2	Prepare and update construction estimates, schedule, and material and manpower report.							E		E	E	D	E
				3	Calculate labor and equipment productivity.							E		E	E	D	E
				4	Read and interpret construction plans, working drawings, contract documents and technical specifications in the preparation of a complete bid for a construction project.							E		E	E	D	E
				5	Identify and present a traditional and new method/theory/practice applied in construction project.							E		E	E	D	E
				6	Investigate, report and propose a solution to a safety or quality issue in a construction project.							E		E	E	D	E
(GTEN 42023) Professional Course – Specialized 2 (Geotechnical Earthquake Engineering)		3	0	1	Classify and evaluate earthquake sources/loads and associated geotechnical earthquake hazards.	E		D		E		E	E			D	
				2	Explain key concepts in the seismic assessment of engineering structures.	E		D		E		E	E			D	
				3	Interpret the behavior of soils and geotechnical structures during earthquakes.	E		D		E		E	E			D	
				4	Apply simplified design-oriented methods for liquefaction analysis, and seismic analysis and design of retaining walls, embankments, bridges and building foundations.	E		D		E		E	E			D	

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(Course Code and Title)	(Course Code and Title)	Lec	Lab	#	CILO	SO1	SO2	SO3	SO4	SO5	SO6	SO7	SO8	SO9	SO10	SO11	SO12
				5	Analyze the response of sites, earth structures and soil-structure systems under earthquake loads.	E		D		E		E	E			D	
(STEN 43023) Professional Course – Specialized 2 (Prestressed Concrete Design)				1	Explain the principle of prestressing.	E		E		E		E				D	
				2	Analyze prestressed concrete beams, columns, piles, slabs and other structures.	E		E		E		E				D	
				3	Design Structural plans, shop drawings, and computation of at least two – storey prestressed concrete structure.	E		E		E		E				D	
(TREN 44023) Professional Course – Specialized 2 (Highway Engineering)				1	Discuss the development of early roads and highway in the Philippines from the early part of 1900 to the present era.	E		D		E		E				E	
				2	Recognize the importance of planning stage of a proposed road construction project and be familiar with the items confronting to highway design standard.	E		D		E		E				E	
				3	Apply the methods of analysis and design relevant to highway engineering.	E		D		E		E				E	
				4	Define the factors of affect the performance of highways and related structures.	E		D		E		E				E	
(WREN 45023) Professional Course – Specialized 2 (Flood Control and Drainage Design)				1	Identify and discuss the different components of rainwater harvesting system.	D				E		E				D	
				2	Identify and discuss the different components of storm water drainage system.	D				E		E				D	
				3	Identify and discuss the different components of flood gates and pumping systems.	D				E		E				D	
				4	Perform calculation related to sizing of gutters, inlets, culverts and channels needed to convey storm water as well as sizing of retention and detention ponds.	D				E		E				D	

Course	Pre-requisite	Units		Course Intended Learning Outcome (CILO)		Student Outcomes															
(Course Code and Title)	(Course Code and Title)	Lec	Lab	#	CILO	SO1	SO2	SO3	SO4	SO5	SO6	SO7	SO8	SO9	SO10	SO11	SO12				
(CMEN 41033) Professional Course – Specialized 3 (Construction Cost Engineering)		3	0	1	Apply the project management information system, cost engineering and accounting concepts in construction projects.							D		E	E	D	E				
				2	To analyze and design an improved cost engineering and accounting system for a construction project.							D		E	E	D	E				
(GTEN 42033) Professional Course – Specialized 3 (Slope Stability and Earth Dams)				1	Evaluate the geotechnical aspects of the causes of slope stability failure.	E		D		E		E	E				D				
				2	Evaluate the structural stability of reinforced slopes and embankments in Earth dams.	E		D		E		E	E				D				
				3	Evaluate the seismic safety and reliability of slope stabilization and repair.	E		D		E		E	E				D				
				4	Evaluate the geotechnical considerations on slope stability in designing Earth dams.	E		D		E		E	E				D				
(STEN 43033) Professional Course – Specialized 3 (Structural Design of Towers and other Vertical Structures)				1	Understand the process involved about the design of towers and other vertical structures.	D		E		E		E						D			
				2	Demonstrate how to design towers structure such as transmission towers.	D		E		E		E							D		
(TREN 44033) Professional Course – Specialized 3 (Railroad Engineering)				1	Discuss the development of early railroads and railways in the Philippines from the early part of 1900 to the present era.	E		E		E		E		E					D		
				2	Recognize the importance of planning stage of a proposed railroad construction project and be familiar with the items confronting to railway design standard.	E		E		E		E								D	
				3	Apply the methods of analysis and design relevant to railroad engineering.	E		E		E		E								D	
				4	Define the factors of affect the performance of railroads and related structures.	E		E		E		E								D	

Course	Pre-requisite	Units		Course Intended Learning Outcome (CILO)		Student Outcomes											
(Course Code and Title)	(Course Code and Title)	Lec	Lab	#	CILO	SO1	SO2	SO3	SO4	SO5	SO6	SO7	SO8	SO9	SO10	SO11	SO12
(WREN 45033) Professional Course – Specialized 3 (Irrigation Engineering)				1	Discuss soil-water relation, enumerate and discuss methods of measuring soil moisture.	E				E		E				D	
				2	Enumerate and discuss the different types of irrigation systems.	E				E		E				D	
				3	Perform calculation related to movement of water into the soil.	E				E		E				D	
				4	Perform calculation to determine the size of irrigation canals/pipes.	E				E		E				D	
				5	Estimate consumptive use of water using evaporation and evapotranspiration formulas.	E				E		E				D	
(CMEN 41043) Professional Course – Specialized 4 (Database Management and Construction)		3	0	1	Identify project information systems.							E		E	E	D	E
				2	Recognize structures and functions of project components.							E		E	E	D	E
				3	Write and present reports in the proper format and in correct English using word processor, spreadsheets.							E		E	E	D	E
				4	Develop teamwork and cooperative learning through group exercises.							E		E	E	D	E
				5	Design and propose a new database management system in an existing information system in the construction industry.							E		E	E	D	E
(GTEN 42043) Professional Course – Specialized 4 (Rock Mechanics)				1	Identify the objectives of geotechnical data collection and rock mass classification methods, and successfully collect and analyze a range of geotechnical datasets for design purposes.	D		E		E		E	E			E	
				2	Describe the theory and analysis of in situ and induced stresses in a rock mass and structurally controlled failure.	D		E		E		E	E			E	
				3	Apply the principles of rock mechanics and excavation design to develop	D		E		E		E	E			E	

Course	Pre-requisite	Units		Course Intended Learning Outcome (CILO)		Student Outcomes												
(Course Code and Title)	(Course Code and Title)	Lec	Lab	#	CILO	SO1	SO2	SO3	SO4	SO5	SO6	SO7	SO8	SO9	SO10	SO11	SO12	
					excavation proposals for a given geologic environment (e.g. stratified massive, blocky, or faulted lithology).													
				4	Explain the principles and techniques of reinforcement design for the primary failure modes in underground rock excavations, and apply these principles and techniques to inform design for tunnels and large excavations.	D		E		E		E	E			E		
				5	Describe the theory, analysis, and control of rock and soil slope stability and rock fall hazards.	D		E		E		E	E			E		
				6	Calculate the Factor of Safety of rock slopes and underground excavations.	D		E		E		E	E			E		
				7	Describe the mechanical implications of various blast types and patterns for a rock mass, and characterize blasting control systems for surface and underground excavations.	D		E		E		E	E			E		
(STEN 43043) Professional Course – Specialized 4 (Bridge Engineering)	1			To understand different types of bridges and its importance to the transportation sector.	D		E		E		E					E		
	2			To select appropriate type of bridge considering different factors such as span length, volume of traffic and design requirements that affecting the design	D		E		E		E					E		
	3			To understand the principles in designing concrete and steel bridges.	D		E		E		E					E		
	4			To design bridge pier and its foundation .	D		E		E		E					E		
(TREN 44043) Professional Course – Specialized 4 (Airport Design)	1			State the history of aviation and requirements of airport design.	E		D		E		E		E				E	
	2			Explain the runway length, airspace requirements, navigation and air traffic control.	E		D		E		E		E				E	

Course	Pre-requisite	Units		Course Intended Learning Outcome (CILO)		Student Outcomes												
(Course Code and Title)	(Course Code and Title)	Lec	Lab	#	CILO	SO1	SO2	SO3	SO4	SO5	SO6	SO7	SO8	SO9	SO10	SO11	SO12	
(WREN 45043) Professional Course – Specialized 4 (Water Supply Planning and Development)		3	0	3	Illustrate airfield capacity and delay runway and taxiway Geometrics, Passenger Terminal Capacity.	E		D		E		E				E		
				4	Explain the Air Traffic Flow Management, Airport Management and economics and airport noise.	E		D		E		E			E			
				1	Identify different water sources and discuss the different treatments.	D				E		E			E	D		
				2	Discuss the environmental impacts of developing the different water sources.	D				E		E			E	D		
				3	Estimate water demands for different users.	D				E		E			E	D		
				4	Provide a typical layout of the water supply system.	D				E		E			E	D		
(CMEN 41053) Professional Course – Specialized 5 (Construction Occupational Safety and Health)		3	0	1	Discuss and explain the importance and value of safety.							D		E	E	E	D	
				2	Describe the different health and hazards and their prevention.							D		E	E	E	D	
				3	Apply the concepts and principles of safety in engineering practice							D		E	E	E	D	
				4	Explain “Safety as the engineer’s professional and moral responsibility.”							D		E	E	E	D	
(GTEN 42053) Professional Course – Specialized 5 (Geotechnical and Geoenvironmental Engineering)				1	Evaluate the fundamental geotechnical concepts and implications of environmental engineering.	E		D		E		E	E				D	
				2	Evaluate the geotechnical and geoenvironmental aspects of ground investigation and compaction/earthwork design.	E		D		E		E	E				D	
				3	Evaluate the geotechnical and geoenvironmental aspects of groundwater, soil permeability, consolidation and settlement.	E		D		E		E	E				D	

Course	Pre-requisite	Units		Course Intended Learning Outcome (CILO)		Student Outcomes											
(Course Code and Title)	(Course Code and Title)	Lec	Lab	#	CILO	SO1	SO2	SO3	SO4	SO5	SO6	SO7	SO8	SO9	SO10	SO11	SO12
				4	Evaluate the geotechnical and geoenvironmental aspects of slope stability and retaining structures.	E		D		E		E	E			D	
				5	Evaluate the geotechnical and geoenvironmental aspects of landfills and waste disposal.	E		D		E		E	E			D	
				6	Create a noise and ground vibration proposal plan considering the geotechnical and geoenvironmental conditions of a site.	E		D		E		E	E			D	
(STEN 43053) Professional Course – Specialized 5 (Dam Engineering)	1			Create a gravity dam structural design using the appropriate calculations.	D		E		E		E				E		
	2			Evaluate the structural integrity of a gravity dam.	D		E		E		E				E		
	3			Create an arch dam structural design using the appropriate calculations	D		E		E		E				E		
	4			Evaluate the structural integrity of an arch dam.	D		E		E		E				E		
	5			Evaluate the structural integrity of a dam using structural dynamics.	D		E		E		E				E		
(TREN 44053) Professional Course – Specialized 5 (Ports and Harbors)	1			Show familiarity in the of concept and importance port operation in the economic development of a nation.	D		E		E		E				E		
	2			Knowledgeably plan ports and harbors.	D		E		E		E				E		
	3			Knowledgeably discuss the developments and operation of ports and harbors.	D		E		E		E				E		
	4			Investigate and analyze areas of improvement in ports and harbors.	D		E		E		E				E		
(WREN 45053) Professional Course – Specialized 5 (River Engineering)	1			Discuss the contribution of rivers in flood control, water supply, environmental conservation and economic activities of the area.	D				E		E				E		
	2			Perform calculation related to measurements, movement, and storages	D				E		E				E		

Course	Pre-requisite	Units		Course Intended Learning Outcome (CILO)		Student Outcomes											
(Course Code and Title)	(Course Code and Title)	Lec	Lab	#	CILO	SO1	SO2	SO3	SO4	SO5	SO6	SO7	SO8	SO9	SO10	SO11	SO12
					in the different processes of the hydrologic cycle.												
G. On-The-Job Training																	
(CIEN 40012) On-the-Job Training 1		0	2	1	To bridge the gap between school and the world of work.				D		D	D	E	E	E	E	D
				2	To incorporate the engineering principles to applied civil engineering				D		D	D	E	E	E	E	D
				3	To address the issues and problems confronting the issues and problems confronting the industry manpower particularly the construction-related professionals				D		D	D	E	E	E	E	D
(CIEN 40022) On-the-Job Training 2	(CIEN 40012) On-the-Job Training 1	0	2	1	To bridge the gap between school and the world of work.				D		D	D	E	E	E	E	D
				2	To incorporate the engineering principles to applied civil engineering				D		D	D	E	E	E	E	D
				3	To address the issues and problems confronting the issues and problems confronting the industry manpower particularly the construction-related professionals				D		D	D	E	E	E	E	D

A.4. Describe the process of curriculum mapping and change. Provide latest samples of curriculum or course improvement and changes. Describe how the improvements support the student outcomes.

As part of the Continuous Quality Improvement (CQI) model, inputs from the Government Industry Advisory Board (GIAB), accrediting bodies and stakeholders (students, parents, faculty members) were solicited for curriculum revision. Said curriculum will be mapped along with the concurrent revision of Program Educational Objectives (PEO), Student Outcomes (SO), Course Intended Learning Outcomes (CILO) – presented as the Curriculum Map. The curriculum map is also concurrently re-aligned internally and then externally to university mandates such as University Mission and Vision, Institutional Learning Outcomes (ILO), Graduate Attributes, etc. The final approved alignment and congruency are then presented in the Outcomes-Based Teaching and Learning Plan (OBTLP) syllabi revision.

Upon approval, the revised curriculum is then implemented starting on the next full academic year. After the implementation, direct and indirect assessment and evaluation methods are designed to measure the level of attainment of the expected objectives of the curriculum on different levels such as institutional level (ILO), program level (PEO and SO), course level (CILO). The collected data from the assessment are then used as feedback system for improvement on the next curriculum revision. Such improvements are: (1) introduction of new courses; (2) rationalization of similar courses; and (3) enhancement of courses in the curriculum. Course syllabi and curriculum mapping are also concurrently updated. Laboratory equipment and facilities are also upgraded to conform with comments from accrediting bodies or minimum requirements. See criterion 9 for the full details of the CQI plan.

For the 2018 curriculum revision, a major reduction of units was implemented due to the transition from 5-year to being a 4-year program as a direct result of the implementation of the Senior High School program. Industry partners J.F. Cancio Structural & Associates and Filipinas Dravo Corporation were consulted for input. Following the minimum requirements established in CHED Memo No. 92 s. 2017, the Curriculum Development Committee approved a four-year 211 course credit unit curriculum program. Said curriculum was mapped along with the concurrent revision of PEO, SO, CILO and presented in the 2018 Curriculum Map. The final approved alignment and congruency are then presented in the 2018 OBTLP syllabi.

The 2018 curriculum was initially implemented on the first-year level during A.Y. 2018 – 2019. During its implementation, direct and indirect assessment and evaluation methods are

designed to measure the level of attainment of the expected objectives of the curriculum on different levels such as institutional level (ILO), program level (PEO and SO), course level (CILO). Course syllabi and curriculum mapping are also concurrently updated. Additional laboratory equipment and facilities are also upgraded and procured to conform with comments from accrediting bodies or minimum requirements such as the latest software for specialized professional courses. The data collected from the assessment were used during the stakeholder and committee meetings during the revision. The key milestone dates were presented in Table 5-4.1.

(Attachment C5-13 – Summary of the Processes Underwent During the 2022 BSCE Curriculum Revision)

Table 5-4.2. Summary of Course Improvement and Changes between the 2018 and 2022 PUP BSCE Curricula

2018 PUP BSCE Curriculum		2022 PUP BSCE Curriculum		Course Improvement and Changes	
Course Code and Title	Credit Units	Course Code and Title	Credit Units	Change	Justification
ENSC 20112 - Engineering Drawing and Plans	2	ENSC 011 - Engineering Drawing and Plans	1	Reduction in Course Credit	Change in course credit following CHED Memorandum Order 92 s. 2017
ENSC 20013 - Computer-Aided Drafting	3	ENSC 014 - Computer-Aided Drafting	1	Reduction in Course Credit	Change in course credit following CHED Memorandum Order 92 s. 2017
CIEN 30013 - Fundamentals of Surveying 1	3	CIEN 201 - Fundamentals of Surveying 1	2	Merged Courses	Merged Fundamentals of Surveying 1 and Fundamentals of Surveying 2 following CHED Memorandum Order 92 s. 2017 minimum course credit requirement
CIEN 30033 - Fundamentals of Surveying 2	3				
CIEN 30053 - Route Surveying and Earthworks	3	CIEN 202 - Fundamentals of Surveying 2 (Applied Surveying)	2	Reduction in Course Credit	Change in course credits for the total units of surveying subjects to match the total CHED Memorandum Order 92 s. 2017 requirement for Fundamentals of Surveying
CIEN 30043 - Structural Theory 1	3	CIEN 306 - Structural Theory	4	Merged Courses	Merged Structural Theory 1 and Structural Theory 2 following CHED Memorandum Order 92 s. 2017
CIEN 30103 - Structural Theory 2	3				
CIEN 30183 - Principles of Reinforced Concrete	3	CIEN 308 - Principles of Reinforced Concrete	4	Merged Courses	Merged courses CHED Memorandum Order 92 s. 2017 to lessen the total no. of units as per recommendation of CHED RQAT. Reinforced Concrete Design was then moved to be a Structural Engineering Specialized Course Deleted course following CHED Memorandum Order 92 s. 2017 and to lessen the total no. of units as per recommendation of CHED RQAT
CIEN 30193 - Reinforced Concrete Design	3		0	Deleted Course	
CIEN 30222 - Timber Design	2				
CIEN 30063 - Hydraulics 1	3	CIEN 311 - Hydraulics 1	2	Reduction in Course Credit	Change in course credit to match the total CHED Memorandum Order 92 s. 2017 unit requirement for Hydraulics
CIEN 30083 - Geotechnical Engineering 1	3	CIEN 301 - Geotechnical Engineering 1 (Soil Mechanics)	2	Reduction in Course Credit	Change in course credit to match the total CHED Memorandum Order 92 s. 2017 unit requirement for Geotechnical Engineering

2018 PUP BSCE Curriculum		2022 PUP BSCE Curriculum		Course Improvement and Changes	
Course Code and Title	Credit Units	Course Code and Title	Credit Units	Change	Justification
CIEN 40012 - On-the-Job Training 1 (300 hours)	2	CIEN 314 - On-the-Job Training 1 (300 hours)	3	Change in Course Credit	Repositioned into 3rd Year, Summer Semester to ensure that students have gained enough competencies for OJT
CIEN 40022 - On-the-Job Training 2 (300 hours)	2	CIEN 412 - On-the-Job Training 2 (200 hours)	1	Change in Course Credit	Repositioned into 4th Year, 2nd Semester to enable students to take OJT based on their specialization
CIEN 10113 - Pagsasalin sa Kontekstong Filipino	3		0	Deleted Course	Deleted course to lessen the total no. of units as per recommendation of CHED RQAT and as per UCEC meeting last May 12, 2022
	0	ENGL 012 - Technical Communication	3	Added Course	Added course to equip the students the necessary technical writing and oral communications skills in preparation for their thesis projects and documentation/presentations for their jobs/on-the-job training programs per Stakeholder Consultation Meeting
TOTAL CREDIT UNITS OF CHANGES	41		25		
TOTAL CREDIT UNITS OF UNCHANGED COURSE SUBJECTS	170		170		
TOTAL CURRICULUM CREDIT UNITS	211		195		

A.5 Attach a flowchart or worksheet that illustrates the prerequisite structure of the program's required courses.

Figures 5.5.1 and 5.5.2 present the course pre-requisite map for the 2018 and 2022 curriculum programs, respectively. The prerequisite flowchart for the BSCE program provides a clear, outcomes-supporting sequence of all required courses and demonstrates that the curriculum is structured in accordance with the CHED CMO for the Civil Engineering program. It visually presents how students' progress from foundational learning to advanced application by organizing courses into a logical pathway where each subject builds on previously completed prerequisites.

At the entry level, the flowchart shows the foundational requirements, particularly Mathematics, Basic Sciences, and introductory engineering courses, positioned early in the program. These courses establish the analytical, scientific, and technical bases required by CHED and serve as the initial preparation for higher-level civil engineering work. As students move forward, the flowchart reflects the transition into Engineering Sciences and core Civil Engineering professional courses (e.g., structural, geotechnical, transportation, water resources, and construction-related courses), ensuring that learners acquire essential competencies before undertaking specialized subjects.

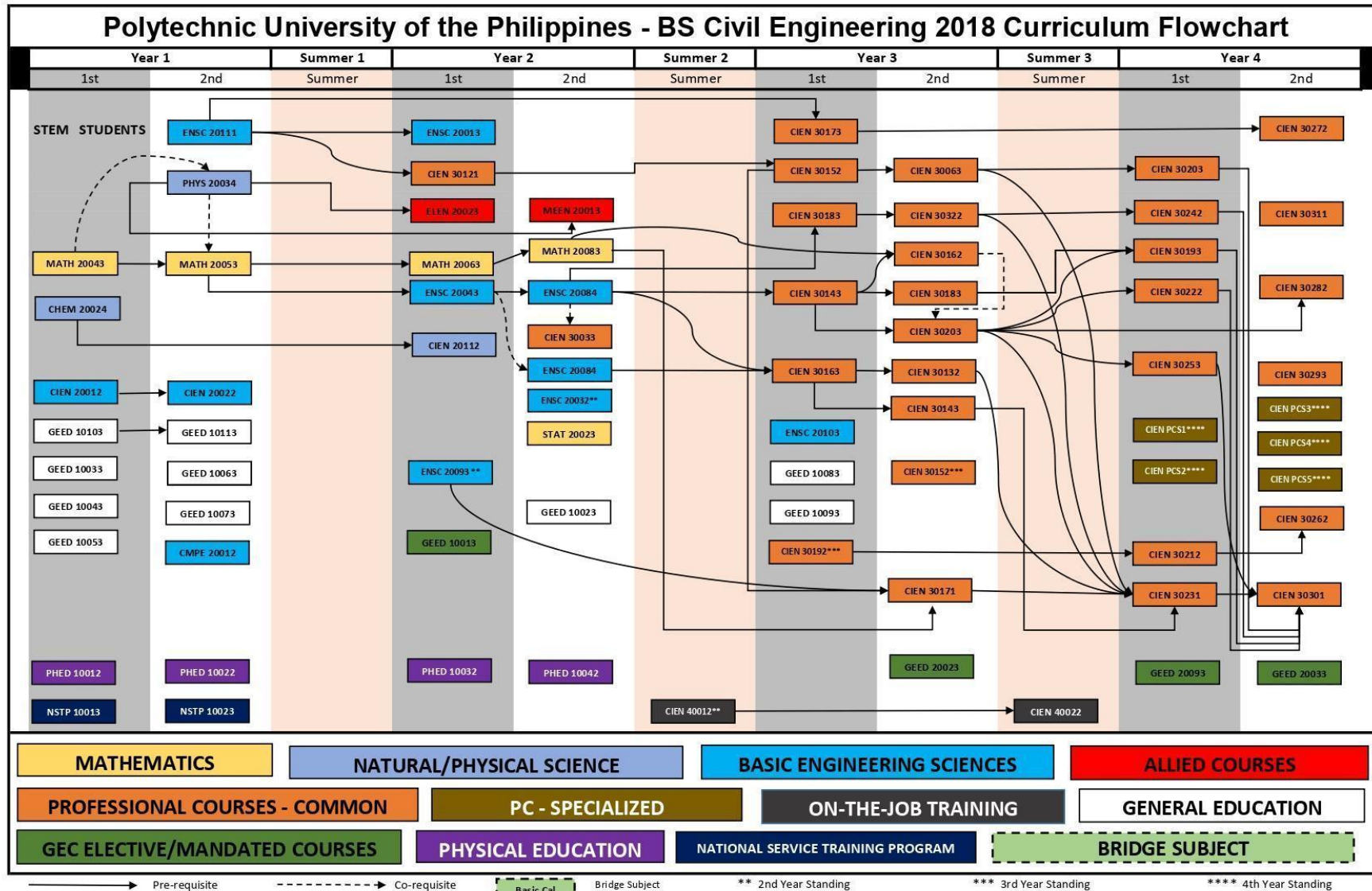
The flowchart also highlights the vertical and horizontal alignment of courses. Vertically, prerequisites ensure that essential concepts and skills are mastered before students proceed to more complex design and analysis courses. Horizontally, related courses are sequenced within the same term or adjacent terms to reinforce integration across civil engineering domains. This arrangement supports progressive learning and prevents gaps in competency development.

Further, the flowchart indicates where laboratory, fieldwork, and integrative learning components are embedded across the curriculum, ensuring that theoretical understanding is reinforced through practical application. By placing advanced design courses, capstone/synthesis experiences, and culminating requirements near the end of the program, the prerequisite structure ensures that students are adequately prepared to demonstrate expected competencies and program outcomes prior to graduation.

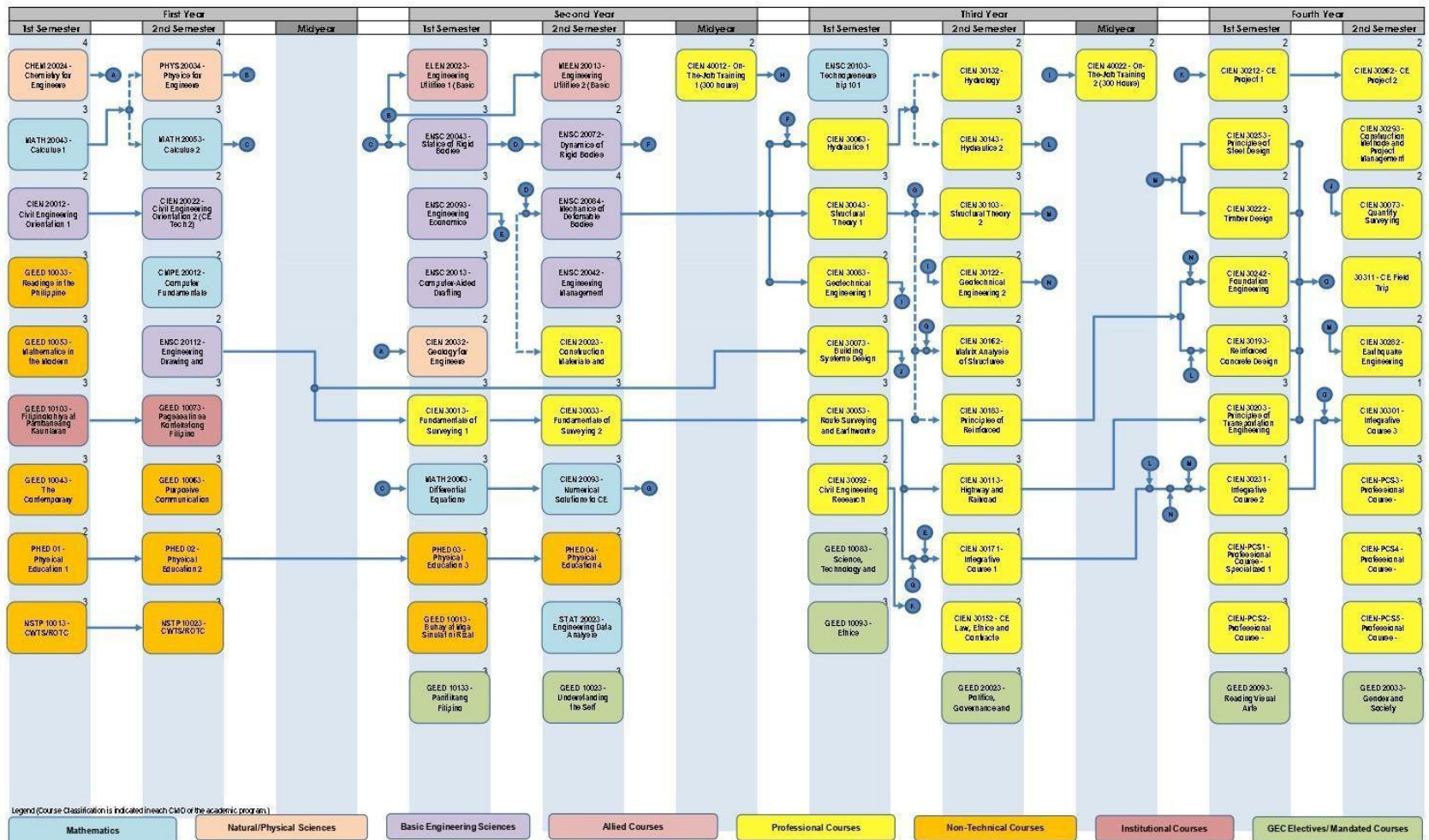
Overall, the prerequisite flowchart serves as evidence that the BSCE curriculum is compliant with CHED CMO requirements and that the program's course sequencing is

intentionally designed to support coherent progression, knowledge scaffolding, and readiness for higher-order civil engineering practice.

Figure 5.5.1. BSCE Prerequisite Flowchart - 2018 Curriculum



BSCE 2018 CURRICULUM COURSE PRE-REQUISITE MAP



Lecture Units
Lab Units
Total

24	21	0
6	18	0
30	39	0

Prepared by:

ENGR. KENNETH BRYAN J. I. TANA
Signature over Printed Name
Program Chair
Date: _____

Reviewed by:

DR. REILLEDIOS G. ADO
Signature over Printed Name
Dean
Date: _____

Recommending Approval by:

DR. FLORDELIZA E. ALVIENDA
Signature over Printed Name
Registrar
Date: _____

Approved by:

DR. ELIANUEL C. DE GUZMAN
Signature over Printed Name
Vice President for Academic Affairs
Date: _____

MATHEMATICS

NATURAL/PHYSICAL SCIENCES

BASIC ENGINEERING SCIENCES

ALLIED COURSES

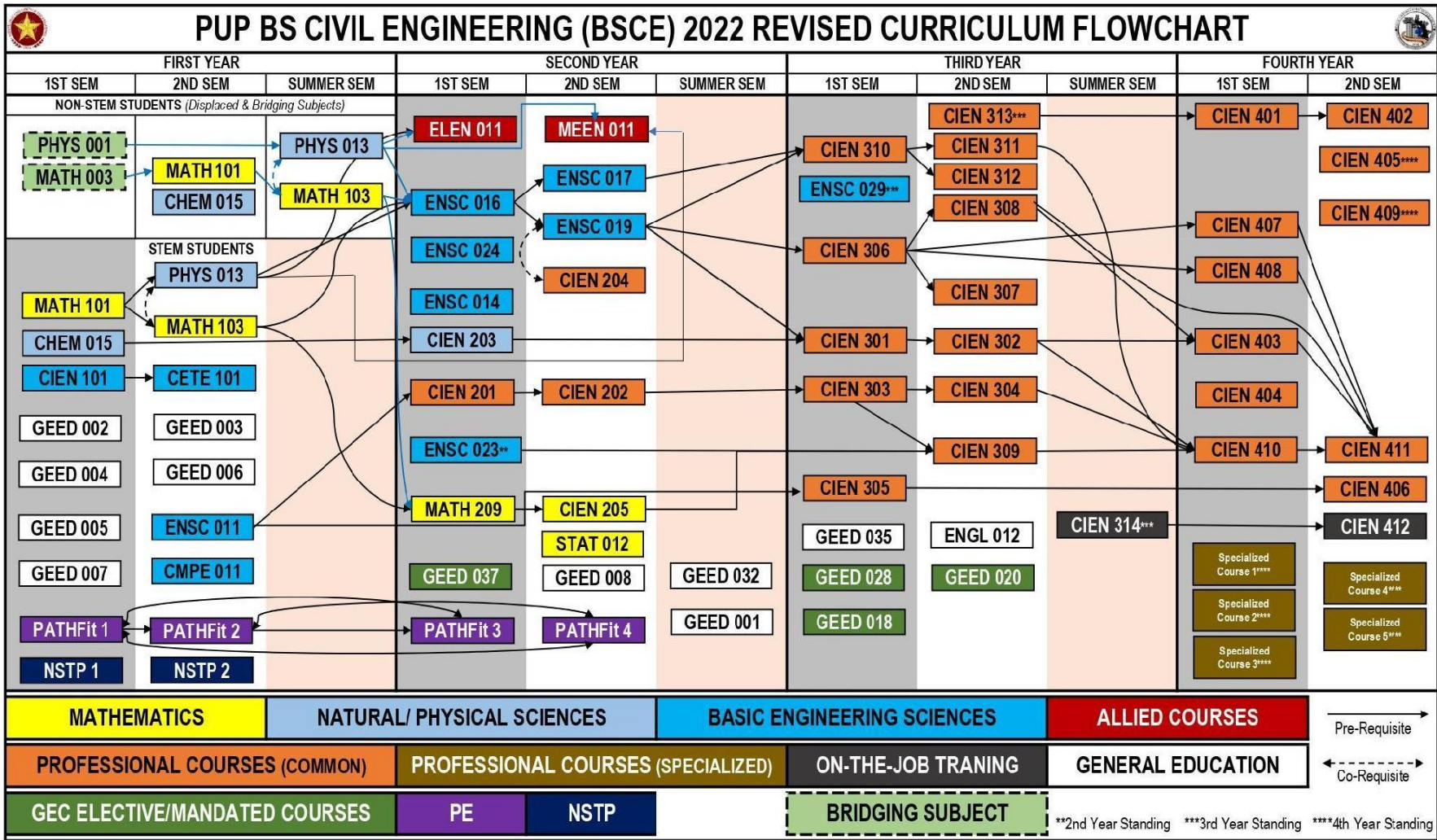
PROFESSIONAL COURSES

NON-TECHNICAL COURSES

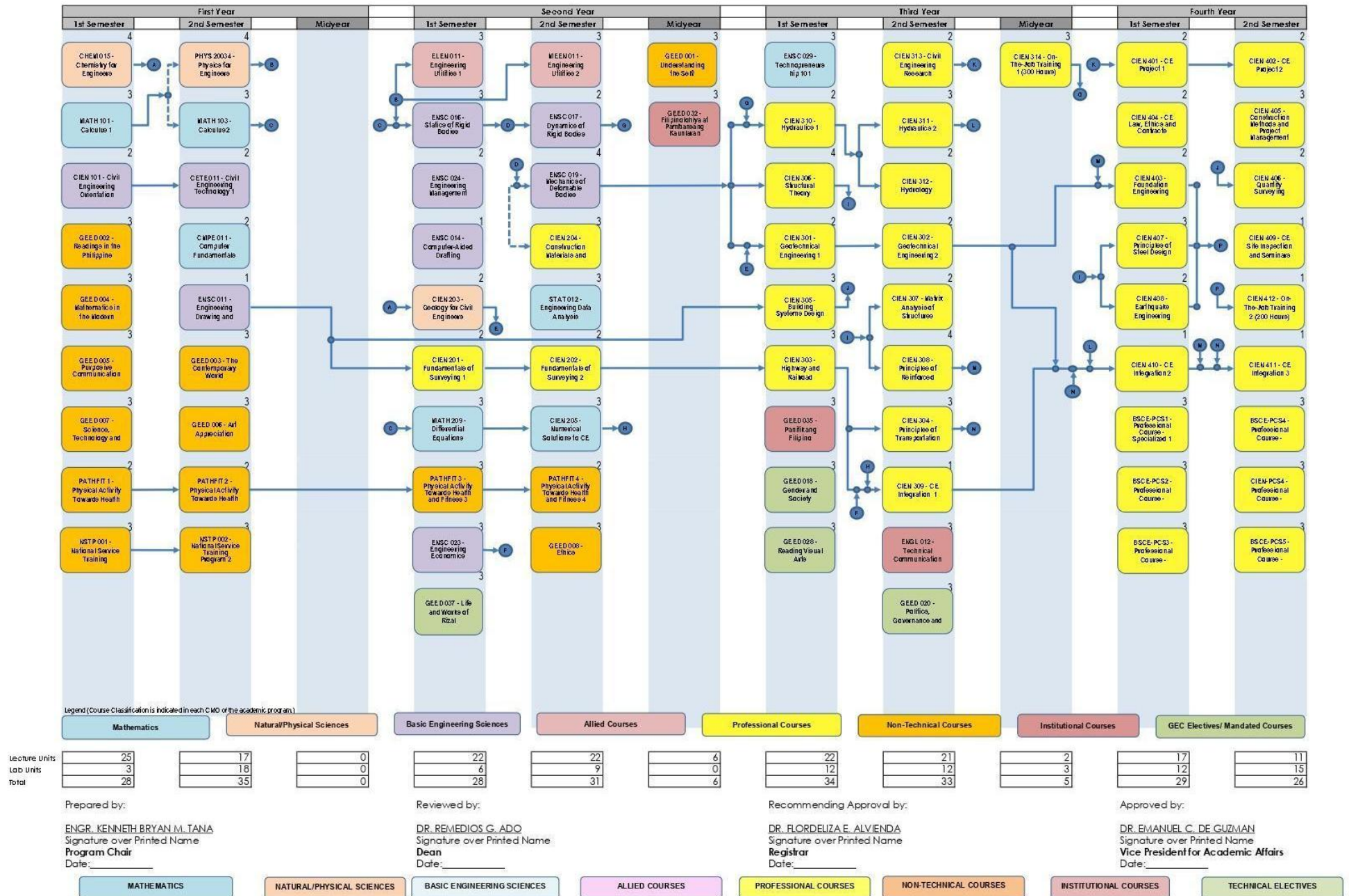
INSTITUTIONAL COURSES

TECHNICAL ELECTIVES

Figure 5.5.2. BSCE Prerequisite Flowchart - 2022 Curriculum



BSCE 2018 CURRICULUM COURSE PRE-REQUISITE MAP



A6. For each curricular area specifically addressed by either the general criteria or the specific program criteria as shown in Table 5-1, describe how your program meets the specific requirements for this program area.

Both 2018 and 2022 BSCE curricula are 4-year programs with two semesters per year level and two mandatory summer/third semesters during the second- and third-year levels. The 2018 curriculum was first implemented on the first-year level during A.Y. 2018 – 2019 and lastly implemented on the fourth-year level during A.Y. 2024 – 2025. On the other hand, the 2022 curriculum was first implemented on the first-year level during A.Y. 2022 – 2023 and is in full implementation across all four-year levels as of this current 2nd Semester of A.Y. 2025 – 2026.

The 211 total credit units of the 2018 BSCE Curriculum comply with the minimum required 171 credit units. 152 technical course units meets the 121 unit requirement and distributed as: 15 out of the required 15 units for Mathematics, 10 out of the required 10 units for Physical Sciences, 28 vs the required 23 units for Basic Engineering Sciences, 6 out of the required 6 units for Allied Courses, 4 out of 4 required units for On-the-Job Training, 74 vs the 49 required units for Common Professional Courses and 15 out of the 15 required units for Specialized Professional Courses.

The 195 total credit units of the 2022 BSCE Curriculum comply with the minimum required 171 credit units. 136 technical course units meets the 121 unit requirement and distributed as: 15 out of the required 15 units for Mathematics, 10 out of the required 10 units for Physical Sciences, 25 vs the required 23 units for Basic Engineering Sciences, 6 out of the required 6 units for Allied Courses, 4 out of 4 required units for On-the-Job Training, 61 vs the 49 required units for Common Professional Courses and 15 out of the 15 required units for Specialized Professional Courses.

Both Table 5-6.1 and Attachment Figure 5.6.1 show the comparison of units and course subjects for each curricular area between the 2018 curriculum, 2022 curriculum, and the minimum requirements. Figure 5.6.1 shows the breakdown and visual comparison of the no. of units for each curricular area. Attachment Figure 5.6.2 and 5.6.3 shows the percent distribution of credit units among each curricular area for the 2018 and 2022 BSCE curricula, respectively.

[\(Attachment C5-14 – Figure 5.6.1. Comparison of the 2018 and 2022 BSCE Curriculum Credit Units with the Minimum Requirements\)](#)

(Attachment C5-15 – Figure 5.6.2. Percentage Distribution of Credit Units – 2018 BSCE Curriculum)

(Attachment C5-16 – Figure 5.6.3. Percentage Distribution of Credit Units – 2022 BSCE Curriculum)

Table 5-6.1. Comparison of the 2018 and 2022 BSCE Curriculum Credit Units with the Minimum Requirements

Minimum BSCE Curriculum Courses per CHED CMO 92 s.2017	Minimum Credit Units	2018 PUP BSCE Curriculum		No. of Credit Units	2022 PUP BSCE Curriculum		No. of Credit Units
I. TECHNICAL COURSES		I. TECHNICAL COURSES			I. TECHNICAL COURSES		
<i>A. Mathematics</i>		<i>A. Mathematics</i>			<i>A. Mathematics</i>		
Calculus 1	3	MATH 20043	Calculus 1	3	MATH 101	Calculus 1	3
Calculus 2	3	MATH 20053	Calculus 2	3	MATH 103	Calculus 2	3
Differential Equations	3	MATH 20063	Differential Equations	3	MATH 209	Differential Equations	3
Numerical Solutions to CE Problems	3	MATH 20093	Numerical Solutions to CE Problems	3	CIEN 205	Numerical Solutions to CE Problems	3
Engineering Data Analysis	3	STAT 20023	Engineering Data Analysis	3	STAT 012	Engineering Data Analysis	3
	15			15			15
<i>B. Natural/Physical Sciences</i>		<i>B. Natural/Physical Sciences</i>			<i>B. Natural/Physical Sciences</i>		
Chemistry for Engineers	4	CHEM 20024	Chemistry for Engineers	4	CHEM 015	Chemistry for Engineers	4
Physics for Engineers (Calculus-based)	4	PHYS 20034	Physics for Engineers (Calculus-based)	4	PHYS 013	Physics for Engineers (Calculus-based)	4
Geology for Civil Engineers	2	CIEN 20032	Geology for Civil Engineers	2	CIEN 203	Geology for Civil Engineers	2
	10			10			10
<i>C. Basic Engineering Sciences</i>		<i>C. Basic Engineering Sciences</i>			<i>C. Basic Engineering Sciences</i>		
Civil Engineering Orientation	2	CIEN 20012	Civil Engineering Orientation 1	2	CIEN 101	Civil Engineering Orientation	2
		CIEN 20022	Civil Engineering Orientation 2	2	CETE 011	Civil Engineering Technology	2
Engineering Drawing and Plans	1	ENSC 20112	Engineering Drawing and Plans	2	ENSC 011	Engineering Drawing and Plans	1

Minimum BSCE Curriculum Courses per CHED CMO 92 s.2017	Minimum Credit Units	2018 PUP BSCE Curriculum		No. of Credit Units	2022 PUP BSCE Curriculum		No. of Credit Units
Computer Fundamentals and Programming	2	CMPE 20012	Computer Fundamentals and Programming	2	CMPE 011	Computer Fundamentals and Programming	2
Computer-Aided Drafting	1	ENSC 20013	Computer-Aided Drafting	3	ENSC 014	Computer-Aided Drafting	1
Statics of Rigid Bodies	3	ENSC 20043	Statics of Rigid Bodies	3	ENSC 016	Statics of Rigid Bodies	3
Dynamics of Rigid Bodies	2	ENSC 20072	Dynamics of Rigid Bodies	2	ENSC 017	Dynamics of Rigid Bodies	2
Mechanics of Deformable Bodies	4	ENSC 20084	Mechanics of Deformable Bodies	4	ENSC 019	Mechanics of Deformable Bodies	4
Engineering Economics	3	ENSC 20093	Engineering Economics	3	ENSC 023	Engineering Economics	3
Engineering Management	2	ENSC 20042	Engineering Management	2	ENSC 024	Engineering Management	2
Technopreneurship 101	3	ENSC 20103	Technopreneurship 101	3	ENSC 029	Technopreneurship 101	3
	23			28			25
<i>D. Allied Courses</i>							
Engineering Utilities 1	3	ELEN 20023	Engineering Utilities 1 (Basic Electrical Engineering)	3	ELEN 011	Engineering Utilities 1	3
Engineering Utilities 2	3	MEEN 20013	Engineering Utilities 2 (Basic Mechanical Engineering)	3	MEEN 011	Engineering Utilities 2	3
	6			6			6
<i>E. Professional Courses - Common</i>		<i>E. Professional Courses - Common</i>			<i>E. Professional Courses - Common</i>		
Fundamentals of Surveying	4	CIEN 30013	Fundamentals of Surveying 1	3	CIEN 201	Fundamentals of Surveying 1	2
		CIEN 30033	Fundamentals of Surveying 2	3			
		CIEN 30053	Route Surveying and Earthworks	3	CIEN 202	Fundamentals of Surveying 2 (Applied Surveying)	2

Minimum BSCE Curriculum Courses per CHED CMO 92 s.2017	Minimum Credit Units	2018 PUP BSCE Curriculum		No. of Credit Units	2022 PUP BSCE Curriculum		No. of Credit Units
Highway and Railroad Engineering	3	CIEN 30113	Highway and Railroad Engineering	3	CIEN 303	Highway and Railroad Engineering	3
Principles of Transportation Engineering	3	CIEN 30203	Principles of Transportation Engineering	3	CIEN 304	Principles of Transportation Engineering	3
Structural Theory	4	CIEN 30043	Structural Theory 1	3	CIEN 306	Structural Theory	4
		CIEN 30103	Structural Theory 2	3			
		CIEN 30162	Matrix Analysis of Structures	2	CIEN 307	Matrix Analysis of Structures	2
Principles of Steel Design	3	CIEN 30253	Principles of Steel Design	3	CIEN 407	Principles of Steel Design	3
		CIEN 30222	Timber Design	2			
Principles of Reinforced Concrete	4	CIEN 30183	Principles of Reinforced Concrete	3	CIEN 308	Principles of Reinforced Concrete	4
		CIEN 30193	Reinforced Concrete Design	3			
		CIEN 30282	Earthquake Engineering	2	CIEN 408	Earthquake Engineering	2
Hydraulics	5	CIEN 30063	Hydraulics 1	3	CIEN 310	Hydraulics 1	2
		CIEN 30143	Hydraulics 2	3	CIEN 311	Hydraulics 2	3
Hydrology	2	CIEN 30132	Hydrology	2	CIEN 312	Hydrology	2
Geotechnical Engineering	4	CIEN 30083	Geotechnical Engineering 1	3	CIEN 301	Geotechnical Engineering 1 (Soil Mechanics)	2
		CIEN 30122	Geotechnical Engineering 2	2	CIEN 302	Geotechnical Engineering 2	2
		CIEN 30242	Foundation Engineering	2	CIEN 403	Foundation Engineering	2

Minimum BSCE Curriculum Courses per CHED CMO 92 s.2017	Minimum Credit Units	2018 PUP BSCE Curriculum		No. of Credit Units	2022 PUP BSCE Curriculum		No. of Credit Units
Construction Materials and Testing	3	CIEN 30023	Construction Materials and Testing	3	CIEN 204	Construction Materials and Testing	3
Building Systems Design	3	CIEN 30073	Building Systems Design	3	CIEN 305	Building Systems Design	3
Quantity Surveying	2	CIEN 30272	Quantity Surveying	2	CIEN 406	Quantity Surveying	2
Construction Methods and Project Management	3	CIEN 30293	Construction Methods and Project Management	3	CIEN 405	Construction Methods and Project Management	3
CE Law, Ethics and Contracts	2	CIEN 30152	CE Law, Ethics and Contracts	2	CIEN 404	CE Law, Ethics and Contracts	2
		CIEN 30092	Civil Engineering Research	2	CIEN 313	Civil Engineering Research	2
CE Project 1	2	CIEN 30212	CE Project 1	2	CIEN 401	CE Project 1	2
CE Project 2	2	CIEN 30262	CE Project 2	2	CIEN 402	CE Project 2	2
		CIEN 30171	Integrative Course 1	1	CIEN 309	CE Integration 1	1
		CIEN 30231	Integrative Course 2	1	CIEN 410	CE Integration 2	1
		CIEN 30301	Integrative Course 3	1	CIEN 411	CE Integration 3	1
		CIEN 30311	CE Field Trip	1	CIEN 409	CE Site Inspection and Seminars	1
	49			74			61
<i>E. Professional Courses - Specialized</i>		<i>E. Professional Courses - Specialized</i>			<i>E. Professional Courses - Specialized</i>		
Professional Course-Specialized 1	3		Professional Course-Specialized 1	3		Professional Course-Specialized 1	3
Professional Course-Specialized 2	3		Professional Course-Specialized 2	3		Professional Course-Specialized 2	3
Professional Course-Specialized 3	3		Professional Course-Specialized 3	3		Professional Course-Specialized 3	3

Minimum BSCE Curriculum Courses per CHED CMO 92 s.2017	Minimum Credit Units	2018 PUP BSCE Curriculum		No. of Credit Units	2022 PUP BSCE Curriculum		No. of Credit Units
Professional Course-Specialized 4	3		Professional Course-Specialized 4	3		Professional Course-Specialized 4	3
Professional Course-Specialized 5	3		Professional Course-Specialized 5	3		Professional Course-Specialized 5	3
	15			15			15
<i>G. On-the-Job Training (minimum of 240 hours)</i>		<i>G. On-the-Job Training</i>			<i>G. On-the-Job Training</i>		
On-the-Job Training	3	CIEN 40012	On-The-Job Training 1 (300 hours)	2	CIEN 314	On-The-Job Training 1 (300 hours)	3
		CIEN 40022	On The Job Training 2 (300 hours)	2	CIEN 412	On-The-Job Training 2 (200 hours)	1
	3			4			4
TOTAL TECHNICAL COURSES	121	TOTAL TECHNICAL COURSES		152	TOTAL TECHNICAL COURSES		136
II. NON-TECHNICAL COURSES		II. NON-TECHNICAL COURSES			II. NON-TECHNICAL COURSES		
<i>A. General Education</i>		<i>A. General Education</i>			<i>A. General Education</i>		
Science, Technology and Society	3	GEED 10083	Science, Technology and Society	3	GEED 007	Science, Technology and Society	3
Readings in Philippine History	3	GEED 10033	Readings in Philippine History	3	GEED 002	Readings in Philippine History	3
Mathematics in the Modern World	3	GEED 10053	Mathematics in the Modern World	3	GEED 004	Mathematics in the Modern World	3
The Contemporary World	3	GEED 10043	The Contemporary World	3	GEED 003	The Contemporary World	3
Understanding the Self	3	GEED 10023	Understanding the Self	3	GEED 001	Understanding the Self	3
Purposive Communication	3	GEED 10063	Purposive Communication	3	GEED 005	Purposive Communication	3

Minimum BSCE Curriculum Courses per CHED CMO 92 s.2017	Minimum Credit Units	2018 PUP BSCE Curriculum		No. of Credit Units	2022 PUP BSCE Curriculum		No. of Credit Units
Art Appreciation	3	GEED 10073	Art Appreciation	3	GEED 006	Art Appreciation	3
Ethics	3	GEED 10093	Ethics	3	GEED 008	Ethics	3
	24			24			24
<i>B. GEC Elective/Mandated Courses</i>		<i>B. GEC Elective/Mandated Courses</i>			<i>B. GEC Elective/Mandated Courses</i>		
GE Elective	9	GEED 20023	Politics, Governance and Citizenship	3	GEED 020	Politics, Governance and Citizenship	3
		GEED 20093	Reading Visual Arts	3	GEED 028	Reading Visual Arts	3
		GEED 20033	Gender and Society	3	GEED 018	Gender and Society	3
Life and Works of Rizal	3	GEED 10013	Buhay at Mga Sinulat ni Rizal	3	GEED 037	Life and Works of Rizal	3
	12			12			12
		<i>Institutional GE Courses</i>			<i>Institutional GE Courses</i>		
		GEED 10103	Filipinolohiya at Pambansang Kaunlaran	3	GEED 032	Filipinolohiya at Pambansang Kaunlaran	3
		GEED 10133	Panitikang Filipino	3	GEED 035	Panitikang Filipino	3
		GEED 10113	Pagsasalin sa Kontekstong Filipino	3			
					ENGL 012	Technical Communication	3
				9			9
<i>C. Physical Education</i>		<i>C. Physical Education</i>			<i>C. PATHFit</i>		
Physical Education 1, 2, 3 and 4	8	PHED 01, 02, 03, 04	Physical Education 1, 2, 3 and 4	8	PATHFit 01, 02, 03, 04	Physical Activity Towards Health and Fitness 1, 2, 3 and 4	8
<i>D. National Service Training Program</i>		<i>D. National Service Training Program</i>			<i>D. National Service Training Program</i>		

Minimum BSCE Curriculum Courses per CHED CMO 92 s.2017	Minimum Credit Units	2018 PUP BSCE Curriculum		No. of Credit Units	2022 PUP BSCE Curriculum		No. of Credit Units
NSTP 1 and 2	6	NSTP/ROT C 10013, 10023	NSTP 1 and 2	6	NSTP/RO TC	NSTP 1 and 2	6
TOTAL NON-TECHNICAL COURSES	50	TOTAL NON-TECHNICAL COURSES		59	TOTAL NON-TECHNICAL COURSES		59
GRAND TOTAL	171	GRAND TOTAL		211	GRAND TOTAL		195

Table 5-6.2. Summary of Comparison per Classification among the 2018 and 2022 BSCE Curriculum Credit Units with the Minimum Requirements

Classification	CHED CMO No. 92 s. 2017 Minimum Requirements	PUP 2018 BSCE Curriculum		PUP 2022 BSCE Curriculum	
Mathematics	14	14	22	14	22
Natural/Physical Sciences	8	8		8	
Basic Engineering Sciences	19	21	27	19	25
Allied Courses	6	6		6	
Common Professional CE Courses	38	51	66	42	57
Specialized Professional CE Courses	15	15		15	
General Education Courses	24	24	45	24	45
Mandated and Institutional GE Courses	12	21		21	
Laboratory and Fieldwork	18	33	33	28	28
On-the-Job Training	3	4	4	4	4
PE	8	8	14	8	14
NSTP	6	6		6	
	171	211		195	

The non-technical courses and learning experiences embedded within the program curriculum play a vital role in achieving the University's Vision and Mission by fostering the holistic development of students beyond their disciplinary and technical expertise. These courses complement the technical components of the curriculum and ensure that graduates possess the knowledge, values, attitudes, and competencies necessary to become responsible professionals, productive citizens, and lifelong learners.

As reflected in Table 5-6.3., the general education, institutional, and mandated courses contribute significantly to the attainment of the University's vision of producing globally competitive graduates who are ethically grounded, socially responsive, and committed to sustainable development. Courses in communication, humanities, social sciences, ethics, environmental studies, entrepreneurship, leadership, and civic engagement cultivate critical thinking, effective communication, cultural awareness, and ethical decision-making. These competencies enable students to navigate complex societal challenges and function effectively in diverse local and global environments.

[\(Attachment C5-17 – Table 5-6.3. Alignment of the 2018 BSCE Curriculum with the PUP Mission and Vision Statements\)](#)

[\(Attachment C5-18 – Table 5-6.4. Alignment of the 2022 BSCE Curriculum with the PUP Mission and Vision Statements\)](#)

The inclusion of values education, ethics, and citizenship-related subjects reinforces the institution's mission of developing graduates who demonstrate integrity, accountability, and social responsibility. Likewise, courses emphasizing cultural understanding, national identity, and community engagement encourage students to become active contributors to nation-building and community development.

These non-technical components are intentionally integrated into the curriculum to support the achievement of program educational objectives and student outcomes. The curriculum demonstrates a balanced approach to education by combining technical competence with broad-based learning experiences that promote ethical conduct, social awareness, lifelong learning, environmental stewardship, and professional responsibility.

Portion of Table 5-6.3. Alignment of the 2018 BSCE Curriculum with the PUP Mission and Vision Statements showing Non-technical Courses

Course	Course Intended Learning Outcome (CILO)		VISION (2024 - Present)	VISION (2022)	MISSION					
(Course Code and Title)	#	CILO	<i>PUP: A Leading Comprehensive Polytechnic University in Asia</i>	<i>The Country's First Polytechnic U</i>	<i>Provide democratized access to educational opportunities for the holistic development of individuals with global perspective</i>	<i>Offer industry-oriented curricula that produce highly skilled professionals with managerial and technical capabilities and a strong sense of public service for nation building</i>	<i>Embed a culture of research and innovation</i>	<i>Continuously develop faculty and employees with the highest level of professionalism</i>	<i>Engage public and private institutions and other stakeholders for the attainment of social development goal</i>	<i>Establish a strong presence and impact in the international academic community</i>
II. NON-TECHNICAL COURSES										
A. General Education										
(GEED 10083) Science, Technology, and Society	1	Explain how science and technology affect society and the environment and their role in nation- building.	✓	✓	✓	✓	✓	✓	✓	✓
	2	Analyze the human condition in order to deeply reflect and express philosophical ramifications that are meaningful to the students as part of society.	✓	✓	✓	✓	✓	✓	✓	✓
	3	Critique human flourishing vis-à-vis the progress of science and technology such that the student may be able to define for himself/herself the meaning of good life.	✓	✓	✓	✓	✓	✓	✓	✓
(GEED 10033) Readings in the Philippine History	1	Distinguish different interpretations of and approaches to globalization	✓	✓	✓	✓		✓	✓	
	2	Describe the emergence of global economic, political, social, and cultural systems	✓	✓	✓	✓		✓	✓	

Course	Course Intended Learning Outcome (CILO)		VISION (2024 - Present)	VISION (2022)	MISSION					
(Course Code and Title)	#	CILO	PUP: A Leading Comprehensive Polytechnic University in Asia	The Country's First Polytechnic U	Provide democratized access to educational opportunities for the holistic development of individuals with global perspective	Offer industry-oriented curricula that produce highly skilled professionals with managerial and technical capabilities and a strong sense of public service for nation building	Embed a culture of research and innovation	Continuously develop faculty and employees with the highest level of professionalism	Engage public and private institutions and other stakeholders for the attainment of social development goal	Establish a strong presence and impact in the international academic community
	3	Analyze the various contemporary drivers of globalization	✓	✓	✓	✓		✓	✓	
(GEED 10053) Mathematics in the Modern World	1	Develop innovative methods and remedies that are guided by ethical standards.	✓	✓	✓	✓	✓		✓	
	2	Utilize computational tools to efficiently process information	✓	✓	✓	✓	✓		✓	
	3	Create solutions to problems in various fields	✓	✓	✓	✓	✓		✓	
(GEED 10043) The Contemporary World	1	Assess the effects of globalization on different social units and their responses.	✓	✓	✓	✓	✓	✓	✓	✓
	2	Distinguish different interpretations of and approaches to globalization.	✓	✓	✓	✓	✓	✓	✓	✓
	3	Analyze the various contemporary drivers of globalization.	✓	✓	✓	✓	✓	✓	✓	✓
(GEED 10023) Understanding the Self	1	Discuss the different representations and conceptualizations of the self from various disciplinary perspectives.	✓	✓	✓	✓	✓	✓	✓	✓
	2	Demonstrate critical, reflective thought in integrating the various aspects of self	✓	✓	✓	✓	✓	✓	✓	✓

Course	Course Intended Learning Outcome (CILO)		VISION (2024 - Present)	VISION (2022)	MISSION					
(Course Code and Title)	#	CILO	PUP: A Leading Comprehensive Polytechnic University in Asia	The Country's First Polytechnic U	Provide democratized access to educational opportunities for the holistic development of individuals with global perspective	Offer industry-oriented curricula that produce highly skilled professionals with managerial and technical capabilities and a strong sense of public service for nation building	Embed a culture of research and innovation	Continuously develop faculty and employees with the highest level of professionalism	Engage public and private institutions and other stakeholders for the attainment of social development goal	Establish a strong presence and impact in the international academic community
	3	Apply these new skills to oneself and functioning for a better quality of life	✓	✓	✓	✓	✓	✓	✓	✓
(GEED 10063) Purposive Communication	1	Describe the models, nature, elements, and functions of verbal and non-verbal	✓	✓	✓	✓		✓	✓	
	2	Determine culturally appropriate terms, expressions, and images	✓	✓	✓	✓		✓	✓	
	3	Create clear, coherent, and effective business correspondence and letters	✓	✓	✓	✓		✓	✓	
(GEED 10073) Art Appreciation	1	Demonstrate an understanding and appreciation of arts in general, including their function, value, and historical significance.	✓	✓	✓	✓	✓	✓	✓	
	2	Discover their identity through art with respect to their nationality, culture, and religion	✓	✓	✓	✓	✓	✓	✓	
	3	Create their own works of art and curate their own production or exhibit.	✓	✓	✓	✓	✓	✓	✓	
(GEED 10093) Ethics	1	Differentiate between moral and non-moral problems	✓	✓	✓	✓		✓	✓	
	2	Explain the influence of Filipino culture on the way students look at moral experiences and solve moral dilemmas	✓	✓	✓	✓		✓	✓	

Course	Course Intended Learning Outcome (CILO)		VISION (2024 - Present)	VISION (2022)	MISSION					
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	3	Internalize the ethical norms of contemporary society at the individual, societal, and in interactions with the environment and other shared resources levels.	✓	✓	✓	✓		✓	✓	
B. GEC Elective/Mandated and Institutional GE Courses										
(GEED 20093) Reading Visual Arts	1	Determine the different ways of reading visual arts according to experts and from the perspective of an audience.	✓	✓	✓	✓	✓	✓	✓	
	2	Interpret artworks based on the elements and principles of art organization.	✓	✓	✓	✓	✓	✓	✓	
	3	Appreciate varied cultures, history, people's lives, and the experiences of others through arts	✓	✓	✓	✓	✓	✓	✓	
(GEED 20023) Politics, Governance and Citizenship	1	Discuss the overview of the nature of Philippine institutions and Philippine politics.	✓	✓	✓	✓	✓	✓	✓	
	2	Demonstrate the various political actors in the Philippine society and how they shape key political institutions.	✓	✓	✓	✓	✓	✓	✓	

Course	Course Intended Learning Outcome (CILO)		VISION (2024 - Present)	VISION (2022)	MISSION					
(Course Code and Title)	#	CILO	<i>PUP: A Leading Comprehensive Polytechnic University in Asia</i>	<i>The Country's First Polytechnic U</i>	<i>Provide democratized access to educational opportunities for the holistic development of individuals with global perspective</i>	<i>Offer industry-oriented curricula that produce highly skilled professionals with managerial and technical capabilities and a strong sense of public service for nation building</i>	<i>Embed a culture of research and innovation</i>	<i>Continuously develop faculty and employees with the highest level of professionalism</i>	<i>Engage public and private institutions and other stakeholders for the attainment of social development goal</i>	<i>Establish a strong presence and impact in the international academic community</i>
	3	Understanding of citizenship and territoriality beyond legal definitions	✓	✓	✓	✓	✓	✓	✓	
(GEED 20033) Gender and Society	1	Gain an understanding on the concepts and theories of Gender and Development.	✓	✓	✓	✓	✓	✓	✓	
	2	Develop competence in the application of tools and analysis applied to person, family, church and community.	✓	✓	✓	✓	✓	✓	✓	
	3	Demonstrate an ability to draw out categories from life experiences utilizing the process of analysis to each gender.	✓	✓	✓	✓	✓	✓	✓	
(GEED 10013) Life and Works of Rizal	1	Learn and understand the life of Dr. Jose P. Rizal from his birth to his martyrdom.	✓	✓	✓	✓	✓	✓	✓	✓
	2	Critically understand Rizal's heroism and realize the significance of his ideas in the present time	✓	✓	✓	✓	✓	✓	✓	✓
	3	Interpret the selected poem, essays and novels and its impact to socio-political situation during the past and present time.	✓	✓	✓	✓	✓	✓	✓	✓

Course	Course Intended Learning Outcome (CILO)		VISION (2024 - Present)	VISION (2022)	MISSION					
(Course Code and Title)	#	CILO	<i>PUP: A Leading Comprehensive Polytechnic University in Asia</i>	<i>The Country's First Polytechnic U</i>	<i>Provide democratized access to educational opportunities for the holistic development of individuals with global perspective</i>	<i>Offer industry-oriented curricula that produce highly skilled professionals with managerial and technical capabilities and a strong sense of public service for nation building</i>	<i>Embed a culture of research and innovation</i>	<i>Continuously develop faculty and employees with the highest level of professionalism</i>	<i>Engage public and private institutions and other stakeholders for the attainment of social development goal</i>	<i>Establish a strong presence and impact in the international academic community</i>
(GEED 10103) Filipinolohiya at Pambansang Kaunlaran	1	Maipaliwanag nang may husay ang kahulugan at kalikasan ng Filipinolohiya na nakaugnay sa Wikang Filipino at nasyunalismo, siyensya at mamamayan	✓	✓	✓	✓		✓	✓	
	2	Makapaglahad ng kalagayan at halaga ng Filipinolohiya sa pamamagitan ng paglalata ng sitwasyong pangkultura, pangwika at panlipunan sa bansa at ibang bansa.	✓	✓	✓	✓		✓	✓	
	3	Makapag-analisa ng kalagayan ng Pambansang Industriya sa Pilipinas at sa ibang Bansa.	✓	✓	✓	✓		✓	✓	
(GEED 10113) Pagsasalin sa Kontekstong Filipino	1	Mailahad ang kronolohikal na kasaysayan ng pagsasalin sa bansa at daigdig	✓	✓	✓	✓		✓	✓	
	2	Maipaliwanag ang uri, kahulugan at katangian ng pagsasaling teknikal at pampanitikan.	✓	✓	✓	✓		✓	✓	
	3	Makapag-analisa ng kalagayan at halaga ng pagsasalin sa industriya ng bansa na nakasalig sa wikang katutubo	✓	✓	✓	✓		✓	✓	

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(GEED 10133) Panitikang Filipino	1	Makapagtamo ng kaalaman sa mga umiiral na panitikan sa Pilipinas ayon sa historikal na pag-unlad nito;	✓	✓	✓	✓		✓	✓	
	2	Nakapagtatalakay ng mga akdang Pampanitikan na sumasalamin sa iba't ibang aspekto ng pamumuhay, kultura, lipunan, pamahalaan at kasaysayan ng sambayanang Pilipino.	✓	✓	✓	✓		✓	✓	
	3	Nakapagpapahayag ng may kabisaan ukol sa pasalita at pasulat na tradisyon ng panitikang Pilipino.	✓	✓	✓	✓		✓	✓	
C. Physical Education										
(PHED 01) Physical Education 1	1	Discover the different exercise workouts that are best suitable to their own exercise program	✓	✓	✓	✓	✓	✓	✓	✓
	2	Display movement competence and confidence using a variety of exercise forms (e.g., cardiorespiratory, functional, resistance training, etc.)	✓	✓	✓	✓	✓	✓	✓	✓
	3	Design, plan, implement, and evaluate a personal exercise program.	✓	✓	✓	✓	✓	✓	✓	✓

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	4	Demonstrate an appreciation of the importance of regular engagement in moderate to vigorous exercise programs.	✓	✓	✓	✓	✓	✓	✓	✓
	5	Share learned knowledge and skills through active engagement in a service- learning project.	✓	✓	✓	✓	✓	✓	✓	✓
(PHED 02) Physical Education 2	1	Discover the different exercise workouts that are best suitable to their own exercise program	✓	✓	✓	✓	✓	✓	✓	✓
	2	Display movement competence and confidence using a variety of exercise forms (e.g., cardiorespiratory, functional, resistance training, etc.)	✓	✓	✓	✓	✓	✓	✓	✓
	3	Design, plan, implement, and evaluate a personal exercise program.	✓	✓	✓	✓	✓	✓	✓	✓
	4	Demonstrate an appreciation of the importance of regular engagement in moderate to vigorous exercise programs.	✓	✓	✓	✓	✓	✓	✓	✓

Course	Course Intended Learning Outcome (CILO)		VISION (2024 - Present)	VISION (2022)	MISSION					
(Course Code and Title)	#	CILO	<i>PUP: A Leading Comprehensive Polytechnic University in Asia</i>	<i>The Country's First Polytechnic U</i>	<i>Provide democratized access to educational opportunities for the holistic development of individuals with global perspective</i>	<i>Offer industry-oriented curricula that produce highly skilled professionals with managerial and technical capabilities and a strong sense of public service for nation building</i>	<i>Embed a culture of research and innovation</i>	<i>Continuously develop faculty and employees with the highest level of professionalism</i>	<i>Engage public and private institutions and other stakeholders for the attainment of social development goal</i>	<i>Establish a strong presence and impact in the international academic community</i>
	5	Share learned knowledge and skills through active engagement in a service- learning project.	✓	✓	✓	✓	✓	✓	✓	✓
(PHED 03) Physical Education 3	1	Participate in moderate to vigorous physical activities (MVPAs) in accordance with national and global recommendations on physical activities for health.	✓	✓	✓	✓	✓	✓	✓	✓
	2	Adapt movement competencies to independent physical activity pursuits that are health- enhancing and personally rewarding.	✓	✓	✓	✓	✓	✓	✓	✓
	3	Monitor progress and evaluate achievement of personal fitness and dietary goals.	✓	✓	✓	✓	✓	✓	✓	✓
	4	Practice empathy, fair play; show respect for differing abilities (i.e., diversity) through interpersonal communication skills and emotional regulation during PA participation.	✓	✓	✓	✓	✓	✓	✓	✓

Course	Course Intended Learning Outcome (CILO)		VISION (2024 - Present)	VISION (2022)	MISSION					
(Course Code and Title)	#	CILO	<i>PUP: A Leading Comprehensive Polytechnic University in Asia</i>	<i>The Country's First Polytechnic U</i>	<i>Provide democratized access to educational opportunities for the holistic development of individuals with global perspective</i>	<i>Offer industry-oriented curricula that produce highly skilled professionals with managerial and technical capabilities and a strong sense of public service for nation building</i>	<i>Embed a culture of research and innovation</i>	<i>Continuously develop faculty and employees with the highest level of professionalism</i>	<i>Engage public and private institutions and other stakeholders for the attainment of social development goal</i>	<i>Establish a strong presence and impact in the international academic community</i>
	5	For sports, apply basic tactics (e.g. defense, offense, transition) in game situations.	✓	✓	✓	✓	✓	✓	✓	✓
(PHED 04) Physical Education 4	1	Participate in moderate to vigorous physical activities (MVPAs) in accordance with national and global recommendations on physical activities for health.	✓	✓	✓	✓	✓	✓	✓	✓
	2	Adapt movement competencies to independent physical activity pursuits that are health- enhancing and personally rewarding.	✓	✓	✓	✓	✓	✓	✓	✓
	3	Monitor progress and evaluate achievement of personal fitness and dietary goals.	✓	✓	✓	✓	✓	✓	✓	✓
	4	Practice empathy, fair play; show respect for differing abilities (i.e., diversity) through interpersonal communication skills and emotional regulation during PA participation.	✓	✓	✓	✓	✓	✓	✓	✓

Course	Course Intended Learning Outcome (CILO)		VISION (2024 - Present)	VISION (2022)	MISSION					
(Course Code and Title)	#	CILO	PUP: A Leading Comprehensive Polytechnic University in Asia	The Country's First Polytechnic U	Provide democratized access to educational opportunities for the holistic development of individuals with global perspective	Offer industry-oriented curricula that produce highly skilled professionals with managerial and technical capabilities and a strong sense of public service for nation building	Embed a culture of research and innovation	Continuously develop faculty and employees with the highest level of professionalism	Engage public and private institutions and other stakeholders for the attainment of social development goal	Establish a strong presence and impact in the international academic community
	5	For sports, apply basic tactics (e.g. defense, offense, transition) in game situations.	✓	✓	✓	✓	✓	✓	✓	✓
D. National Service Training Program										
(NSTP 10013) NSTP 1	1	Develop self and internalize positive Filipino values for harmonious and productive relationship with other workers in community activities.	✓	✓	✓	✓	✓	✓	✓	✓
	2	Enhance leadership, teamwork, decision-making, and conflict-resolution skills while participating actively in student organizations and supporting national efforts for nation-building.	✓	✓	✓	✓	✓	✓	✓	✓
	3	Recognize the importance of environment protection and disaster risk reduction.	✓	✓	✓	✓	✓	✓	✓	✓
	4	Practice good citizenship through involvement in social concerns and in projects designed to attain national development goals.	✓	✓	✓	✓	✓	✓	✓	✓
	5	Use talents and education in field of specialization for the improvement of	✓	✓	✓	✓	✓	✓	✓	✓

Course	Course Intended Learning Outcome (CILO)		VISION (2024 - Present)	VISION (2022)	MISSION					
(Course Code and Title)	#	CILO	<i>PUP: A Leading Comprehensive Polytechnic University in Asia</i>	<i>The Country's First Polytechnic U</i>	<i>Provide democratized access to educational opportunities for the holistic development of individuals with global perspective</i>	<i>Offer industry-oriented curricula that produce highly skilled professionals with managerial and technical capabilities and a strong sense of public service for nation building</i>	<i>Embed a culture of research and innovation</i>	<i>Continuously develop faculty and employees with the highest level of professionalism</i>	<i>Engage public and private institutions and other stakeholders for the attainment of social development goal</i>	<i>Establish a strong presence and impact in the international academic community</i>
		the community, its facilities, and for national service.								
(NSTP 10023) NSTP 2	1	Explain the concept and processes of values education and transformational leadership within the framework of national development	✓	✓	✓	✓	✓	✓	✓	✓
	2	Interpret the basic concepts of Development	✓	✓	✓	✓	✓	✓	✓	✓
	3	Management and its application in civic welfare projects for a given community	✓	✓	✓	✓	✓	✓	✓	✓
	4	Identify develop and/or design appropriate development approaches for implementing projects to benefit a given community	✓	✓	✓	✓	✓	✓	✓	✓
	5	Apply the knowledge and skills acquired by carrying out the projects identified	✓	✓	✓	✓	✓	✓	✓	✓
	6	Evaluate the project/s undertaken by the respective groups in the assigned community	✓	✓	✓	✓	✓	✓	✓	✓
	7	Recognize their vital role in nation-building, promote civic consciousness	✓	✓	✓	✓	✓	✓	✓	✓

Course	Course Intended Learning Outcome (CILO)		VISION (2024 - Present)	VISION (2022)	MISSION					
(Course Code and Title)	#	CILO	<i>PUP: A Leading Comprehensive Polytechnic University in Asia</i>	<i>The Country's First Polytechnic U</i>	<i>Provide democratized access to educational opportunities for the holistic development of individuals with global perspective</i>	<i>Offer industry-oriented curricula that produce highly skilled professionals with managerial and technical capabilities and a strong sense of public service for nation building</i>	<i>Embed a culture of research and innovation</i>	<i>Continuously develop faculty and employees with the highest level of professionalism</i>	<i>Engage public and private institutions and other stakeholders for the attainment of social development goal</i>	<i>Establish a strong presence and impact in the international academic community</i>
		among them. Express ideas effectively in different team building and bonding activities								
	8	Participate actively and develop their physical, moral, spiritual, intellectual and social well-being.	✓	✓	✓	✓	✓	✓	✓	✓

A7. Describe how the requirements of Knowledge Profile, Range of Complex Problem Solving and complex civil engineering Activities have been addressed.

As part of the Student Outcome (SO) plan, CE Project 1 & 2 are designed as the culmination of the knowledge learned from the earlier course subjects in a specialized field of Civil Engineering practice.

This effectively bridges academic preparation and professional engineering practice, necessary to ensure that graduates meet international standards for engineering education.

Knowledge Profile (WK1–WK8)

CE Project 1 focuses on the proposal and design of their research framework, whether descriptive or experimental, compliant with the Knowledge Profile (WK1–WK4) of the Washington Accord as specified by certain examples:

WK1 – Natural Sciences

Students utilize concepts from mathematics and physical sciences in the creation of their research frameworks, such as: application of statistics in data gathering, hydrologic modeling, or traffic analysis, application of calculus for complex structural load analysis

WK2 – Engineering Fundamentals

Students define the statement of the problem and establish the theoretical framework of the study using fundamental engineering concepts.

WK3 – Specialized Engineering Knowledge

With the students conducting CE Project under a specific specialized field, specific design standards are implemented such as (ASTM, ACI, AASHTO, NSCP, DPWH etc.)

WK4 – Investigation and Research Methods

The research process further requires students to engage in systematic investigation, including literature review, data collection, and critical analysis.

CE Project 2 focuses on the design implementation and data analysis of the approved research framework from CE Project 1, compliant with the Knowledge Profile (WK5–WK8) of the Washington Accord, as specified by certain examples:

WK5 – Engineering Design

Design considerations are incorporated as students develop solutions that meet specified needs while accounting for safety, functionality, and efficiency.

WK6 – Engineering Practice

The application of engineering practice is reinforced through the use of applicable codes, standards, and professional judgment.

WK7 – Comprehension of Impacts

Consideration of societal, environmental, and economic impacts is embedded in the evaluation of proposed solutions, ensuring that students appreciate the broader implications of engineering decisions.

WK8 – Ethics and Professional Responsibility

Ethical responsibility is likewise emphasized through adherence to academic integrity, proper attribution of sources, and the prioritization of public safety and welfare.

Range of Complex Problem Solving (WP1–WP7)

CE Project 1 focuses on the proposal and design of their research framework, whether descriptive or experimental, compliant with the Range of Complex Problem Solving (WP1–WP4) of the Washington Accord as specified:

WP1 – Depth of Knowledge Required

Students are required to define and refine engineering problems that do not have straightforward or predetermined solutions. These problems typically involve the integration of multiple sub-disciplines within civil engineering.

WP2 – Conflicting Requirements

Research design requires the compromise and compliance of competing requirements such as cost, safety, and sustainability.

WP3 – Depth of Analysis Required

The analytical process involves the use of appropriate modeling techniques, simulation tools, or experimental methods, often necessitating assumptions that must be justified and validated. Students engage in higher-level analysis, including sensitivity evaluation and interpretation of results, to ensure the robustness of their solutions.

WP4 – Familiarity of Issues

The problems addressed are generally unfamiliar in nature, requiring independent judgment and adaptation of existing knowledge rather than reliance on standard procedures.

CE Project 2 focuses on the design implementation and data analysis of the approved research framework from CE Project 1, compliant with the Range of Complex Problem Solving (WP5–WP7) as specified:

WP5 – Extent of Applicable Codes

The application of engineering codes and standards goes beyond routine use, as students must interpret and modify these guidelines to suit specific conditions.

WP6 – Stakeholder Involvement

Consideration of stakeholders is incorporated through the identification of user needs, regulatory requirements, and potential societal impacts.

WP7 – Interdependence

The interdependence of various system components is also addressed, as solutions often involve interactions among structural, environmental, geotechnical, or transportation elements.

Complex Civil Engineering Activities (EA1–EA5)

EA1 – Range of Resources

Students utilize a wide range of resources, including modern engineering software, laboratory or field data, and relevant technical literature, to support their work.

EA2 – Level of Interaction

The process involves interaction with faculty advisers and, where applicable, external stakeholders, fostering an environment of professional collaboration.

EA3 – Innovation and Creativity

Innovation and creativity are encouraged as students are expected to propose original or context-sensitive solutions to identified problems.

EA4 – Consequences to Society and Environment

The consequences of engineering decisions are carefully evaluated, particularly in terms of their impact on public safety, environmental sustainability, and economic feasibility.

EA5 – Accountability and Responsibility

Students are required to take full responsibility for their work, as evidenced by the preparation of comprehensive technical reports and the defense of their findings before a panel of evaluators. This process ensures accountability and the ability to communicate technical information effectively.

Moreover, the BS Civil Engineering program utilizes assessment instruments and analytic rubrics to determine whether selected course activities, design outputs, research

projects, laboratory tasks, and capstone requirements address **Complex Engineering Problems** and **Complex Engineering Activities** consistent with Washington Accord expectations. These instruments provide a structured basis for evaluating the complexity of student work and for ensuring that learning activities require the application of in-depth engineering knowledge, engineering judgment, creativity, stakeholder consideration, sustainability awareness, and integration of multiple technical and non-technical constraints.

For **Complex Engineering Problems**, the Department uses a complexity analysis matrix and analytic rubric that examine the nature of the problem based on defined characteristics such as depth of knowledge required, range of conflicting requirements, depth of analysis, familiarity of issues, extent of applicable codes, stakeholder involvement, and interdependence of sub-problems. These criteria allow faculty members to determine whether a given problem requires advanced engineering analysis, application of specialized civil engineering knowledge, engagement with standards and codes of practice, and consideration of diverse stakeholder needs. The rubric also supports the identification of problems that require abstract thinking, originality in analysis, and formulation of appropriate engineering models.

For **Complex Engineering Activities**, the Department uses a corresponding complexity analysis matrix and analytic rubric that assess the extent to which an activity involves diverse resources, interactions among technical and non-technical issues, innovation, consequences to society and the environment, and application of principles beyond previous experiences. This ensures that major activities are not limited to routine exercises but require students to integrate engineering principles, research-based knowledge, appropriate technologies, and sustainable decision-making. The criteria also guide faculty members in evaluating whether students are exposed to activities that involve realistic constraints, multidisciplinary considerations, and potential impacts on communities, infrastructure systems, and the environment ([Attachment C5-19 – Complex Engineering Activities](#)).

The rubrics use defined performance scales ranging from low to very high complexity, allowing faculty evaluators to classify the level of complexity demonstrated in student outputs and activities. This provides consistency in assessment and ensures that student work is evaluated using transparent, measurable, and outcomes-aligned criteria. The scoring process supports faculty judgment in determining whether the activity or problem is appropriate for the expected level of the course, particularly in advanced professional courses, research courses, internship, and capstone or major design experiences.

These rubrics are used alongside course-level assessment tools aligned with the **Course Intended Learning Outcomes (CILOs)**, **Student Outcomes (SOs)**, and performance indicators. Through this alignment, the Department is able to verify that students are not only completing required outputs but are also demonstrating the level of complexity, analysis, professional responsibility, and engineering judgment expected of graduating civil engineering students. Assessment results generated from these rubrics provide direct evidence of student capability in addressing complex civil engineering problems and undertaking complex engineering activities.

The use of these rubrics also supports the program's **Continuous Quality Improvement (CQI)** process. Results from the assessment of complex problems and activities are reviewed to identify whether students require stronger preparation in areas such as problem formulation, design integration, stakeholder analysis, sustainability considerations, application of codes and standards, technical communication, and use of modern engineering tools. Findings from these assessments may inform revisions in Teaching-Learning Activities (TLAs), Assessment Tasks (ATs), laboratory exercises, design projects, research requirements, and capstone implementation.

Through the systematic use of complexity analysis matrices and analytic rubrics, the BS Civil Engineering program ensures that students are provided with appropriate opportunities to engage with problems and activities that reflect the nature of professional civil engineering practice. These instruments strengthen the evidence-based assessment of student learning and demonstrate that the curriculum supports the development of graduates capable of addressing complex engineering problems and performing complex engineering activities in accordance with accepted professional and accreditation standards.

[\(Attachment C5-21 – Complexity Analysis Matrix of Complex Engineering Problems\)](#)

[\(Attachment C5-21 – Complexity Analysis Matrix of Complex Engineering Activities\)](#)

[\(Attachment C5-22 – Analytic Rubric for Complexity of Engineering Problems\)](#)

[\(Attachment C5-23 – Analytic Rubric for Complexity of Engineering Activities\)](#)

A8. Describe the major design experience that prepares students for engineering practice. Describe how this experience is based upon the knowledge and skills acquired in earlier coursework, specialized courses, and incorporate appropriate engineering standards and multiple design constraints.

For both 2018 and 2022 BSCE Curricular programs, CE Project 1 and CE Project 2 are designed to be a culminating major engineering design experience on the last two semesters which requires students to integrate knowledge acquired from earlier course subjects in mathematics, natural sciences, basic engineering sciences, allied courses, professional and specialized civil engineering courses.

The research project is intentionally structured to build upon and integrate the knowledge, skills, and professional competencies acquired throughout the curriculum. It provides students with an opportunity to demonstrate the attainment of the BSCE Student Outcomes through the application of engineering principles to complex, real-world civil engineering problems.

Students are required to synthesize concepts from these disciplines to investigate engineering problems, develop technically sound solutions, and evaluate project outcomes using appropriate engineering methodologies. The research project experience simulates professional engineering practice by requiring students to address complex civil engineering problems while considering engineering design codes and standards, evidence-based decision-making supported by data and engineering analysis, use of modern engineering software and technologies, safety and risk mitigation measures, environmental and sustainability assessments, economic and financial feasibility, ethical and professional responsibilities, and social, cultural, and community impact.

Students with fourth year standing are allowed to take CE Project 1, where the final requirement is the proposal and design of their research framework, whether descriptive or experimental, of their research study. CE Project 2 then focuses on CE Project 2 focuses on the design implementation and data analysis of the approved research framework from CE Project 1.

With Civil Engineering being a multi-specialized field, students are allowed to conduct research studies on the following specialized fields:

- Construction Engineering and Management
- Geotechnical Engineering

- Structural Engineering
- Transportation Engineering
- Water Resources Engineering

Each design stage and addressed Student Outcome of each step of the culminating CE Project 1 and 2 are explained in the following table:

Course Subject	Research/ Design Stage	Description of Activity	Student Outcome (SO) Addressed	Demonstration of Student Outcome
CE Project 1	Identification of Engineering Problem	Identification of a real-world civil engineering issue requiring investigation or design intervention.	SO2. Conduct investigations of complex civil engineering problems using research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.	Students define engineering problems and establish technical requirements based on actual societal and industry needs.
	Review of Literature, Codes, and Standards	Examination of previous studies, technical references, engineering codes, and professional standards.		Students independently obtain, evaluate, and synthesize technical information from diverse sources.
	Formulation of Research Objectives and Scope	Development of measurable objectives, assumptions, constraints, and expected outcomes.		Students formulate engineering questions and establish systematic approaches to solving them.
	Planning of Research Methodology and Experimental Design	Selection of procedures, testing protocols, analytical methods, and computational approaches.	SO3. Design solutions for complex civil engineering problems and design systems, components, or processes to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, cultural, and sustainability, in accordance with standards	Students design an investigation or engineering approach that addresses realistic project constraints and objectives.
	Proposal Defense	Students formally present and defend the technical merit, feasibility, ethical considerations, safety implications, sustainability aspects, methodology, and compliance with engineering standards before a panel of evaluators. Students respond to	SO7. Communicate effectively on complex civil engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.	Students effectively communicate technical concepts and defend engineering decisions.

Course Subject	Research/ Design Stage	Description of Activity	Student Outcome (SO) Addressed	Demonstration of Student Outcome
		technical questions and incorporate recommendations for improvement.		
CE Project 2	Data Collection, Fieldwork, and Laboratory Testing	Gathering engineering data through field surveys, site investigations, experiments, and laboratory testing.	SO2. Conduct investigations of complex civil engineering problems using research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.	Students perform experiments, collect data, and evaluate the validity and reliability of results.
	Application of Engineering and Computational Tools	Utilization of software, simulation tools, GIS, modeling applications, and statistical analyses.	SO11. Create, select, and apply appropriate techniques, skills, resources, and modern engineering and IT tools, including prediction and modeling, to complex civil engineering problems with an understanding of the limitations	Students apply appropriate engineering and computing tools while understanding their limitations.
	Engineering Analysis and Interpretation of Results	Analysis of collected data using engineering principles and mathematical models.	SO2. Conduct investigations of complex civil engineering problems using research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.	Students apply engineering science, mathematics, and technical knowledge to interpret findings and solve engineering problems.
	Development of Alternative Engineering Solutions	Generation of multiple feasible design alternatives or engineering interventions.	SO3. Design solutions for complex civil engineering problems and design systems, components, or processes to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, cultural, and sustainability, in accordance with standards	Students develop innovative and technically viable alternatives considering engineering requirements.
	Evaluation of Sustainability, Safety, and Environmental Impacts	Assessment of environmental, health, safety, and sustainability implications of proposed solutions.	SO8. Understand and evaluate the sustainability and impact of professional engineering work in the solution of complex civil engineering problems in a	Students incorporate sustainability, safety, environmental stewardship, and public welfare into engineering decisions.

Course Subject	Research/ Design Stage	Description of Activity	Student Outcome (SO) Addressed	Demonstration of Student Outcome
	Economic and Constructability Assessment	Evaluation of cost-effectiveness, resource utilization, and implementation feasibility.	global, economic, societal, and environmental context.	Students consider economic, operational, and practical constraints in engineering design.
	Consideration of Ethical and Professional Responsibilities	Analysis of ethical implications, legal requirements, and professional obligations.	SO6. Apply ethical principles and commit to professional ethics, and responsibilities and norms of engineering practice.	Students demonstrate understanding of professional ethics and responsibilities in engineering practice.
	Integration of Applicable Engineering Codes and Standards	Verification that designs comply with local and international engineering standards.	SO8. Understand and evaluate the sustainability and impact of professional engineering work in the solution of complex civil engineering problems in a global, economic, societal, and environmental context.	Students develop designs that satisfy regulatory, technical, and professional requirements.
	Team Coordination and Project Management Activities	Planning, scheduling, delegation of tasks, and resource management among project members.	SO12. Demonstrate knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.	Students function effectively as members or leaders of multidisciplinary teams.
	Preparation of Research Manuscript	Development of a comprehensive technical document detailing methods, analyses, and recommendations.	SO7. Communicate effectively on complex civil engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.	Students communicate engineering information clearly and professionally through written reports.
	Oral Presentation and Research Defense	Formal presentation and defense of findings before faculty, industry experts, and stakeholders.		Students effectively communicate technical concepts and defend engineering decisions.
	Assessment of Societal, Cultural, and Global Impacts	Evaluation of how engineering solutions affect communities, infrastructure, and stakeholders.	SO8. Understand and evaluate the sustainability and impact of professional engineering work in the solution of complex civil engineering problems in a global, economic, societal, and environmental context.	Students recognize the broader societal consequences of engineering decisions.
	Reflection, Conclusions, and Recommendations	Evaluation of project outcomes, limitations, and opportunities for future work.	SO2. Conduct investigations of complex civil engineering problems using research-based knowledge and research methods including design of experiments, analysis and interpretation of data,	Students demonstrate reflective practice and identify areas for continuous professional development.
	Completion of the Research Project	Integration of all technical,		Students demonstrate readiness for

Course Subject	Research/ Design Stage	Description of Activity	Student Outcome (SO) Addressed	Demonstration of Student Outcome
		professional, and societal considerations into a final engineering solution.	and synthesis of information to provide valid conclusions.	professional engineering practice through a comprehensive major engineering design experience.

Through this research project, students demonstrate readiness for professional practice by producing technically sound, socially responsible, economically viable, and sustainable engineering solutions.

In addition to this, research studies on each of the CE specializations can address on or more of the local and foreign codes and standards listed below:

Civil Engineering Specialization	Common Local Codes and Standards	Common International Codes and Standards
Construction Engineering and Management	National Building Code of the Philippines (PD 1096) Philippine Contractors Accreditation Board (PCAB) DPWH Standard Specifications (Blue Book) Occupational Safety and Health Standards (DOLE)	Project Management Institute PMBOK, International Organization for Standardization 9001, ISO 45001, FIDIC Conditions of Contract
Geotechnical Engineering	NSCP Geotechnical Provisions, DPWH Design Guidelines DENR Regulations	American Association of State Highway and Transportation Officials (AASHTO) American Society for Testing and Materials (ASTM) International Standards
Structural Engineering	National Structural Code of the Philippines (NSCP), National Building Code of the Philippines (PD 1096), DPWH Standards	American Concrete Institute (ACI 318), American Institute of Steel Construction (AISC) Specifications Eurocode EN 1990–1999
Transportation Engineering	DPWH Highway Safety Design Standards Philippine Road Safety Design Manual	American Association of State Highway and Transportation Officials (AASHTO) Green Book, Transportation Research Board HCM, MUTCD, Austroads Standards
Water Resources Engineering	National Plumbing Code of the Philippines, DENR Water Quality Guidelines, NWRB Regulations, DPWH Hydraulic Design Criteria	American Water Works Association Standards, Water Environment Federation Manuals, American Society of Civil Engineers (ASCE) Hydraulic Standards

The use of these standards ensures that students become familiar with industry practices and develop the ability to design solutions that are technically valid, safe, compliant, and implementable.

The Civil Engineering department also implement a rubric for evaluation of research paper anchored to the Student Outcomes of the BSCE program. It ensures that the proposed research or design project is technically sound, feasible, ethical, safe, sustainable, and aligned with applicable engineering standards before significant resources are committed to implementation. Through this activity, students demonstrate their ability to communicate engineering concepts effectively, justify design decisions using engineering principles, address realistic constraints, and respond professionally to technical critiques. The said rubric is shown on the table on the following page:

Table 5-8.1. Rubrics for Evaluation of CE Project 1 and 2 Research Paper

STUDENT OUTCOMES (SO)		PERFORMANCE INDICATOR (PI)		COURSE INTENDED LEARNING OUTCOMES (CILO)		CRITERIA	WEIGHT	PERFORMANCE LEVEL				SCORE				WEIGHTED SCORE
								INSUFFICIENT (1)	DEVELOPING (2)	PROFICIENT (3)	ADVANCED (4)	1	2	3	4	
SO2	Conduct investigations of complex civil engineering problems using research-based knowledge and research methods including design of experiments, analysis, and interpretation of data, and synthesis of the information to provide valid conclusions.	4	Interpret the collected data from laboratory experiment and field works and draw out conclusions using appropriate methods in accordance to engineering principles	1. Implement the approved research methodology by executing the planned procedures and documenting deviations, limitations, and quality controls. 2. Collect and organize research data using defined protocols and prepare the dataset for analysis with clear traceability.	Design Completion	10.00%	Does not complete any process design	Completes designing process, but needs revisions.	Completes designing process according to the task description.	Completes designing process according to the task description; develops design strategies						
SO3	Design solutions for complex civil engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.	2	Create a design project/ prototype/ system/ component/ process that meets the relevant design codes and standards		Constraints	5.00%	No consideration of economics, safety, and environment.	Includes only minor or cursory consideration of economic, safety, and environmental constraints.	Develops a solution that includes economic, safety, environmental and other realistic constraints.	Develops and applies solutions that includes economic, safety, environmental and other realistic constraints.						

STUDENT OUTCOMES (SO)		PERFORMANCE INDICATOR (PI)		COURSE INTENDED LEARNING OUTCOMES (CILO)		CRITERIA	WEIGHT	PERFORMANCE LEVEL				SCORE				WEIGHTED SCORE
								INSUFFICIENT (1)	DEVELOPING (2)	PROFICIENT (3)	ADVANCED (4)	1	2	3	4	
SO5	An ability to identify, formulate, and solve engineering problems	1	identify complex civil engineering problems and identify the key issues, factors, and concepts affecting the problem;	4. Evaluate the research outcomes against the research questions and objectives, discuss validity and limitations, and present recommendations for improvement and future work in a final technical report and oral presentation.		Identification of Problems	10.00%	Insufficient identification of problems, inadequate objectives.	Partial identification of problems; lack of specific does impair solution of design.	Adequate identification of problem; any lack of specific does not impair solution or design.	Clear and complete identification of design goals and objectives.					
		2	analyze and formulate solutions to complex civil engineering problems;			Strategy	5.00%	Has no coherent strategies for problem solving.	Has some strategies for problem-solving, but does not apply them consistently.	Formulates strategies for solving problems.	Formulates a number of strategies and apply to solve problems.					
SO6	Identify, formulate, research literature and analyze complex civil engineering problems reaching substantiated conclusions using the first principles of mathematics, natural sciences and engineering sciences.	2	Demonstrate professionalism by complying to relevant codes and standards			Ethics	5.00%	There is evidence of plagiarism.	The team did not quote all the source of information that they used.	The team quoted nearby all the source of information that they used.	Avoid plagiarism, does not use information without giving credit to the appropriate source.					
SO7	Communicate effectively on complex civil engineering activities with the engineering community and with society at large, such	1	use effective written communication by writing grammatically correct and logical reports			Organization & Content of Manuscript	10.00%	Sequence of information is difficult to follow; No grasp of information. Clearly no	Work is hard to follow as there is very little continuity; Uncomfortable with	Information is presented in a logical manner, which is easily followed; At	Information is presented in a logical, interesting way, which is easy to					

STUDENT OUTCOMES (SO)		PERFORMANCE INDICATOR (PI)		COURSE INTENDED LEARNING OUTCOMES (CILO)		CRITERIA	WEIGHT	PERFORMANCE LEVEL				SCORE				WEIGHTED SCORE
								INSUFFICIENT (1)	DEVELOPING (2)	PROFICIENT (3)	ADVANCED (4)	1	2	3	4	
	as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions		using appropriate graphics and technical style.					<i>knowledge of subject matter.</i>	<i>content. Only basic concepts are demonstrated and interpreted.</i>	<i>ease with content and able to elaborate and explain to some degree.</i>	<i>follow; Demonstration of full knowledge of the subject with explanations and elaboration.</i>					
						Format & Aesthetics of Manuscript	10.00%	<i>Work is illegible, format changes throughout.</i>	<i>Mostly consistent format. Figures and tables are legible, but not convincing.</i>	<i>Format is generally consistent including heading styles and captions. Figures and tables are neatly done and provide intended information.</i>	<i>Format is consistent Throughout including heading styles and captions. Figures and tables are presented logically and reinforce the text.</i>					
SO8	Understand and evaluate the sustainability and impact of professional engineering work in the solution of complex civil	1	explain the impacts of civil engineering solutions in a global, economic, environmental, and societal context			Literature Search	10.00%	<i>Information is gathered from a single source.</i>	<i>Information is gathered from a limited number of sources.</i>	<i>Information is gathered from multiple sources.</i>	<i>Information is gathered from multiple, research-based sources.</i>					

STUDENT OUTCOMES (SO)		PERFORMANCE INDICATOR (PI)		COURSE INTENDED LEARNING OUTCOMES (CILO)		CRITERIA	WEIGHT	PERFORMANCE LEVEL				SCORE				WEIGHTED SCORE
								INSUFFICIENT (1)	DEVELOPING (2)	PROFICIENT (3)	ADVANCED (4)	1	2	3	4	
	engineering problems in societal and environmental contexts.	2	establish the significance of the solutions in terms of beneficiaries			Significance	10.00%	Information is gathered from a single source.	Information is gathered from a limited number of sources.	Information is gathered from multiple sources.	Information is gathered from multiple, research-based sources.					
SO9	Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.	1	The student can demonstrate self-directed learning through acquired new knowledge from various modes and strategies (research, attendance in conferences and exhibits, site visits)			Current Issues	5.00%	Has no clue about issues in the world.	Has some knowledge of current issues.	Has knowledge of current issues in the engineering discipline and in society.	Has broad and in depth knowledge of current issues in the engineering discipline and in society.					
SO11	Create, select and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modelling, to complex civil engineering problems with an understanding of the limitations.	3	Application of the techniques, resources, skills, and modern engineering tools to address complex civil engineering problem			Skills	10.00%	Is not able to interpret and understand information from a variety of techniques, skills and tools.	Is somewhat able to interpret and understand information from a variety of techniques, skills and tools.	Is able to interpret and understand information from a variety of techniques, skills and tools.	Is able to fully interpret and understand information from a variety of techniques, skills and tools.					

A9. Describe the industry-academe linkage, such as industry immersion to satisfy curricular requirements specifically addressed by either the general or program criteria. Describe the academic component of this experience and how the program evaluates this.

The Government-Industry Advisory Board (GIAB) has an instrumental role in the Industry-Academe Linkage of the BS Civil Engineering program. As reflected in Criterion 8, activities with industry linkage and involvements are well-based on the Program Educational Objectives that were formed together with the committee. The GIAB meets periodically to evaluate the program educational objectives, student outcomes and curricula to ensure that these are relevant and attuned with the current societal and professional requirements and developments in the industry ([Attachment C5-24 – Minutes of the Government Industry Advisory Board Meetings](#)).

The following are the objectives of the BSCE linkage program:

1. Upgrading of the status of the civil engineering education and its practice so that industries will be able to draw top caliber civil engineers into various industrial fields.
2. Incorporation into the curricula of the BSCE program and practice applicable to the industry.
3. Development of mechanism for the exchange of technical knowledge of expertise between the civil engineering program and the industry as well as with other academic institutions for their mutual benefit; and
4. Strengthen the efforts for potential collaborative research works between the academe and the industry.

Table 5-9.1. shows the relationship among the CE's Industry-Academe Linkage Objectives to its Program Educational Objectives.

Table 5-9.1. Relationship of Industry-Linkage Program Objectives to the Program Educational Objectives

Industry-Academe Linkage Program Objectives	Program Educational Objectives		
	PEO 1. Demonstrating Engineering Profession <i>Employed and/or practiced in the field of Civil Engineering, such as but not limited to Construction Engineering and Management, Geotechnical Engineering, Structural Engineering, Transportation Engineering, and Water Resources Engineering, or other related fields.</i>	PEO 2. Social and Ethical Responsibility <i>Committed members of professional and other allied organizations engaging in community development or nation-building.</i>	PEO 3. Lifelong Learning <i>Involved in continuous training and development in current trends and advancements in their fields of specialization.</i>
1. Upgrading the status of civil engineering education and its practice so that industries will be able to draw top-caliber civil engineers into various industrial fields.	✓		
2. Incorporation into the curricula of the CE program and practice applicable to the industry.	✓		
3. Development of a mechanism for the exchange of technical knowledge and expertise between the civil engineering program and the industry, as well as with other academic institutions, for their mutual benefit; and	✓		✓
4. Strengthen the efforts for potential collaborative research works between the academe and the industry.	✓	✓	

Both the 2018 and 2022 BSCE curriculum specifies 2 on-the-job training courses to be completed in two different semesters. OJT 1 aims to be actual work experience in the broad civil engineering industry, while OJT 2 aims to provide actual work experience within a field of

specialization of civil engineering practice. The details of the curricular requirements are summarized in Table 5-9.2.

Table 5-9.2. Comparative Summary of the Curricular Requirements for Industry Immersion among the 2018 Curriculum, 2022 Curriculum and the Minimum CHED Requirements

Minimum Requirements per CHED Memo No.92 s.2017	Credit Units	2018 BSCE Curriculum			2022 BSCE Curriculum		
		Course Code and Title	Credit Units	Semester and Year Level	Course Code and Title	Credit Units	Semester and Year Level
On-The-Job Training (minimum of 240 hours)	3	CIEN 40012 On-The-Job Training 1 (300 hours)	2	2 nd Year, Summer/Third Semester	CIEN 314 On-The-Job Training 1 (300 hours)	3	3 rd Year, Summer/Third Semester
		CIEN 40022 On The Job Training 2 (300 hours)	2	3 rd Year, Summer/Third Semester	CIEN 412 On The Job Training 2 (200 hours)	1	4 th Year, Second Semester

One of the major discussions and concerns during the 2022 BSCE Curriculum Revision is the evaluation and changes to the OJT course requirements and implementation. Aside from the Government-Industry Advisory Board, comments and suggestions were also taken from stakeholder consultations (students, student organizations and alumni), faculty members, Curriculum Development Committee (CDC) members, and Curriculum Evaluation Committee (CEC) members. The approved changes are listed and explained as follows:

1. The implementation of separate OJT 1 and OJT 2 courses on different year levels and semesters are retained. With Civil Engineering being a multi-specialized field, it is imperative that OJT1 will serve as the student's immersion with the general civil engineering field, while OJT2 will be the student's immersion with one of the civil engineering specializations: Construction Engineering and Management, Geotechnical Engineering, Structural Engineering, Transportation Engineering, and Water Resources Engineering
2. Semestral Repositioning of OJT1 and OJT2 courses. One of the most notable concerns for the 2018 BSCE curriculum during the revision is the readiness of outgoing 2nd year students for OJT 1. It was agreed that those outgoing 2nd year students have not yet taken most Civil Engineering professional courses that will provide technical knowledge for their industry immersion such as Building Systems Design and Structural Theory. Additionally for the 2018 curriculum, incoming 4th year students who are designed to take the OJT2 within a Civil Engineering specialization are yet to take their specialized examination. OJT 1, which is an industry immersion for the general

Civil Engineering field, was repositioned from 2nd Year, Summer/Third Semester to 3rd Year, Summer/Third Semester. OJT, which is an industry immersion for a Civil Engineering specialization, was repositioned from 3rd Year, Summer/Third Semester to 4th Year, Second Semester

3. *Reduction of minimum required hours for OJT2.* With the repositioning of OJT 2 4th Year, Second Semester, students are required to take on-the-job training during regular semestral offering along major and specialized courses such as CE Project 2 and CE Integration 3.

To allow the students a healthier study and OJT coursework for the 4th Year, 2nd Semester, OJT2 was reduced from 300 to 200 minimum required rendered hours.

The On-the-Job Training Advisers assist students in identifying OJT opportunities and in applying for available positions. The student can choose from a wide selection of engineering companies within and outside of Metro Manila, where PUP – Sta Mesa has an OJT linkage. This is accomplished upon the recommendation of the department and college dean and the approval of the company officials. Trainings may be related and not constricted to field operations and maintenance of the entities, production/quality assurance/product research and development and instrumentation/automation.

At the end of the training, the student will be evaluated by his/her immediate supervisor at the workplace and their OJT adviser. The OJT Adviser is the person in charge for the whole duration of the training who will regularly monitor the progress of the trainee and will submit the accomplishment reports of the student on the department for monitoring.

Students are then graded using the following rubrics aligned with the Student Outcomes (SO) and Course Intended Learning Outcomes (CILO) for both their actual OJT performance and their submitted OJT Portfolio.

The BS Civil Engineering program also maintains industry–academe linkages through partnerships with **Host Training Establishments (HTEs)** that provide students with opportunities for industry immersion, on-the-job training, field exposure, and professional practice-based learning. These linkages support the curricular requirement for internship and enable students to apply civil engineering theories, methods, tools, and professional standards in actual workplace settings. The Department coordinates with partner agencies, construction firms, engineering consultancies, government offices, and other relevant institutions to ensure

that student placements are aligned with the competencies expected of BS Civil Engineering students.

The Department maintains a record of its HTE partners to monitor the availability, suitability, and relevance of training opportunities provided to students. The HTE tracking mechanism includes information on partner institutions, type of industry or agency, specialization area, number of students accommodated, deployment period, status of partnership, and feedback received from students and supervisors. This allows the Department to determine whether existing partners continue to provide appropriate training environments and whether additional partnerships are needed to support student placement in areas such as structural engineering, construction engineering and management, transportation engineering, geotechnical engineering, water resources engineering, and allied fields ([Attachment C5-25 – Host Training Establishment Distribution and Visits](#)).

The academic component of the industry immersion experience is implemented through internship orientation, approval of deployment, monitoring of student performance, submission of required reports, and evaluation of student outputs. Prior to deployment, students are oriented on internship requirements, workplace conduct, safety protocols, documentation procedures, expected learning outcomes, and professional responsibilities. During the internship period, students are expected to perform tasks relevant to civil engineering practice, document their activities, reflect on their learning experiences, and relate workplace assignments to the applicable Course Intended Learning Outcomes (CILOs) and Student Outcomes (SOs).

The Department also conducts **HTE visits** to monitor student interns and verify the adequacy of the training environment. These visits provide an opportunity for faculty coordinators or authorized Department representatives to confer with HTE supervisors, assess student performance, identify concerns encountered during deployment, and determine whether the assigned tasks are consistent with the intended academic objectives of the internship. HTE visits also strengthen communication between the Department and partner institutions and provide feedback for improving future internship implementation.

Student performance during internship is evaluated using multiple sources of evidence, including HTE supervisor evaluations, faculty monitoring reports, internship journals, accomplishment reports, attendance records, final internship reports, and oral or written reflections. These assessment tools allow the Department to evaluate students' technical competence, communication skills, teamwork, professionalism, ethical conduct,

adaptability, and ability to apply civil engineering knowledge in actual practice. The evaluation results are reviewed to determine whether students have attained the intended learning outcomes of the internship and whether the HTE provided an appropriate learning environment.

Feedback from HTE supervisors and students is also used as input for Continuous Quality Improvement (CQI). Identified concerns, such as mismatch between student assignments and specialization, limited exposure to technical work, documentation issues, or the need for additional pre-deployment preparation, are considered in revising internship guidelines, improving orientation activities, strengthening partner screening, and refining monitoring procedures. Through this process, industry immersion becomes not only a curricular requirement but also an evidence-based mechanism for assessing student readiness for civil engineering practice.

Table 5-9.3. On-The-Job Training Performance Evaluation Rubrics for On-The-Job Training 1/ On-The-Job Training 2

STUDENT OUTCOMES (SO)		PERFORMANCE INDICATOR (PI)		COURSE INTENDED LEARNING OUTCOMES (CILO)		CRITERIA	WEIGHT	<i>Exemplary (4)</i>	<i>Capable (3)</i>	<i>Developing (2)</i>	<i>Beginning (1)</i>	SCORE	WEIGHTED SCORE
4	Function effectively as an individual and as a member or leader in diverse teams and in multi-disciplinary settings.	4.1	Participate, express ideas freely, and execute member tasks towards attainment of the objectives.	CILO 1	Contribute effectively to OJT team activities by proposing relevant ideas and accomplishing assigned responsibilities in accordance with agreed objectives and work plans.	COLLABORATIVE PARTICIPATION AND TASK ENGAGEMENT <i>(Assesses the trainee's ability to participate productively in team activities, propose relevant ideas, and complete assigned responsibilities in support of agreed workplace objectives and team plans.)</i>	10.00%	<i>Consistently demonstrates active and constructive participation in team activities, proposes relevant and well-considered ideas, and accomplishes assigned responsibilities beyond expected standards in support of agreed objectives and work plans.</i>	<i>Demonstrates active participation in team activities, contributes relevant ideas, and accomplishes assigned responsibilities in accordance with agreed objectives and work plans.</i>	<i>Participates in team activities when prompted and accomplishes assigned responsibilities with occasional guidance, but contributions to objectives and work plans require improvement.</i>	<i>Shows limited participation in team activities, rarely contributes ideas, and fails to accomplish assigned responsibilities in accordance with agreed objectives and work plans.</i>		

STUDENT OUTCOMES (SO)		PERFORMANCE INDICATOR (PI)		COURSE INTENDED LEARNING OUTCOMES (CILO)		CRITERIA	WEIGHT	<i>Exemplary (4)</i>	<i>Capable (3)</i>	<i>Developing (2)</i>	<i>Beginning (1)</i>	SCORE	WEIGHTED SCORE
		4.2	Integrate inputs from team members and make decisions based on objective criteria.	CILO 2	Integrate inputs from team members and make OJT task decisions using explicit, objective, and defensible criteria.	EVIDENCE-BASED TEAM DECISION-MAKING (Assesses the trainee's ability to consider team inputs, evaluate alternatives, and make sound OJT-related decisions using clear, objective, and defensible bases.)	10.00%	Consistently integrates relevant inputs from team members, evaluates alternatives carefully, and makes well-justified decisions based on explicit, objective, and defensible criteria.	Integrates inputs from team members and makes appropriate OJT task decisions based on objective and reasonable criteria.	Considers some team inputs but requires guidance in applying explicit, objective, and defensible criteria in making OJT task decisions.	Rarely considers team inputs and makes decisions without clear, objective, or defensible criteria.		
6	Apply ethical principles and commit to professional ethics, responsibilities, and norms of engineering practice.	6.1	Demonstrate ethical and professional behavior during practice.	CILO 3	Demonstrate ethical and professional conduct throughout OJT consistent with the responsibilities, norms, and standards of engineering practice.	ETHICAL PRACTICE AND PROFESSIONAL ACCOUNTABILITY (Assesses the trainee's observance of ethical conduct, professional responsibility, accountability, workplace discipline, and compliance with applicable engineering norms, policies, codes, and standards.)	15.00%	Consistently demonstrates exemplary ethical behavior, professionalism, accountability, respect, and compliance with workplace policies, responsibilities, norms, codes, and standards	Demonstrates ethical and professional behavior and generally complies with workplace policies, responsibilities, norms, codes, and standards of engineering practice.	Demonstrates acceptable professional behavior but requires reminders or guidance in consistently observing ethical responsibilities, workplace	Demonstrates behavior inconsistent with ethical and professional expectations and fails to comply with responsibilities, workplace norms, codes, or standards.		
		6.2	Demonstrate professionalism by complying with relevant codes and standards.										

STUDENT OUTCOMES (SO)		PERFORMANCE INDICATOR (PI)		COURSE INTENDED LEARNING OUTCOMES (CILO)		CRITERIA	WEIGHT	<i>Exemplary (4)</i>	<i>Capable (3)</i>	<i>Developing (2)</i>	<i>Beginning (1)</i>	SCORE	WEIGHTED SCORE
								<i>of engineering practice.</i>		<i>norms, codes, or standards.</i>			
7	Communicate effectively on complex civil engineering activities with the engineering community and with society at large.	7.1	Use effective written communication by writing grammatically correct and logical reports using appropriate graphics and technical style.	CILO 4	Prepare OJT documentation that is grammatically correct, logically structured, and supported by appropriate graphics and professional technical style.	TECHNICAL WRITING AND PROFESSIONAL DOCUMENTATION (Assesses the trainee's ability to prepare clear, organized, grammatically correct, and technically appropriate OJT documentation supported by relevant visuals, data, or graphics.)	10.00%	<i>Prepares highly organized, grammatically correct, and technically sound OJT documentation with appropriate graphics, clear structure, accurate content, and professional technical style.</i>	<i>Prepares clear and logical OJT documentation with minor errors in grammar, organization, graphics, or technical style.</i>	<i>Prepares OJT documentation with noticeable errors in grammar, structure, graphics, or technical style that require revision.</i>	<i>Prepares incomplete, unclear, poorly organized, or grammatically weak documentation that does not meet professional technical reporting standards.</i>		
		7.2	Use effective oral communication by presenting and defending ideas with sound reasoning and in an articulate manner.	CILO 5	Present and defend OJT outputs with sound technical reasoning and clear, professional delivery in workplace settings.	ORAL PRESENTATION AND TECHNICAL DEFENSE (Assesses the trainee's ability to communicate OJT outputs orally, explain technical ideas clearly, justify decisions, and respond to questions with sound reasoning and professional delivery.)	10.00%	<i>Presents and defends OJT outputs with excellent clarity, confidence, technical reasoning, and professional delivery; responds to questions</i>	<i>Presents and defends OJT outputs clearly and professionally, with sound reasoning and generally appropriate responses to questions.</i>	<i>Presents OJT outputs with limited clarity or confidence and requires guidance in defending ideas using sound technical reasoning.</i>	<i>Has difficulty presenting or defending OJT outputs and provides unclear, unsupported, or unprofessional responses.</i>		

STUDENT OUTCOMES (SO)		PERFORMANCE INDICATOR (PI)		COURSE INTENDED LEARNING OUTCOMES (CILO)		CRITERIA	WEIGHT	<i>Exemplary (4)</i>	<i>Capable (3)</i>	<i>Developing (2)</i>	<i>Beginning (1)</i>	SCORE	WEIGHTED SCORE
								<i>accurately and articulately.</i>					
8	Understand and evaluate the sustainability and impact of professional engineering work in the solution of complex civil engineering problems in a global, economic, societal, and environmental context.	8.1	Explain the impacts of civil engineering solutions in a global, economic, environmental, and societal context.	CILO 6	Explain the global, economic, environmental, and societal implications of civil engineering solutions encountered during OJT.	CONTEXTUAL EVALUATION OF ENGINEERING IMPACTS <i>(Assesses the trainee's ability to recognize and explain the broader implications of civil engineering work in relation to global, economic, environmental, and societal considerations.)</i>	10.00%	<i>Provides comprehensive and well-reasoned explanations of the global, economic, environmental, and societal implications of civil engineering solutions observed during OJT.</i>	<i>Explains the major global, economic, environmental, and societal implications of civil engineering solutions encountered during OJT.</i>	<i>Identifies some implications of civil engineering solutions but explanations are limited, general, or require further development.</i>	<i>Shows limited understanding of the broader implications of civil engineering solutions encountered during OJT.</i>		

STUDENT OUTCOMES (SO)		PERFORMANCE INDICATOR (PI)		COURSE INTENDED LEARNING OUTCOMES (CILO)		CRITERIA	WEIGHT	<i>Exemplary (4)</i>	<i>Capable (3)</i>	<i>Developing (2)</i>	<i>Beginning (1)</i>	SCORE	WEIGHTED SCORE
		8.2	Establish the significance of the solutions in terms of beneficiaries.	CILO 7	Establish the significance of observed or proposed solutions by identifying beneficiaries and articulating value to relevant stakeholders.	STAKEHOLDER VALUE AND BENEFICIARY ANALYSIS (Assesses the trainee's ability to identify beneficiaries and explain the relevance, value, and significance of engineering solutions to clients, communities, end users, or other stakeholders.)	10.00%	Clearly identifies beneficiaries and provides a well-supported explanation of the value, relevance, and significance of observed or proposed solutions to stakeholders.	Identifies beneficiaries and explains the value and significance of observed or proposed solutions to relevant stakeholders.	Identifies some beneficiaries but provides limited explanation of the value or significance of the solutions to stakeholders.	Fails to identify relevant beneficiaries or explain the significance of the solutions to stakeholders.		
12	Demonstrate knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.	12.1	Use appropriate tools and theories to manage multidisciplinary projects in a team.	CILO 8	Apply appropriate tools and project management concepts to coordinate and manage multidisciplinary tasks within a team environment.	PROJECT COORDINATION AND MANAGEMENT APPLICATION (Assesses the trainee's ability to apply appropriate engineering management tools, theories, and project management concepts in coordinating multidisciplinary tasks within a team setting.)	12.50%	Effectively applies appropriate tools, theories, and project management concepts to coordinate and manage multidisciplinary tasks with minimal supervision and strong team awareness.	Applies appropriate tools and project management concepts to support coordination and management of multidisciplinary tasks within a team.	Applies some tools or project management concepts but requires guidance in coordinating and managing multidisciplinary tasks.	Shows limited ability to apply tools or project management concepts in coordinating or managing multidisciplinary team tasks.		

STUDENT OUTCOMES (SO)		PERFORMANCE INDICATOR (PI)		COURSE INTENDED LEARNING OUTCOMES (CILO)		CRITERIA	WEIGHT	<i>Exemplary (4)</i>	<i>Capable (3)</i>	<i>Developing (2)</i>	<i>Beginning (1)</i>	SCORE	WEIGHTED SCORE
		12.2	Identify and devise solutions to real-world problems as a member/leader of a cross-functional team.	CILO 9	Identify real-world problems encountered during OJT and devise practicable solutions as a member/leader of a cross-functional team, supported by objective criteria and aligned with project objectives.	PRACTICAL PROBLEM-SOLVING IN CROSS-FUNCTIONAL SETTINGS (Assesses the trainee's ability to identify actual workplace problems and formulate practical, objective, and project-aligned solutions while functioning as a member or leader of a cross-functional team.)	12.50%	Accurately identifies real-world OJT problems and develops practical, well-justified solutions supported by objective criteria and aligned with project objectives.	Identifies real-world OJT problems and proposes practical solutions generally supported by objective criteria and aligned with project objectives.	Identifies some real-world problems but proposed solutions need improvement in practicality, justification, objectivity, or alignment with project objectives.	Fails to identify relevant real-world problems or propose practicable solutions aligned with project objectives.		
TOTAL:							100.00%						
							PERCENTAGE EQUIVALENT:						

Table 5-9.4. On-The-Job Training Portfolio Evaluation Rubrics for On-The-Job Training 1/ On-The-Job Training 2

STUDENT OUTCOMES (SO)		PERFORMANCE INDICATOR (PI)		COURSE INTENDED LEARNING OUTCOMES (CILO)		CRITERIA	WEIGHT	Highly Evident (4)	Evident (3)	Partially Evident (2)	Not Evident (1)	SCORE	WEIGHTED SCORE
4	Function effectively as an individual and as a member or leader in diverse teams and in multi-disciplinary settings.	4.1	Participate, express ideas freely, and execute member tasks towards attainment of the objectives.	CILO 1	Contribute effectively to OJT team activities by proposing relevant ideas and accomplishing assigned responsibilities in accordance with agreed objectives and work plans.	TEAM PARTICIPATION AND TASK ACCOMPLISHMENT (Evaluates how well the portfolio documents the student's participation in team activities, assigned responsibilities, daily tasks, and contributions to agreed workplace objectives and work plans.)	8.00%	The portfolio presents complete, detailed, and well-organized evidence that the student actively participated in team activities, proposed relevant ideas, and accomplished assigned responsibilities beyond expected standards in accordance with workplace objectives and work plans.	The portfolio presents sufficient evidence that the student participated in team activities, contributed relevant ideas, and accomplished assigned responsibilities in accordance with workplace objectives and work plans.	The portfolio presents limited evidence of team participation, idea contribution, and task accomplishment. Some daily tasks or assigned responsibilities are incomplete, unclear, or insufficiently supported.	The portfolio provides little or no evidence of team participation, idea contribution, or accomplishment of assigned responsibilities during OJT.		

STUDENT OUTCOMES (SO)		PERFORMANCE INDICATOR (PI)		COURSE INTENDED LEARNING OUTCOMES (CILO)		CRITERIA	WEIGHT	<i>Highly Evident (4)</i>	<i>Evident (3)</i>	<i>Partially Evident (2)</i>	<i>Not Evident (1)</i>	SCORE	WEIGHTED SCORE
7	Communicate effectively on complex civil engineering activities with the engineering community and with society at large.	7.1	Use effective written communication by writing grammatically correct and logical reports using appropriate graphics and technical style.	CIL O 4	Prepare OJT documentation that is grammatically correct, logically structured, and supported by appropriate graphics and professional technical style.	WRITTEN DOCUMENTATION AND PORTFOLIO ORGANIZATION (Evaluates the clarity, organization, grammar, format, technical style, and logical presentation of the OJT portfolio, including company profile, daily task narratives, reports, captions, tables, figures, appendices, and supporting documents.)	15.00 %	The portfolio is highly organized, grammatically correct, logically structured, and professionally written. The company profile, task descriptions, reports, captions, tables, figures, appendices, and supporting documents are clear, complete, properly labeled, and technically appropriate.	The portfolio is organized and generally well-written, with minor errors in grammar, structure, format, labeling, graphics, or technical style. Required sections are understandable and properly presented.	The portfolio shows basic organization but contains noticeable errors in grammar, structure, formatting, labeling, graphics, or technical style. Some sections are unclear, incomplete, or poorly presented.	The portfolio is poorly organized, grammatically weak, incomplete, or difficult to understand and does not demonstrate acceptable technical documentation.		

STUDENT OUTCOMES (SO)		PERFORMANCE INDICATOR (PI)		COURSE INTENDED LEARNING OUTCOMES (CILO)		CRITERIA	WEIGHT	Highly Evident (4)	Evident (3)	Partially Evident (2)	Not Evident (1)	SCORE	WEIGHTED SCORE
		7.2	Use effective oral communication by presenting and defending ideas with sound reasoning and in an articulate manner.	CIL O 5	Present and defend OJT outputs with sound technical reasoning and clear, professional delivery in workplace settings.	ORAL PRESENTATION AND DEFENSE OF PORTFOLIO OUTPUTS (Evaluates the student's ability to present, explain, and defend the contents of the OJT portfolio, including daily tasks, accomplishments, learning experiences, case study, and proposed solutions.)	10.00 %	The student presents and defends the portfolio contents clearly, confidently, and professionally, with sound technical reasoning and accurate explanation of tasks, outputs, learning experiences, case study findings, and proposed solutions.	The student presents and explains the portfolio contents clearly and professionally, with generally sound reasoning and appropriate responses to questions.	The student presents the portfolio contents with limited clarity or confidence and needs guidance in explaining tasks, outputs, learning experiences, case study findings, or proposed solutions.	The student has difficulty presenting or defending the portfolio contents and provides unclear, incomplete, or unsupported explanations.		
8	Understand and evaluate the sustainability and impact of professional engineering work in the solution of complex civil engineering problems in a global, economic, societal, and environmental context.	8.1	Explain the impacts of civil engineering solutions in a global, economic, environmental, and societal context.	CIL O 6	Explain the global, economic, environmental, and societal implications of civil engineering solutions encountered during OJT.	REFLECTION ON ENGINEERING IMPACTS (Evaluates the student's written reflection on the broader implications of the engineering tasks, projects, or solutions encountered during OJT, particularly in relation to global, economic, environmental, and societal contexts.)	10.00 %	The portfolio includes a comprehensive, well-written, and experience-based reflection explaining the global, economic, environmental, and societal implications of engineering tasks, projects, or solutions encountered during OJT.	The portfolio includes a clear reflection explaining the major global, economic, environmental, and societal implications of engineering tasks, projects, or solutions encountered during OJT.	The portfolio includes a limited or general reflection on the impacts of engineering work, but the discussion lacks depth, specificity, or direct connection to actual OJT experiences.	The portfolio does not adequately explain the broader impacts of engineering tasks, projects, or solutions encountered during OJT.		

STUDENT OUTCOMES (SO)		PERFORMANCE INDICATOR (PI)		COURSE INTENDED LEARNING OUTCOMES (CILO)		CRITERIA	WEIGHT	Highly Evident (4)	Evident (3)	Partially Evident (2)	Not Evident (1)	SCORE	WEIGHTED SCORE
		8.2	Establish the significance of the solutions in terms of beneficiaries.	CIL O 7	Establish the significance of observed or proposed solutions by identifying beneficiaries and articulating value to relevant stakeholders.	STAKEHOLDER AND BENEFICIARY VALUE OF THE CASE STUDY SOLUTION <i>(Evaluates how well the portfolio identifies the beneficiaries and stakeholders of the observed or proposed engineering solution, especially the solution developed in the OJT case study.)</i>	8.00%	<i>The portfolio clearly identifies relevant beneficiaries and stakeholders and provides a well-supported explanation of the value, usefulness, and significance of the observed or proposed solution to the company, project, clients, community, or end users.</i>	<i>The portfolio identifies relevant beneficiaries and stakeholders and explains the general value or significance of the observed or proposed solution.</i>	<i>The portfolio identifies some beneficiaries or stakeholders, but the explanation of value or significance is limited, unclear, or weakly connected to the case study solution.</i>	<i>The portfolio does not identify relevant beneficiaries or stakeholders and does not explain the significance of the observed or proposed solution.</i>		

STUDENT OUTCOMES (SO)		PERFORMANCE INDICATOR (PI)		COURSE INTENDED LEARNING OUTCOMES (CILO)		CRITERIA	WEIGHT	Highly Evident (4)	Evident (3)	Partially Evident (2)	Not Evident (1)	SCORE	WEIGHTED SCORE
12	Demonstrate knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.	12.1	Use appropriate tools and theories to manage multidisciplinary projects in a team.	CIL O 8	Apply appropriate tools and project management concepts to coordinate and manage multidisciplinary tasks within a team environment.	APPLICATION OF PROJECT MANAGEMENT TOOLS AND CONCEPTS (Evaluates how the portfolio documents the student's use of appropriate tools, methods, schedules, work plans, coordination practices, and project management concepts in completing OJT tasks or conducting the case study.)	7.00%	The portfolio provides strong evidence that the student applied appropriate tools, methods, schedules, work plans, coordination practices, and project management concepts to manage multidisciplinary OJT tasks and support the development of the case study.	The portfolio provides adequate evidence that the student used appropriate tools, methods, schedules, work plans, or project management concepts in completing OJT tasks or supporting the case study.	The portfolio provides limited evidence of the use of tools, methods, schedules, work plans, coordination practices, or project management concepts in completing OJT tasks or developing the case study.	The portfolio does not show evidence of task planning, work coordination, use of tools, or project management concepts during OJT or in the case study.		

STUDENT OUTCOMES (SO)		PERFORMANCE INDICATOR (PI)		COURSE INTENDED LEARNING OUTCOMES (CILO)		CRITERIA	WEIGHT	Highly Evident (4)	Evident (3)	Partially Evident (2)	Not Evident (1)	SCORE	WEIGHTED SCORE
		12.2	Identify and devise solutions to real-world problems as a member/leader of a cross-functional team.	CIL O 9	Identify real-world problems encountered during OJT and devise practicable solutions as a member/leader of a cross-functional team, supported by objective criteria and aligned with project objectives.	OJT CASE STUDY: PROBLEM IDENTIFICATION AND PROPOSED SOLUTION (Evaluates the quality of the OJT case study, particularly the identification of an actual workplace or company problem, analysis of the problem, use of objective criteria, and development of a practical solution that may benefit the company or training site.)	20.00 %	The case study clearly identifies a relevant real-world problem encountered in the company or training site, analyzes it using objective and defensible criteria, and presents a practical, well-justified, project-aligned solution that can provide clear benefit to the company, project, stakeholders, or end users.	The case study identifies a real-world workplace problem and presents a practical solution that is generally supported by objective criteria and aligned with company or project needs.	The case study identifies a workplace problem, but the analysis or proposed solution is limited, insufficiently supported, or not clearly aligned with company or project needs.	The case study does not clearly identify a real-world workplace problem or does not provide a practical and objectively supported solution.		
TOTAL:							100.00 %						
							PERCENTAGE EQUIVALENT:						

A10. Describe the adequacy of laboratory courses in the curriculum. Include a discussion of class sizes, number of identical experimental set-ups, and number of students in an experiment group. Describe how it supports the applicable student outcomes.

For a field which is one of the foundational pillars of modern civilization, it is crucial to bridge the gap between theoretical concepts and real-world application. Both 2018 and 2022 BSCE curricular programs have various laboratory courses and components, specifically for the specialized fields: Construction Engineering and Management, Geotechnical Engineering, Structural Engineering, Transportation Engineering, and Water Resources Engineering.

37 out of 211 course credit units of the 2018 BSCE curriculum are for laboratory/non-lecture courses and components. While for the 2022 BSCE curriculum, there are 30 out of 195 course credit units are for laboratory/non-lecture courses and components.

The non-lecture units on the BSCE curriculum, are classed as laboratory, fieldwork, computational or integrative. Laboratory courses are designed to provide students with hands-on exposure to the concepts discussed during lectures in civil engineering courses. These laboratory courses are integrated across multiple years of the curriculum to reinforce theoretical knowledge and develop practical engineering competencies.

To facilitate the achievement of student outcomes covered by the laboratory courses, the Civil Engineering Department have the following laboratory rooms and provisions:

- 1. Civil Engineering Surveying Laboratory (CEA Room 104A)**
- 2. Civil Engineering Soil Mechanics Laboratory (CEA Room 105A)**
- 3. Civil Engineering Construction Material & Testing Laboratory (CEA 106B)**
- 4. Hydraulics Laboratory (2nd Floor, ESRC Building)**
- 5. Civil Engineering Computer Laboratory (CEA Room 211)**
- 6. College of Engineering Drafting Rooms (CEA Rooms 207 & 208)**
- 7. College of Engineering Chemistry and Physics Laboratories (CEA Rooms 404, 405, 419 and 420)**
- 8. Virtual Laboratory for the BS Civil Engineering Program**

Along with its lecture counterparts, laboratory courses are organized into manageable sections to ensure effective supervision and meaningful student participation, allowing instructors and laboratory technicians to provide close guidance.

Laboratory courses of the BSCE program are equipped with multiple identical experimental set-ups for major experiments. The availability of identical equipment ensures that experiments can be conducted simultaneously by different groups, minimizing idle time and maximizing hands-on engagement. This approach supports the requirement that laboratories and equipment be sufficient and systematically maintained to enable students to attain program outcomes. Presented in Table 5-10.1. and 5.10.2. are the summarized data of laboratory adequacy of laboratory courses for both 2018 and 2022 BSCE curriculum.

Table 5-10.1. Summary of Class Size, and Set-up Adequacy for Laboratory Components of the 2018 BSCE Curriculum

Course Code	Course Title	Credit Lab Units	Type of Lab Component	Class Size	Number of Identical Set-Up Available	Number of Students per Group
CHEM 20024	Chemistry for Engineers	1	Laboratory	52.6	8	5
PHYS 20034	Physics for Engineers	1	Laboratory	45.8	10	5
CIEN 20012	Civil Engineering Orientation 1	1	Laboratory	52.6	10	5
CIEN 20022	Civil Engineering Orientation 2 (CE Tech 2)	1	Laboratory	49.4	10	5
ENSC 20112	Engineering Drawing and Plans	2	Laboratory	48.5	50	-
CMPE 20012	Computer Fundamentals and Programming	2	Laboratory	45.7	50	-
ENSC 20013	Computer Aided Drafting	1	Laboratory	46.3	50	-
CIEN 30013	Fundamentals of Surveying 1	1	Fieldwork	46.1	5	5
CIEN 30023	Construction Materials and Testing	1	Laboratory	44.5	4	5

Course Code	Course Title	Credit Lab Units	Type of Lab Component	Class Size	Number of Identical Set-Up Available	Number of Students per Group
CIEEN 30033	Fundamentals of Surveying 2	1	Fieldwork	44.4	16	5
MATH 20093	Numerical Solutions to CE Problems	1	Computational	46.1	50	-
CIEEN 40012	On-The-Job Training 1 (300 hours)	2	Fieldwork	44	-	-
CIEEN 30092	Civil Engineering Research	1	Fieldwork, Laboratory, Computational	43.8	-	5
CIEEN 30083	Geotechnical Engineering 1	1	Laboratory	38.3	3	5
CIEEN 30073	Building System Design	1	Laboratory	45.2	50	-
CIEEN 30063	Hydraulics 1	1	Laboratory	37.7	4	5
CIEEN 30053	Route Surveying and Earthworks	1	Fieldwork	46.4	50	-
CIEEN 30043	Structural Theory 1	1	Computational	38.7	50	-
CIEEN 30103	Structural Theory 2	1	Computational	36	50	-
CIEEN 30143	Hydraulics 2	1	Laboratory	32.2	5	5
CIEEN 30171	Integrative Course 1	1	Integrative/Correlation	39.5	-	-
CIEEN 30183	Principles of Reinforced Concrete	1	Computational	35.9	50	-
CIEEN 40022	On The Job Training 2 (300 hours)	2	Fieldwork	43	-	-
CIEEN 30193	Reinforced Concrete Design	1	Computational	23.1	50	-
CIEEN 30212	CE Project 1	1	Fieldwork, Laboratory, Computational	31.3	-	5
CIEEN 30222	Timber Design	1	Computational	37.4	50	-
CIEEN 30231	Integrative Course 2	1	Integrative/Correlation	27.2	-	-
CIEEN 30242	Foundation Engineering	1	Computational	24.1	50	-
CIEEN 30253	Principles of Steel Design	1	Computational	36.7	50	-
CIEEN 30262	CE Project 2	1	Fieldwork, Laboratory, Computational	30.8	-	5
CIEEN 30272	Quantity Surveying	1	Laboratory	26.8	50	-
CIEEN 30301	Integrative Course 3	1	Integrative/Correlation	23.9	-	-
CIEEN 30311	CE Field Trip	1	Fieldwork	52	-	-

Course Code	Course Title	Credit Lab Units	Type of Lab Component	Class Size	Number of Identical Set-Up Available	Number of Students per Group
TOTAL		37				

Table 5-10.2. Summary of Class Size, and Set-up Adequacy for Laboratory Components of the 2022 BSCE Curriculum

Course Code	Course Title	Credit Lab Units	Type of Lab Component	Class Size	Number of Identical Set-Up Available	Number of Students per Group
CHEM 015	Chemistry for Engineers	1	Laboratory	53.79	8	5
CETE 011	Civil Engineering Technology	2	Laboratory	43.61	10	5
PHYS 013	Physics for Engineers (Calculus-based)	1	Laboratory	37.78	10	5
CMPE 011	Computer Fundamentals and Programming	2	Laboratory	42.67	50	-
ENSC 011	Engineering Drawing and Plans	1	Laboratory	42.39	50	-
CIEN 201	Fundamentals of Surveying 1	1	Fieldwork	45.33	5	5
ENSC 014	Computer-Aided Drafting	1	Laboratory	41.72	50	-
CIEN 202	Fundamentals of Surveying 2 (Applied Surveying)	1	Fieldwork	46	16	5
CIEN 204	Construction Materials and Testing	1	Laboratory	28.17	4	5
CIEN 205	Numerical Solutions to CE Problems	1	Laboratory	46.22	50	-
CIEN 301	Geotechnical Engineering 1 (Soil Mechanics)	1	Laboratory	31.45	3	5
CIEN 305	Building Systems Design	1	Laboratory	48	50	-
CIEN 306	Structural Theory	1	Computational	32.54	50	-
CIEN 310	Hydraulics 1	1	Laboratory	26.01	4	5
CIEN 308	Principles of Reinforced Concrete	1	Computational	34.19	50	-
CIEN 309	CE Integration 1	1	Integrative/Correlation	41.9	-	5
CIEN 311	Hydraulics 2	1	Laboratory	25.9	50	-

Course Code	Course Title	Credit Lab Units	Type of Lab Component	Class Size	Number of Identical Set-Up Available	Number of Students per Group
CEN 313	Civil Engineering Research	1	Fieldwork, Laboratory, Computational	44.63	-	5
CEN 314	On-The-Job Training 1 (300 Hours)	1	Fieldwork	48.5	-	-
CEN 401	CE Project 1	1	Fieldwork, Laboratory, Computational	51.67	-	5
CEN 403	Foundation Engineering	1	Computational	21.83	50	-
CEN 407	Principles of Steel Design	1	Computational	31.67	50	-
CEN 410	CE Integration 2	1	Integrative/Correlation	29.83	-	-
CEN 402	CE Project 2	1	Fieldwork, Laboratory, Computational	50.5	-	5
CEN 406	Quantity Surveying	1	Laboratory	26.67	50	-
CEN 409	CE Site Inspection and Seminars	1	Fieldwork	45.17	-	-
CEN 411	CE Integration 3	1	Integrative/Correlation	27	-	-
CEN 412	On-The-Job Training 2 (200 Hours)	1	Fieldwork	50.67	-	-
TOTAL		30				

As shown in the presented tables, the laboratory, fieldwork, integrative and computational components of both the 2018 and 2022 BS Civil Engineering curricula demonstrate adequate instructional support for the development of engineering competencies through a balanced combination of hands-on experimentation, field exposure, computer-based analysis, and integrative learning experiences. The laboratory courses are generally offered with class sizes ranging from approximately between 20 to 50 students, while experimental activities are organized into small groups of five students per set-up. This arrangement ensures that each student has meaningful opportunities to participate in data gathering, equipment operation, observation, analysis, and reporting activities. The relatively small group size promotes active engagement and collaborative problem-solving rather than passive observation, thereby strengthening students' ability to apply engineering principles to practical situations.

Table 5-10.3. Summary of Student Outcomes Primarily addressed by the different types of laboratory credit units of the 2018 and 2022 BSCE Curricula

Laboratory Component	Description	Course Implementation in BSCE Curriculum	Primary Competencies Developed	Student primarily Outcomes Addressed
Laboratory Units	Learning experiences involving experimentation, testing, measurement, and validation of engineering concepts through controlled laboratory activities.	Construction Materials and Testing, Hydraulics 1 & 2, Geotechnical Engineering, Engineering Drawing and Plans, Computer-Aided Drafting, Civil Engineering Technology, Building Systems Design, Quantity Surveying, Laboratory testing and experiments in CE Research, Project 1 and 2 studies	Experimental design, testing procedures, data interpretation, critical thinking, scientific inquiry	SO1. Apply knowledge of mathematics, natural science, engineering fundamentals, and a civil engineering specialization to the solution of complex civil engineering problems. SO2. Conduct investigations of complex civil engineering problems using research-based knowledge and research methods including design of experiments, analysis, and interpretation of data, and synthesis of the information to provide valid conclusions. SO4. Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings. SO7. Communicate effectively on complex civil engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
Computational Units	Learning experiences utilizing computer applications, engineering software, simulations, modeling tools, and numerical methods	Numerical Solutions to CE Problems, Structural Theory 1 & 2, Principles of Reinforced Concrete, Reinforced Concrete Design, Foundation Engineering, Principles of Steel Design, Timber Design, Software analysis and simulation	Digital literacy, engineering analysis, simulation, design, modeling, modern engineering tool utilization	SO1. Apply knowledge of mathematics, natural science, engineering fundamentals, and a civil engineering specialization to the solution of complex civil engineering problems.

Laboratory Component	Description	Course Implementation in BSCE Curriculum	Primary Competencies Developed	Student primarily Outcomes Addressed
	for engineering analysis and design.	in CE Research, Project 1 and 2 studies		SO2. Conduct investigations of complex civil engineering problems using research-based knowledge and research methods including design of experiments, analysis, and interpretation of data, and synthesis of the information to provide valid conclusions. SO3. Design solutions for complex civil engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations. SO5. An ability to identify, formulate, and solve engineering problems SO11. Create, select and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modelling, to complex civil engineering problems with an understanding of the limitations.
Field Units	Learning experiences conducted in actual engineering environments where students perform measurements, observations, inspections, surveys, and site investigations under real-world conditions.	Fundamentals of Surveying, Route Surveying and Earthworks, On-the-job Training 1 & 2, Site Inspection and Seminars, Field exploration and data gathering in CE Research, Project 1 and 2 studies	Practical application of engineering knowledge, engineering judgment, field investigation, teamwork, professional awareness	SO1. Apply knowledge of mathematics, natural science, engineering fundamentals, and a civil engineering specialization to the solution of complex civil engineering problems. SO2. Conduct investigations of complex civil engineering problems using research-based knowledge and research methods including design of experiments, analysis, and

Laboratory Component	Description	Course Implementation in BSCE Curriculum	Primary Competencies Developed	Student primarily Outcomes Addressed
				interpretation of data, and synthesis of the information to provide valid conclusions. SO4. Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings. SO5. An ability to identify, formulate, and solve engineering problems SO7. Communicate effectively on complex civil engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions. SO9. Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change. SO10. Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues, and the consequent responsibilities relevant to professional engineering practice and solutions to complex civil engineering problems. SO12. Demonstrate knowledge and understanding of engineering management principles and economic

Laboratory Component	Description	Course Implementation in BSCE Curriculum	Primary Competencies Developed	Student primarily Outcomes Addressed
				decision-making and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments
Integrative/ Correlation Units	Learning experiences designed to synthesize, integrate, correlate, and reinforce knowledge acquired from various civil engineering disciplines. These units focus on comprehensive problem-solving and preparation for professional practice and licensure examinations.	Integrative Course 1/ CE Integration 1, Integrative Course 2/ CE Integration 2, Integrative Course 3/ CE Integration 3	Knowledge integration, multidisciplinary thinking, professional preparation, systems perspective, lifelong learning, engineering judgment	SO1. Apply knowledge of mathematics, natural science, engineering fundamentals, and a civil engineering specialization to the solution of complex civil engineering problems. SO5. An ability to identify, formulate, and solve engineering problems

The adequacy of laboratory and experiential learning provisions in the BS Civil Engineering program is demonstrated through the availability of instructional facilities, laboratory stations, field equipment, and computing resources that support the attainment of the applicable Course Intended Learning Outcomes (CILOs) and Student Outcomes (SOs). Laboratory and field activities are designed to reinforce theoretical instruction, develop technical proficiency, and provide students with opportunities to apply engineering principles through experimentation, measurement, analysis, design, and documentation.

Foundational natural and physical science courses, such as **Chemistry for Engineers** and **Physics for Engineers**, are supported by approximately eight to ten identical laboratory stations. These laboratory set-ups allow several student groups to conduct experiments simultaneously while maintaining manageable student-to-equipment ratios. This arrangement reduces waiting time, increases actual experimentation time, and enables students to repeatedly perform observations, measurements, computations, and analysis under faculty supervision. As a result, students are able to strengthen their understanding of scientific principles that serve as the foundation for succeeding engineering science and professional civil engineering courses.

Similarly, courses such as **Civil Engineering Technology**, **Civil Engineering Orientation**, and **Surveying** are supported by appropriate instruments and equipment necessary for laboratory and field-based exercises. These provisions allow students to gain early exposure to engineering tools, measurement procedures, field documentation, and technical observation. The availability of sufficient equipment enables students to participate actively in hands-on tasks rather than merely observe the procedure, thereby improving their confidence, accuracy, and technical competence in the use of engineering instruments.

Laboratory courses in professional civil engineering subjects, including **Construction Materials and Testing**, **Geotechnical Engineering**, and **Hydraulics**, are implemented through class sizes and group arrangements appropriate to the nature of the experiments being conducted. Although these courses may have fewer identical experimental set-ups due to the specialized nature of the equipment, the reduced enrollment in upper-level courses results in acceptable utilization rates. Groupings of approximately five students allow each learner to participate in specimen preparation, equipment operation, testing procedures, data collection, interpretation of results, and preparation of technical laboratory reports.

Through these professional laboratory activities, students develop competencies in conducting experiments, applying standard testing procedures, analyzing engineering data, drawing evidence-based conclusions, and evaluating the behavior and performance of engineering materials and systems under varying conditions. These experiences directly support the development of engineering judgment, technical communication, teamwork, and adherence to safety and quality control procedures. They also provide direct evidence for assessing student performance in relation to the applicable CILOs, SOs, and performance indicators.

Fieldwork courses, such as **Fundamentals of Surveying**, **Route Surveying and Earthworks**, **Site Inspection and Seminars**, and **On-the-Job Training**, provide authentic learning environments that extend beyond the classroom and laboratory. The provision of multiple surveying instruments and field equipment enables students to work in teams while assuming rotating responsibilities related to instrument operation, measurement, computation, documentation, field verification, and quality checking. These activities expose students to actual site conditions and practical constraints encountered in civil engineering projects.

The field-based learning experiences strengthen students' ability to collect reliable field data, identify engineering problems, apply appropriate procedures, and exercise engineering judgment in real or simulated project environments. They also develop non-technical competencies, including teamwork, communication, leadership, professional responsibility, safety awareness, and ethical conduct. These competencies are essential in preparing students for internship, professional practice, and multidisciplinary civil engineering work.

Computational laboratory courses and design-oriented subjects are supported by computer laboratories with approximately fifty workstations. These facilities provide nearly one-to-one access for enrolled students, particularly in courses such as **Computer Fundamentals and Programming**, **Computer-Aided Drafting**, and **Numerical Solutions to Civil Engineering Problems**. The availability of dedicated workstations allows students to perform drafting, numerical analysis, modeling, simulation, and design-related computations with adequate access to digital tools.

Higher year-level professional and design courses, including **Structural Theory**, **Reinforced Concrete Design**, **Foundation Engineering**, and **Steel Design**, benefit from these computing resources by enabling students to apply modern engineering software and computational methods in analysis and design tasks. Access to computer laboratories supports independent work, technical accuracy, and the development of digital competencies required in contemporary civil engineering practice. These resources also contribute to students' ability to analyze complex engineering problems, evaluate design alternatives, and prepare professional engineering outputs.

The inclusion of **Civil Engineering Research**, **CE Project 1**, **CE Project 2**, and other integrative courses further enhances the adequacy of laboratory and experiential learning provisions. These courses require students to combine laboratory investigations, field observations, computational analysis, design principles, and research methods in addressing civil engineering problems. Working in small project teams, students formulate research or design problems, gather and evaluate data, apply relevant standards and methods, and develop outputs that consider technical, economic, environmental, safety, and societal constraints.

The project-based structure of these courses promotes independent learning, critical thinking, innovation, professional communication, and multidisciplinary integration. Student outputs are commonly presented through technical reports, project documentation, oral

presentations, and project defenses, which provide additional evidence of students' readiness for engineering practice. Collectively, the laboratory, fieldwork, computational, research, and capstone components of the curriculum provide sufficient and appropriate learning environments for developing the technical and professional competencies expected of BS Civil Engineering graduates.

Laboratory Rooms, Facilities, and Provisions for the BS Civil Engineering Program

The Department of Civil Engineering continuously identifies, reviews, and requests the procurement of laboratory equipment and instructional resources necessary to support the delivery of the Bachelor of Science in Civil Engineering (BSCE) program. These requests form part of the Department's effort to ensure that laboratory facilities remain adequate, functional, and responsive to the requirements of outcomes-based engineering education and current professional practice.

Modern and properly maintained laboratory facilities enable students to engage in hands-on learning activities that complement theoretical instruction. Through laboratory work, students are able to operate engineering instruments, conduct material and systems testing, collect and analyze data, interpret results, and prepare technical reports. These activities strengthen the attainment of applicable Course Intended Learning Outcomes (CILOs) and Student Outcomes (SOs), particularly those related to experimentation, problem analysis, engineering design, teamwork, communication, and the use of modern engineering tools.

The availability of updated equipment also allows students to become familiar with testing methods, procedures, and technologies that reflect actual civil engineering practice. In areas such as construction materials testing, soil mechanics, hydraulics, surveying, structural engineering, and computer-aided analysis and design, laboratory facilities provide students with opportunities to develop technical competence, engineering judgment, and professional responsibility. These experiences support students' transition from classroom-based learning to fieldwork, internship, research, capstone projects, and eventual professional practice.

The Department's continuing efforts to improve laboratory provisions also support the program's Continuous Quality Improvement (CQI) initiatives. Feedback from faculty members, students, laboratory personnel, industry partners, and accreditation activities is considered in identifying needed equipment, facility upgrades, maintenance actions, and instructional improvements. These mechanisms help ensure that the laboratory environment remains

aligned with curriculum requirements, safety standards, and the evolving demands of the civil engineering profession.

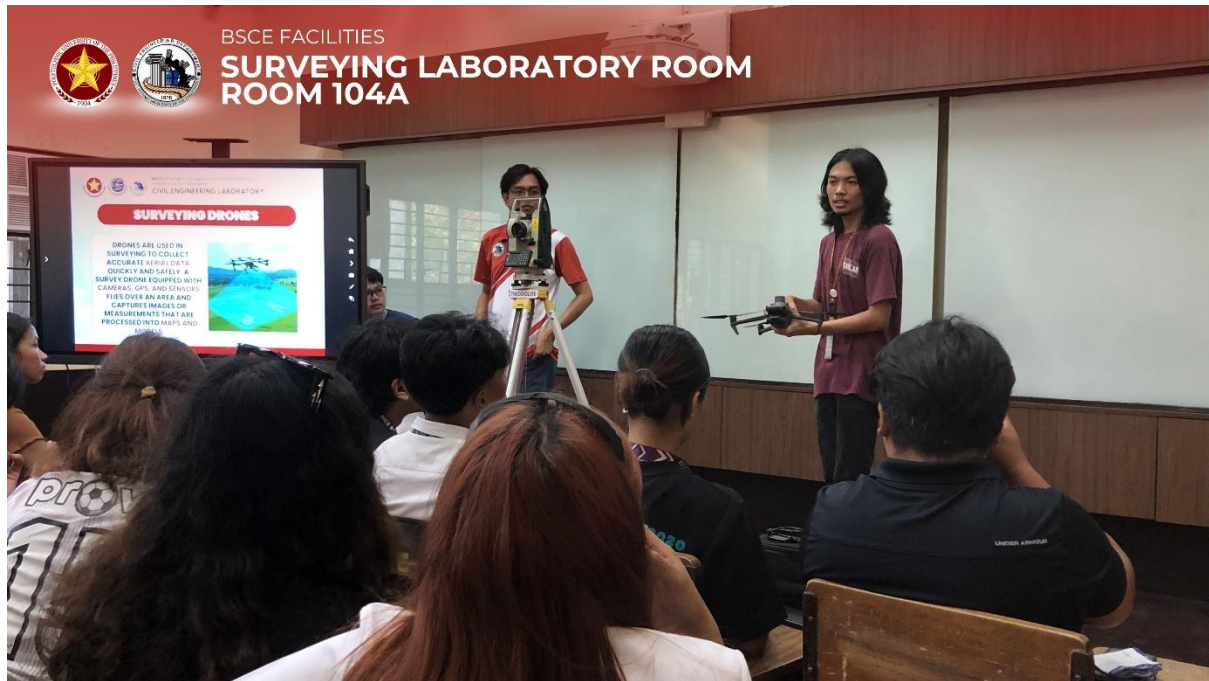
Presented in the following table are the summarized data on the current laboratory rooms, facilities, equipment provisions, and related instructional resources available to support the implementation of the BS Civil Engineering program.

**Table 5-10.4. Sizes of Laboratory Rooms and Facilities for the
BS Civil Engineering Program**

Laboratory Room	Location	Room dimensions			
		length (m)	width (m)	height (m)	area (m ²)
Surveying Laboratory Room	CEA Room 104A	9.00	7.00	3.66	63.00
Soil Mechanics Laboratory	CEA Room 105A	9.00	7.00	3.66	63.00
Construction Material & Testing Laboratory	CEA Room 106B	9.00	7.00	3.66	63.00
Hydraulics Laboratory	2nd Floor, PUP Economic and Social Research Council (ESRC) Building	9.50	7.20	3.60	67.00
Civil Engineering Computer Laboratory	CEA Room 211A	9.00	7.00	3.66	63.00
College of Engineering Drafting Rooms	CEA Room 207	13.50	7.00	3.66	94.50
	CEA Room 208	13.50	7.00	3.66	94.50
College of Engineering Physics Laboratory Rooms	CEA Room 404	13.50	7.00	3.66	94.50
	CEA Room 405	13.50	7.00	3.66	94.50
College of Engineering Chemistry Laboratory Rooms	CEA Room 419	13.50	7.00	3.66	94.50
	CEA Room 420	13.50	7.00	3.66	94.50
Civil Engineering Laboratory Head Office Room	CEA Room 105B	7.00	4.50	3.66	31.50
Civil Engineering Tools and Equipment Room	CEA Room 106A	7.00	4.50	3.66	31.50
Civil Engineering Dirty Laboratory	Ground Floor, Back of Rooms 104, 105 and 106	36.00	4.00	3.66	144.00

Each of the major laboratory rooms are explained in greater detail in the following pages:

CIVIL ENGINEERING SURVEYING LABORATORY (CEA ROOM 104A)



The **Civil Engineering Surveying Laboratory** provides students with essential hands-on learning experiences in the use of surveying instruments, field measurement procedures, and spatial data collection techniques that form part of the foundation of civil engineering practice. Through laboratory and field activities in **Fundamentals of Surveying 1**, **Fundamentals of Surveying 2**, **Route Surveying**, and **Earthworks**, students develop the technical skills required to collect, process, analyze, and interpret survey data used in planning, design, construction, and infrastructure development, particularly in highway, railway, site development, and other civil engineering projects.

The laboratory enables students to apply mathematical, scientific, and engineering principles in determining distances, elevations, coordinates, alignments, areas, volumes, and land configurations. By conducting actual field measurements and performing related computations, students gain a practical understanding of surveying concepts introduced in classroom instruction. This strengthens their ability to connect theoretical principles with professional engineering applications and prepares them for more advanced courses involving transportation engineering, construction engineering, geotechnical works, and infrastructure design.

Through the operation of surveying equipment such as levels, total stations, measuring tapes, leveling rods, and other related instruments, students acquire competencies in conducting engineering measurements with accuracy, precision, and reliability. Laboratory exercises emphasize

proper equipment handling, field procedures, error checking, data validation, and preparation of field notes. These activities help students appreciate the importance of minimizing measurement errors and ensuring the quality of survey data, which are critical in engineering design, construction layout, quantity estimation, and project implementation.

The surveying laboratory also supports the development of problem-solving and engineering judgment. During field activities, students encounter practical challenges such as uneven terrain, line-of-sight limitations, measurement discrepancies, environmental constraints, and data inconsistencies. These conditions require students to analyze field situations, apply appropriate surveying methods, make necessary adjustments, and arrive at technically sound solutions based on established principles and standards.

Collaborative fieldwork further develops teamwork, communication, leadership, and professional responsibility. Students perform assigned roles within survey teams, coordinate field tasks, document observations, prepare computations, generate technical reports, and present survey results. These activities simulate actual engineering practice, where surveying work requires coordination among engineers, field personnel, designers, contractors, and other stakeholders.

Moreover, the laboratory reinforces ethical and professional practice by emphasizing accuracy, accountability, safety, and responsibility in the conduct of surveying operations. Students are made aware that reliable survey data directly affect infrastructure planning, land development, construction activities, and public safety. Through these laboratory and field experiences, the Civil Engineering Surveying Laboratory contributes to the attainment of relevant Course Intended Learning Outcomes (CILOs) and Student Outcomes (SOs), particularly in problem analysis, application of engineering tools, teamwork, communication, professional responsibility, and preparation for civil engineering practice.

Table 5-10.5. Summary of Supported Laboratory courses and Target Student Outcomes of the Civil Engineering Surveying Laboratory

Laboratory Room	Supported Laboratory Courses in the Curriculum		Student Outcomes Primarily Addressed	Method of Supporting Student Outcomes
	2018 BSCE Curriculum	2022 BSCE Curriculum		
Civil Engineering Surveying Laboratory	CIEN 30013 - Fundamentals of Surveying 1 CIEN 30033 - Fundamentals of Surveying 2 CIEN 30053 - Route Surveying and Earthworks	CIEN 201 - Fundamentals of Surveying 1 CIEN 202 - Fundamentals of Surveying 2	SO1. Apply knowledge of mathematics, natural science, engineering fundamentals, and a civil engineering specialization to the solution of complex civil engineering problems. SO2. Conduct investigations of complex civil engineering problems using research-based knowledge and research methods including design of experiments, analysis, and interpretation of data, and synthesis of the information to provide valid conclusions. SO4. Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings. SO5. An ability to identify, formulate, and solve engineering problems SO7. Communicate effectively on complex civil engineering activities with the engineering community and	Provides hands-on experience in field measurements, topographic surveys, route location, and earthwork computations. Students operate surveying instruments, collect and analyze data, prepare technical reports, and perform collaborative field activities that mirror professional surveying practice

			with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions. SO12. Demonstrate knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments	
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CIVIL ENGINEERING SOIL MECHANICS LABORATORY (CEA ROOM 105A)



The **Soil Mechanics Laboratory** serves as an essential instructional facility that supports the development of student competencies in geotechnical engineering. It provides students with hands-on learning experiences that allow them to investigate the engineering properties and behavior of soils and understand how these properties influence the design, performance, stability, and safety of civil engineering structures. Through laboratory activities in **Geotechnical Engineering** and related courses, students are exposed to standard soil testing procedures that are fundamental to foundation design, slope stability analysis, earthworks, pavement subgrade evaluation, retaining structures, and other geotechnical applications.

Through laboratory experiments, students apply principles from mathematics, physics, geology, mechanics, and engineering science to determine important soil properties such as moisture content, specific gravity, grain size distribution, Atterberg limits, permeability, compaction characteristics, consolidation behavior, and shear strength. These activities strengthen students' ability to analyze soil behavior using scientific methods, standardized testing procedures, and evidence-based interpretation of results. By linking laboratory findings with geotechnical theories discussed in lectures, students develop a deeper understanding of soil classification, stress-strain behavior, settlement, bearing capacity, and stability considerations.

The laboratory provides practical experience in the use of geotechnical testing equipment and instruments. Students learn to prepare soil samples, conduct tests, observe proper laboratory procedures, record measurements, process data, and evaluate results under varying soil conditions. These activities develop their ability to collect reliable data, recognize sources of experimental error, validate findings, and interpret soil behavior with appropriate engineering judgment. Such competencies are essential in addressing real-world engineering problems involving foundations, embankments, excavations, slopes, and other soil-supported structures.

Experimental investigations in the Soil Mechanics Laboratory also enhance students' ability to plan and execute engineering studies. Students are trained to define laboratory objectives, follow prescribed testing standards, organize data, perform computations, analyze trends, and draw conclusions that support engineering decision-making. The preparation of laboratory reports further develops their capacity to present technical information in a clear, organized, and professional manner, including the discussion of results, limitations, assumptions, and implications for civil engineering practice.

The facility also promotes critical thinking and responsible engineering judgment by exposing students to the inherent variability and uncertainty of soil as an engineering material. Through the analysis of laboratory data, students learn to assess risks, identify limitations of test results, and recommend appropriate geotechnical considerations while taking into account safety, constructability, cost, environmental conditions, and project requirements. These experiences prepare students to recognize that geotechnical engineering decisions require both technical analysis and careful consideration of site-specific conditions.

Laboratory activities are commonly conducted through collaborative group work, enabling students to function effectively as members of a technical team. Students coordinate testing procedures, assign responsibilities, document observations, verify computations, and discuss findings with their peers and instructors. This collaborative environment strengthens teamwork, communication, leadership, accountability, and professional responsibility, which are necessary competencies in actual geotechnical investigations and multidisciplinary civil engineering projects.

By providing practical exposure to soil testing, geotechnical investigation, data interpretation, and engineering analysis, the Soil Mechanics Laboratory supports the attainment of relevant Course Intended Learning Outcomes (CILOs) and Student Outcomes (SOs). It contributes particularly to student development in experimentation, problem analysis, use of modern

engineering tools, technical communication, teamwork, engineering design considerations, ethical responsibility, and preparation for professional civil engineering practice.

Table 5-10.6. Summary of Supported Laboratory courses and Target Student Outcomes of the Civil Engineering Soil Mechanics Laboratory

Laboratory Room	Supported Laboratory Courses in the Curriculum		Student Outcomes Primarily Addressed	Method of Supporting Student Outcomes
	2018 BSCE Curriculum	2022 BSCE Curriculum		
Civil Engineering Soil Mechanics Laboratory	CIEN 30083 - Geotechnical Engineering 1	CIEN 301 - Geotechnical Engineering 1 (Soil Mechanics)	SO1. Apply knowledge of mathematics, natural science, engineering fundamentals, and a civil engineering specialization to the solution of complex civil engineering problems. SO2. Conduct investigations of complex civil engineering problems using research-based knowledge and research methods including design of experiments, analysis, and interpretation of data, and synthesis of the information to provide valid conclusions. SO4. Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings. SO5. An ability to identify, formulate, and solve engineering problems	Enables students to perform laboratory testing of soil properties, analyze geotechnical behavior, interpret results for foundation and earthwork applications, and make engineering recommendations based on evidence and established standards

			SO7. Communicate effectively on complex civil engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.	
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CONSTRUCTION MATERIALS & TESTING LABORATORY (CEA ROOM 106B)



The **Construction Materials and Testing Laboratory** provides students with practical learning opportunities to examine the properties, behavior, and performance of materials commonly used in civil engineering projects. Through laboratory activities in **Construction Materials and Testing** and related courses, students develop a comprehensive understanding of how material characteristics affect the quality, safety, durability, and serviceability of infrastructure systems. The facility supports the integration of classroom-based theories with actual testing procedures, enabling students to relate material behavior to design, construction, and quality control requirements in civil engineering practice.

The laboratory enables students to apply scientific and engineering principles in evaluating construction materials such as concrete, cement, aggregates, steel, asphalt, masonry units, and other commonly used engineering materials. Through experimental investigations, students examine material properties such as strength, workability, grading, density, absorption, hardness, durability, and response to loading and environmental conditions. These activities strengthen students' understanding of how materials are selected, tested, and accepted for use in actual construction projects.

Hands-on testing activities allow students to become familiar with standard laboratory procedures and industry-recognized testing methods. Students are trained to properly prepare

specimens, operate testing equipment, collect measurements, record observations, analyze data, and determine whether test results comply with applicable specifications, standards, and project requirements. This exposure helps students develop accuracy, discipline, and technical competence in conducting material tests that are essential to construction quality assurance and quality control.

The laboratory experience also strengthens students' ability to investigate engineering problems through systematic experimentation and data analysis. By comparing laboratory results with theoretical expectations, design assumptions, and standard acceptance criteria, students develop critical thinking and engineering judgment. They learn to evaluate material performance, identify possible sources of error or failure, interpret test results, and understand how laboratory findings influence decisions in design, procurement, construction, inspection, and project implementation.

Exposure to quality control and quality assurance procedures reinforces the importance of safety, reliability, sustainability, and compliance in civil engineering projects. Students gain an appreciation of the role of material testing in ensuring that structures and infrastructure systems meet required performance levels throughout their intended service life. The laboratory also highlights the significance of responsible material selection, efficient resource use, and adherence to established standards in promoting durable and sustainable construction practices.

Laboratory activities are commonly conducted through collaborative exercises, allowing students to function effectively as members of a technical team. Students coordinate testing procedures, assign responsibilities, document observations, verify computations, and prepare technical reports. These activities develop teamwork, communication, accountability, and professional documentation skills, which are essential in construction laboratories, project sites, quality assurance units, and engineering consultancy work.

Overall, the Construction Materials and Testing Laboratory serves as a key facility for developing student competencies in experimentation, material evaluation, engineering analysis, quality assurance, technical communication, teamwork, and professional responsibility. Its activities contribute to the attainment of relevant Course Intended Learning Outcomes (CILOs) and Student Outcomes (SOs) by preparing students to evaluate construction materials, interpret test results, and apply engineering standards in support of safe, reliable, and sustainable civil engineering practice.

Table 5-10.7. Summary of Supported Laboratory courses and Target Student Outcomes of the Civil Engineering Construction Materials & Testing Laboratory

Laboratory Room	Supported Laboratory Courses in the Curriculum		Student Outcomes Primarily Addressed	Method of Supporting Student Outcomes
	2018 BSCE Curriculum	2022 BSCE Curriculum		
Construction Materials & Testing Laboratory	CIEN 30023 - Construction Materials and Testing	CIEN 204 - Construction Materials and Testing	SO1. Apply knowledge of mathematics, natural science, engineering fundamentals, and a civil engineering specialization to the solution of complex civil engineering problems. SO2. Conduct investigations of complex civil engineering problems using research-based knowledge and research methods including design of experiments, analysis, and interpretation of data, and synthesis of the information to provide valid conclusions. SO4. Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings. SO5. An ability to identify, formulate, and solve engineering problems SO7. Communicate effectively on complex civil engineering activities with the engineering community and with	Develops competency in evaluating concrete, steel, aggregates, and other construction materials through standard testing procedures. Students learn quality control, material selection, performance evaluation, and documentation practices essential to safe infrastructure development

			society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.	
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HYDRAULICS LABORATORY (2ND FLOOR, ESRC BUILDING)



The **Hydraulics Laboratory** supports student learning in **Hydraulics 1**, **Hydraulics 2**, and related water resources engineering courses by providing opportunities to observe, measure, and analyze the behavior of fluids under controlled laboratory conditions. The facility serves as an important instructional venue where theoretical principles in fluid mechanics and hydraulics are translated into practical engineering applications. Through laboratory-based activities, students develop a deeper understanding of fluid flow behavior, pressure variation, energy relationships, hydraulic losses, and the operation of hydraulic systems used in civil engineering practice.

Laboratory experiments enable students to apply concepts from mathematics, physics, and engineering sciences in the analysis of hydraulic phenomena. By conducting experiments involving pipe flow, open-channel flow, flow measurement, hydraulic losses, pump performance, and related hydraulic systems, students are able to validate theoretical equations and observe the actual behavior of fluids under different conditions. These activities strengthen students' ability to connect analytical models with physical observations, thereby improving their understanding of the principles used in water resources engineering, drainage design, water supply systems, flood control, and other hydraulic applications.

The laboratory provides hands-on experience in the use of hydraulic apparatus, measuring devices, and experimental setups. Students learn to operate equipment properly, record

observations, collect data, compute hydraulic parameters, and evaluate system performance. These activities develop technical competence in experimentation, accuracy in measurement, and discipline in following standard laboratory procedures. They also allow students to recognize sources of experimental error, assess the reliability of results, and compare measured values with theoretical predictions.

The Hydraulics Laboratory also strengthens students' ability to design and conduct engineering investigations. Students are required to establish experimental objectives, perform procedures systematically, analyze data, interpret results, and draw conclusions based on evidence. Through this process, they develop critical thinking and engineering judgment in evaluating hydraulic behavior and determining the significance of experimental findings. These competencies are essential in solving civil engineering problems that involve water flow, pressure systems, drainage, channel design, and hydraulic infrastructure.

The laboratory environment further promotes analytical thinking by exposing students to practical hydraulic challenges that require evaluation of influencing factors such as flow rate, pipe diameter, slope, roughness, energy loss, and system efficiency. These activities encourage students to identify relationships among variables, assess limitations of theoretical assumptions, and formulate appropriate engineering recommendations. Such experiences prepare students to apply hydraulic principles in planning, design, operation, and evaluation of water-related civil engineering systems.

Collaborative laboratory work also develops teamwork, communication, and professional responsibility. Students conduct experiments in groups, coordinate assigned tasks, document procedures, verify computations, and prepare technical laboratory reports. The preparation and presentation of experimental findings enhance students' ability to communicate engineering concepts, methods, results, and conclusions using appropriate technical language and professional documentation standards.

By integrating experimentation, analysis, and practical application, the Hydraulics Laboratory contributes significantly to the attainment of relevant Course Intended Learning Outcomes (CILOs) and Student Outcomes (SOs). It supports student development in engineering investigation, problem analysis, use of modern engineering tools, design considerations, data interpretation, technical communication, teamwork, and preparation for professional civil engineering practice, particularly in the fields of hydraulics and water resources engineering.

Table 5-10.8. Summary of Supported Laboratory courses and Target Student Outcomes of the Hydraulics Laboratory

Laboratory Room	Supported Laboratory Courses in the Curriculum		Student Outcomes Primarily Addressed	Method of Supporting Student Outcomes
	2018 BSCE Curriculum	2022 BSCE Curriculum		
Hydraulics Laboratory	CIEN 30063 – Hydraulics 1 CIEN 30143 – Hydraulics 2	CIEN 310 – Hydraulics 1 CIEN 311 – Hydraulics 2	SO1. Apply knowledge of mathematics, natural science, engineering fundamentals, and a civil engineering specialization to the solution of complex civil engineering problems. SO2. Conduct investigations of complex civil engineering problems using research-based knowledge and research methods including design of experiments, analysis, and interpretation of data, and synthesis of the information to provide valid conclusions. SO4. Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings. SO5. An ability to identify, formulate, and solve engineering problems SO7. Communicate effectively on complex civil engineering activities with the	Allows students to investigate hydraulic and fluid-flow phenomena through experimentation. Laboratory activities strengthen analytical skills, data interpretation, engineering judgment, and understanding of water-related infrastructure systems

			engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions. SO9. Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change. SO12. Demonstrate knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments	
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CIVIL ENGINEERING COMPUTER LABORATORY (CEA ROOM 211)



The **Civil Engineering Computer Laboratory** provides students with access to modern computing resources and specialized engineering software that support instruction, analysis, design, modeling, visualization, and project development across the major areas of civil engineering. The facility serves as an important learning environment for developing digital and computational competencies, which are increasingly essential in contemporary engineering practice. Through its available hardware, software, and supervised laboratory activities, students are trained to use appropriate technological tools in solving engineering problems and preparing professional engineering outputs.

The laboratory supports courses such as **Computer-Aided Drafting, Numerical Solutions to Civil Engineering Problems, Building Systems Design, Quantity Surveying**, and other design- and computation-based subjects in the curriculum. Through these courses, students develop proficiency in computer-aided drafting, numerical computation, engineering modeling, quantity take-off, cost estimation, design visualization, and technical documentation. These activities strengthen students' ability to apply engineering principles with the aid of modern tools and to produce outputs that reflect current professional practice.

The availability of specialized software packages, including **MIDAS SoilWorks, MIDAS Civil, MIDAS Gen, ArcGIS**, and other relevant engineering applications, enables students to

perform advanced technical tasks in geotechnical engineering, structural engineering, transportation engineering, water resources engineering, and construction engineering and management. Exposure to these software tools allows students to model engineering systems, analyze project conditions, evaluate alternatives, simulate design scenarios, and interpret software-generated results. These experiences enhance students' readiness for industry practice by familiarizing them with digital tools commonly used in engineering offices, project sites, and consultancy work.

The laboratory also supports the development of analytical and computational thinking. Students are trained to combine engineering theory, mathematical methods, design standards, and digital technologies to improve the accuracy, efficiency, and reliability of engineering solutions. By using computational tools, students are able to process large data sets, perform iterative calculations, evaluate design options, and validate results against engineering assumptions and applicable criteria. This strengthens their capacity to solve complex civil engineering problems using both technical judgment and modern engineering tools.

In addition, the facility serves as a critical resource for **CE Project 1** and **CE Project 2**, where students conduct independent or group-based investigations, perform engineering analyses, prepare design outputs, and develop comprehensive project documentation. Access to specialized software enables students to integrate concepts from various civil engineering fields into realistic and multidisciplinary project settings. This supports the curriculum's emphasis on research, design, problem-solving, and application of engineering knowledge to practical and socially relevant problems.

Collaborative project work within the computer laboratory also promotes teamwork, communication, and project management competencies. Students engage in group discussions, data sharing, software-based analysis, design reviews, technical consultations, and presentation of outputs. These activities simulate professional engineering environments where collaboration, documentation, coordination, and effective communication are essential in the successful completion of engineering projects.

Overall, the Civil Engineering Computer Laboratory provides essential technological infrastructure that supports the attainment of relevant **Course Intended Learning Outcomes (CILOs)** and **Student Outcomes (SOs)**. It contributes to student development in engineering analysis, design, investigation, modern tool utilization, technical communication, teamwork, lifelong learning, and professional readiness. By providing students with access to current computing resources and specialized engineering software, the laboratory helps ensure that graduates are

equipped with the digital competencies necessary to respond to evolving industry demands and professional expectations in civil engineering practice.

Table 5-10.9. Summary of Supported Laboratory courses and Target Student Outcomes of the Civil Engineering Computer Laboratory

Laboratory Room	Supported Laboratory Courses in the Curriculum		Student Outcomes Primarily Addressed	Method of Supporting Student Outcomes
	2018 BSCE Curriculum	2022 BSCE Curriculum		
Civil Engineering Computer Laboratory	ENSC 20013 - Computer-Aided Drafting MATH 20093 - Numerical Solutions to CE Problems CIEN 30073 - Building Systems Design CIEN 30272 - Quantity Surveying	ENSC 014 - Computer-Aided Drafting CIEN 205 - Numerical Solutions to CE Problems CIEN 305 - Building Systems Design CIEN 406 - Quantity Surveying	SO1. Apply knowledge of mathematics, natural science, engineering fundamentals, and a civil engineering specialization to the solution of complex civil engineering problems. SO3. Design solutions for complex civil engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations. SO5. An ability to identify, formulate, and solve engineering problems SO7. Communicate effectively on complex civil engineering activities with the engineering community and with society at large, such as being able	Provides access to industry-standard software for structural, geotechnical, transportation, water resources, drafting, quantity surveying, and numerical analysis applications. Students develop digital competencies required for engineering analysis, design, simulation, modeling, and project study work

			to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions. SO10. Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues, and the consequent responsibilities relevant to professional engineering practice and solutions to complex civil engineering problems. SO11. Create, select, and apply appropriate techniques, skills, resources, and modern engineering and IT tools, including prediction and modeling, to complex civil engineering problems with an understanding of the limitations	
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COLLEGE OF ENGINEERING DRAFTING ROOMS (CEA 207 & 208)



The **Drafting Rooms** provide an appropriate and dedicated learning environment for developing the graphical communication and technical visualization skills that are fundamental to civil engineering practice. Through laboratory activities in **Engineering Drawing and Plans** and related courses, students acquire the ability to represent engineering concepts, structural components, site layouts, and construction details accurately through technical drawings and engineering plans. These facilities support the development of students' ability to communicate design intent clearly, systematically, and in accordance with accepted engineering drafting standards.

The facilities enable students to apply engineering standards, drawing conventions, line types, scales, symbols, dimensions, annotations, and design principles in the preparation of plans, elevations, sections, details, and other engineering drawings. These activities establish a strong foundation for effective visualization and documentation of civil engineering works. By preparing drawings manually and through guided drafting exercises, students develop precision, spatial understanding, technical accuracy, and attention to detail, which are essential in the planning, design, construction, and inspection of infrastructure projects.

Students gain hands-on experience in the use of drafting instruments, drawing materials, and established drafting procedures. These activities help them understand the importance of

accuracy, neatness, proportion, scale, and completeness in engineering documentation. Through repeated drafting exercises, students learn to interpret engineering information, convert conceptual ideas into graphical representations, and prepare drawings that can be understood by engineers, architects, contractors, field personnel, and other stakeholders involved in construction and project implementation.

The drafting activities also support the development of problem-solving and engineering judgment. Students are required to analyze project requirements, determine appropriate drawing methods, organize graphical information, and ensure that outputs comply with applicable standards and specifications. These exercises encourage students to consider practical constraints such as constructability, clarity of details, drawing coordination, and consistency of information. As a result, students develop the ability to balance technical requirements with practical considerations in the preparation of engineering plans.

The Drafting Rooms also promote communication, collaboration, and professional responsibility. Students participate in peer discussions, drawing reviews, consultations, and constructive feedback sessions that help improve the quality of their outputs. These activities develop both graphical and written communication skills, as students are expected to prepare, explain, revise, and present engineering plans in a clear and professional manner. The learning environment also reinforces discipline, accuracy, accountability, and proper documentation practices, which are important attributes in engineering design and construction work.

By providing a dedicated environment for technical drawing, engineering visualization, and preparation of construction-related documents, the Drafting Rooms contribute to the attainment of relevant Course Intended Learning Outcomes (CILOs) and Student Outcomes (SOs). These facilities support student development in technical competence, engineering design, communication, problem-solving, professional documentation, and readiness for civil engineering practice.

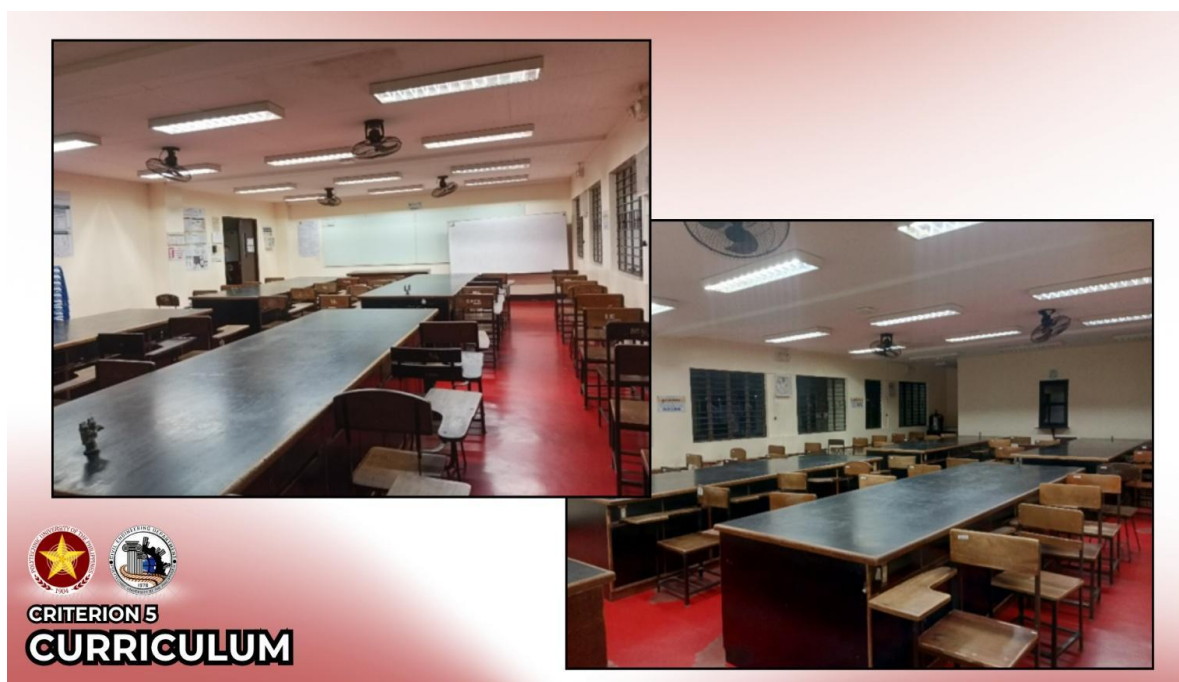
Table 5-10.10. Summary of Supported Laboratory courses and Target Student Outcomes of the College of Engineering Drafting Rooms

Laboratory Room	Supported Laboratory Courses in the Curriculum		Student Outcomes Primarily Addressed	Method of Supporting Student Outcomes
	2018 BSCE Curriculum	2022 BSCE Curriculum		
College of Engineering Drafting Rooms	ENSC 20112 - Engineering Drawing and Plans	ENSC 011 - Engineering Drawing and Plans	SO4. Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings. SO11. Create, select, and apply appropriate techniques, skills, resources, and modern engineering and IT tools, including prediction and modeling, to complex civil engineering problems with an understanding of the limitations	Supports the development of graphical communication skills through preparation and interpretation of engineering drawings and plans. Students learn technical drafting standards, visualization techniques, and engineering documentation practices essential to professional design work

COLLEGE OF ENGINEERING CHEMISTRY (CEA ROOM 419 AND 420) AND PHYSICS (CEA ROOM 404 AND 405) LABORATORIES



Chemistry Laboratories



Physics Laboratories

The **Chemistry and Physics Laboratories** provide foundational scientific learning experiences that support the development of analytical, experimental, and problem-solving competencies among BS Civil Engineering students. Through laboratory courses in **Physics for Engineers** and **Chemistry for Engineers**, students gain a deeper understanding of the scientific principles that serve as the foundation for engineering analysis, material behavior, structural response, energy systems, environmental processes, and other civil engineering applications. These laboratories strengthen the students' ability to connect fundamental scientific concepts with the technical requirements of higher-level engineering courses.

The laboratories allow students to verify theoretical concepts through actual experimentation, observation, and measurement. By conducting physics and chemistry experiments, students are able to examine how scientific laws and principles govern motion, forces, energy transfer, fluid and material behavior, chemical reactions, material composition, and environmental interactions. These experiences provide students with a practical understanding of concepts that are later applied in engineering mechanics, construction materials, hydraulics, geotechnical engineering, structural analysis, and other professional civil engineering courses.

Through laboratory activities, students develop essential skills in data collection, measurement, observation, computation, analysis, and interpretation of experimental results. They are trained to use laboratory apparatus and instruments properly, record observations accurately, process data systematically, and compare experimental findings with theoretical expectations. These activities strengthen their ability to apply the scientific method in investigating engineering-related phenomena and evaluating the reliability and validity of laboratory results.

The facilities also promote critical thinking and problem-solving by requiring students to formulate objectives, follow or design experimental procedures, analyze observed trends, identify possible sources of error, and draw evidence-based conclusions. Such experiences prepare students for more advanced engineering investigations in the BS Civil Engineering curriculum, particularly in laboratory-based and design-oriented courses where accuracy, analytical reasoning, and evidence-based decision-making are essential.

Laboratory activities reinforce the importance of accuracy, safety, ethical conduct, and professional discipline in scientific and engineering work. Students are oriented on proper laboratory procedures, responsible handling of equipment, safe use of materials, and compliance with

established safety protocols. These practices develop accountability and awareness of the importance of safety and reliability in engineering experimentation and professional practice.

Collaborative experiments and laboratory reporting activities further develop teamwork, communication, and technical documentation skills. Students work in groups to conduct experiments, assign responsibilities, verify results, and prepare laboratory reports that present procedures, data, analyses, findings, and conclusions using appropriate technical language and formats. Through these activities, the Chemistry and Physics Laboratories contribute to the attainment of relevant Course Intended Learning Outcomes (CILOs) and Student Outcomes (SOs), particularly in experimentation, problem analysis, communication, teamwork, ethical responsibility, and preparation for advanced civil engineering study.

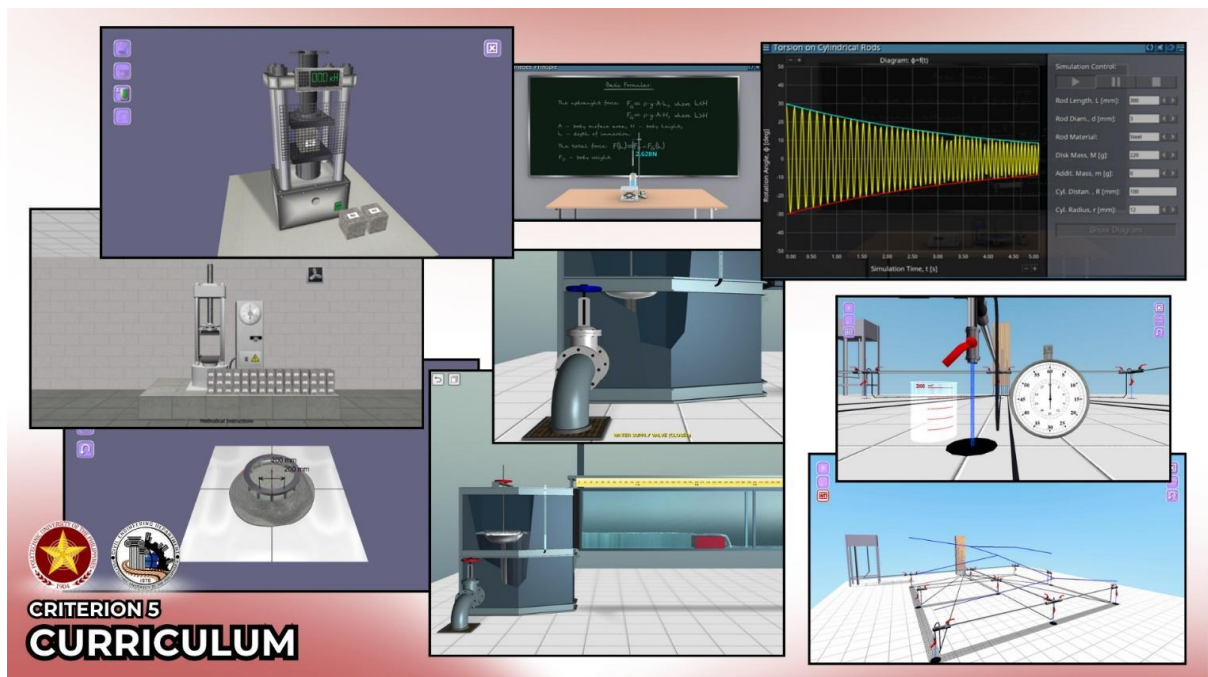
Table 5-10.11. Summary of Supported Laboratory courses and Target Student Outcomes of the College of Engineering Chemistry and Physics Laboratories

Laboratory Room	Supported Laboratory Courses in the Curriculum		Student Outcomes Primarily Addressed	Method of Supporting Student Outcomes
	2018 BSCE Curriculum	2022 BSCE Curriculum		
Chemistry and Physics Laboratories	PHYS 20034 - Physics for Engineers (Calculus-based)	PHYS 013 - Physics for Engineers (Calculus-based)	SO1. Apply knowledge of mathematics, natural science, engineering fundamentals, and a civil engineering specialization to the solution of complex civil engineering problems.	Establishes the scientific foundation required for advanced civil engineering courses. Through experimentation, data analysis, laboratory reporting, and adherence to safety procedures, students develop scientific inquiry, analytical thinking, and professional laboratory practices.
	CHEM 20024 - Chemistry for Engineers	CHEM 015 - Chemistry for Engineers	SO2. Conduct investigations of complex civil engineering problems using research-based knowledge and research methods including design of experiments, analysis, and interpretation of data, and synthesis of the information to provide valid conclusions.	

VIRTUAL LABORATORY FOR THE BS CIVIL ENGINEERING PROGRAM

The availability of virtual laboratory programs provides an effective supplementary learning environment during periods when access to physical laboratories is restricted. Situations such as pandemics, prolonged suspension of face-to-face classes, severe weather disturbances caused by supertypoons, flooding, and extreme heat index conditions can significantly disrupt the conduct of laboratory activities. The adoption of virtual laboratory platforms ensures continuity of learning by allowing students to engage in laboratory-based experiences despite limitations in physical access to campus facilities. Through these digital learning resources listed, students remain actively involved in experimentation, observation, analysis, and interpretation of engineering phenomena that are fundamental to their professional development.

- **Building Materials Virtual Lab**
- **Self-Compacting Concrete Virtual Lab**
- **Open Channel Hydraulics Virtual Lab**
- **Water Supply Networks Virtual Lab**
- **Mechanics of Solids Bodies and Liquids Virtual Lab**
- **Translational and Rotational Motion Virtual Lab**
- **Mechanical Oscillations and Waves Virtual Lab**



The Building Materials Virtual Lab and Self-Compacting Concrete Virtual Lab provide opportunities for students to investigate the behavior, properties, and performance of construction materials in a simulated environment. These platforms allow learners to perform virtual experiments involving material characterization, concrete mix design, testing procedures, and evaluation of engineering properties. By interacting with realistic simulations, students develop an understanding of material selection, quality control, durability considerations, and construction performance. The virtual environment enables repeated experimentation and exploration of various conditions that may not always be feasible within limited laboratory schedules, thereby strengthening analytical skills and reinforcing engineering concepts essential to construction engineering practice.

The Open Channel Hydraulics and Water Supply Networks virtual laboratories support learning in hydraulic and water resources engineering by allowing students to visualize fluid behavior, hydraulic principles, and network performance under varying operating conditions. Through simulation-based exercises, students can investigate flow characteristics, energy losses, channel behavior, pipe network operations, and system responses to changing parameters. These activities help students understand the relationships among theoretical equations, engineering assumptions, and actual system performance. The ability to manipulate variables and immediately observe outcomes enhances students' capacity to evaluate alternatives, interpret engineering data, and formulate technically sound decisions for water-related infrastructure systems.

The virtual laboratory modules on Mechanics of Solid Bodies and Liquids, Translational and Rotational Motion, and Mechanical Oscillations and Waves provide valuable opportunities for students to strengthen their scientific and engineering foundations. These simulations allow students to explore fundamental principles governing force, motion, equilibrium, deformation, fluid behavior, vibrations, and wave phenomena. Through interactive experimentation, learners can observe concepts that may otherwise be difficult to visualize in traditional classroom settings. Such experiences contribute to deeper conceptual understanding and help students establish strong connections between scientific theories and their applications in civil engineering analysis and design.

Table 5-10.12. Summary of Supported Laboratory and Lecture Courses of Civil Engineering Virtual Laboratory Programs

Virtual Laboratory Package	Virtual Experiments	Laboratory Courses	Virtual Laboratory Package
Building Materials Virtual Lab	Bending Strength Bulk Density Compressive Strength	Construction Materials and Testing,	Construction Materials and Testing,

	Non-Destructive Method Normal Density Setting Time True Density	Building Systems and Design	Building Systems and Design
Self-Compacting Concrete Virtual Lab	Compress Strength J-Ring Test L-Box Test Slump-Flow Test V-Funnel Test	Construction Materials and Testing	Construction Materials and Testing
Open Channel Hydraulics	Labwork - 2m Channel Labwork - 15m Channel Surface Editor	Hydraulics 2	Hydraulics 2
Water Supply Networks	Water Supply Networks Virtual Lab	Hydraulics 2	Hydraulics 2
Mechanics of Solids Bodies and Liquids	Lab 1 - Hookes Law Lab 2 - Lever Lab 3 - Parallelogram of Forces Lab 4 - Inclined Plane Lab 5 - Friction	Physics for Engineers (Calculus-based)	Statics of Rigid Bodies, Physics for Engineers (Calculus-based)
	Lab 6 - Bending of Beams Lab 7 - Torsion		Mechanics of Deformable Bodies, Physics for Engineers (Calculus-based)
	Lab 8 - Viscosimeter Lab 9 - Surface Tension Lab 10 - Archimedes Principle	Hydraulics 1 Physics for Engineers (Calculus-based)	Hydraulics 1, Physics for Engineers (Calculus-based)
Translational and Rotational Motion	Lab 1 - Uniform Accel Motion Lab 2 - Motion with Uniform Accel Lab 3 - Laws of Collisions Lab 4 - Free Fall Lab 5 - Inclined Launch Lab 6 - Gyroscope Lab 7 - Rotational Motion Lab 8 - Mom Inertia of Horizontal Rod Lab 9 - Mom Inertia of Various Bodies Lab 10 - Maxwells Wheel	Physics for Engineers (Calculus-based)	Physics for Engineers (Calculus-based) Dynamics of Rigid Bodies
Mechanical Oscillations and Waves	Lab 1 - Harmonic Oscillation Lab 2 - Elliptical Oscillation Lab 3 - Variable g Pendulum Lab 4 - Reversible Pendulum Lab 5 - Simple Harmonic Oscillations Lab 6 - Pohls Torsion Pendulum Lab 7 - Forced Harmonic Oscillation Lab 8 - Coupled Oscillations Lab 9 - Mechanical Waves Lab 10 - Speed of Sound in Air	Physics for Engineers (Calculus-based)	Physics for Engineers (Calculus-based) Dynamics of Rigid Bodies

	Lab 11 - Standing Sound Waves Lab 12 - Sound in Rods		
Building Materials Virtual Lab	Bending Strength Bulk Density Compressive Strength Non-Destructive Method Normal Density Setting Time True Density	Construction Materials and Testing, Building Systems and Design	Construction Materials and Testing, Building Systems and Design
Self-Compacting Concrete Virtual Lab	Compress Strength J-Ring Test L-Box Test Slump-Flow Test V-Funnel Test	Construction Materials and Testing	Construction Materials and Testing
Open Channel Hydraulics	Labwork - 2m Channel Labwork - 15m Channel Surface Editor	Hydraulics 2	Hydraulics 2
Water Supply Networks	Water Supply Networks Virtual Lab	Hydraulics 2	Hydraulics 2
Mechanics of Solids Bodies and Liquids	Lab 1 - Hookes Law Lab 2 - Lever Lab 3 - Parallelogram of Forces Lab 4 - Inclined Plane Lab 5 - Friction	Physics for Engineers (Calculus-based)	Statics of Rigid Bodies, Physics for Engineers (Calculus-based)

Overall, the collection of virtual laboratory programs provides an adequate and effective mechanism for sustaining laboratory-based learning when conventional face-to-face instruction cannot be conducted for extended periods. While virtual laboratories do not completely replace the value of physical experimentation, they offer meaningful opportunities for students to develop competencies in engineering analysis, experimentation, investigation, problem-solving, use of modern engineering tools, communication, teamwork, and professional responsibility. Their integration into the Civil Engineering program ensures that student learning outcomes continue to be achieved even during disruptions caused by public health emergencies, natural disasters, severe weather events, or other circumstances that limit access to physical laboratory facilities.

Through structured laboratory activities, students develop experimental planning skills, operate modern engineering equipment, and interpret results in the context of real-world civil engineering applications. These experiences reinforce the outcome requiring graduates to develop and conduct appropriate experimentation and use engineering judgment.

ATTACHMENTS

[Attachment C5-01 – Technical Laboratory Manual and Reports](#)

[Attachment C5-02 – Laboratory and Field-Based Learning Activities](#)

[Attachment C5-03 – Major Design or Capstone Project Activities](#)

[Attachment C5-04 – Conduct of Integrative and In-House Review Classes](#)

[Attachment C5-05 – Internship and Industry Immersion Activities](#)

[Attachment C5-07 – Professional Development, Certification, and Lifelong Learning Activities](#)

[Attachment C5-08 – Assessment Rubrics](#)

[Attachment C5-09 – Table 5-2.1. Relationship of Civil Engineering Courses to the Program Educational Objectives - 2018 BSCE Curriculum](#)

[Attachment C5-10 – Table 5-2.2. Relationship of Civil Engineering Courses to the Program Educational Objectives - 2022 BSCE Curriculum](#)

[Attachment C5-11 – Table 5-3.1 Mapping of Course Intended Learning Outcomes \(CILO\) and Student Outcomes \(SO\) - 2018 BSCE Curriculum](#)

[Attachment C5-12 – Table 5-3.2 Mapping of Course Intended Learning Outcomes \(CILO\) and Student Outcomes \(SO\) - 2022 BSCE Curriculum](#)

[Attachment C5-13 – Summary of the Processes Underwent During the 2022 BSCE Curriculum Revision](#)

[Attachment C5-14 – Figure 5.6.1. Comparison of the 2018 and 2022 BSCE Curriculum Credit Units with the Minimum Requirements](#)

[Attachment C5-15 – Figure 5.6.2. Percentage Distribution of Credit Units – 2018 BSCE Curriculum](#)

[Attachment C5-16 – Figure 5.6.3. Percentage Distribution of Credit Units – 2022 BSCE Curriculum](#)

[Attachment C5-17 – Table 5-6.3. Alignment of the 2018 BSCE Curriculum with the PUP Mission and Vision Statements](#)

[Attachment C5-18 – Table 5-6.4. Alignment of the 2022 BSCE Curriculum with the PUP Mission and Vision Statements](#)

[Attachment C5-19 – Complex Engineering Activities](#)

[Attachment C5-21 – Complexity Analysis Matrix of Complex Engineering Problems](#)

[Attachment C5-21 – Complexity Analysis Matrix of Complex Engineering Activities](#)

[Attachment C5-22 – Analytic Rubric for Complexity of Engineering Problems](#)

[Attachment C5-23 – Analytic Rubric for Complexity of Engineering Activities](#)

[Attachment C5-24 – Minutes of the Government Industry Advisory Board Meetings](#)

[Attachment C5-25 – Host Training Establishment Distribution and Visits](#)